University College London, 2025-2026
Econ 0039 – Advanced Macro – Franck Portier

Problem set 2: Business Cycles
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I – Real Business Cycles in the UV Model

This problem takes you through the solving of the RBC model in the UV setting of Lecture 4 on Business Cycles. The economy features a representative firm and a representative household. The representative household owns the firm, and receives its profits π. There is a unique good (numéraire) that can be used for consumption c and investment x Technology is $y = c + x = A\ell^\alpha$ with $0 < \alpha < 1$. y is total output.

The representative household utility is of the UV form, and is given by

$$U(c,\ell) + V(x,\theta) = \gamma \ln c - B\ell + \theta \ln x$$

and the household budget constraint is

$$c + x = w\ell + \pi$$

1 – Write the profit maximization problem of the firm and derive the first order condition to this maximization.

2 – Write the utility maximization problem of the household and derive the two first order conditions to this maximization.

3 – Define the general equilibrium of this economy in terms of quantities c, ℓ, x, wage w and profit π.

4 – Use the equilibrium definition to compute the equilibrium levels of c, ℓ, x and y.

5 – For each variable $v \in \{c, \ell, x, y\}$ and each exogenous variable $z \in \{A, \theta, B, \gamma\}$, compute $\frac{\partial v}{\partial z}$ and its sign.

6 – Interpret each shock ∂z and the response of the economy.

7 – How good is that model as a theory of Business Cycles comovements?

II – Gains From Trade Cycles in the UV Model

This problem takes you through the solving of the RBC model in the UV setting of Lecture 4 on Business Cycles. The economy produces two goods, a consumption good c (the numéraire) and an investment good x. There are two firms, each one being the representative firm of its sector. Technology is $c = A\ell^c$ in the consumption sector (price 1) and $x = A\ell^x$ in the investment sector (price q). Wages are w^c and w^x and profits are $\pi^c = A\ell^c - w^c\ell^c$ and $\pi^x = qA\ell^x - w^x\ell^x$. Total profits in the economy are denoted π. GDP is given by $y = c = q\,x$.

There are two households; a representative consumption-worker and a representative investment-worker. Each household works only in one of the two sectors. They both have UV preferences, that are given by $\ln c^c - B^c\ell^c + \theta \ln x^c$ for the c-worker and $\ln(c^x - B^x(\ell^x)^2) + \theta \ln x^x$ for the x-worker.

The c-worker receives a fraction ν^c of total profits and the x-worker a fraction $(1-\nu^c)$. The households budget constraints are $c^c + qx^c = w^c\ell^c + \nu^c\pi$ for the c-worker and $c^x + qx^x = w^x\ell^x + (1-\nu^c)\pi$ for the x-worker.

1 – Write the profit maximization problem of each firms and derive the first order condition to this maximization. Show that profits will be null in equilibrium.

2 – Write the utility maximization problem of each household and derive the two first order conditions to this maximization.

3 – Define the general equilibrium of this economy in terms of quantities $c, x, \ell, c^c, x^c, \ell^c, c^x, x^x, \ell^x$ and prices w^c, w^x, q

4 – Show that the amount of trade between the two agents will be given by $q\,x^c = c^x$. Interpret that equation.

5 – Use the equilibrium definition to compute the equilibrium levels of c, x, ℓ, q and y.

6 – For each variable $v \in \{c, x, \ell, q, y\}$ and each exogenous variable $z \in \{A, \theta\}$, compute $\frac{\partial v}{\partial z}$ and its sign.

7 – Interpret each shock ∂z and the response of the economy.

8 – How good is that model as a theory of Business Cycles comovements?

III – Monetary Policy and Sticky Prices in a "UV" model

This problem considers a monetary model in the UV setting of Lecture 4 on Business Cycles. The economy features a representative firm, a representative household and a Central Bank. The representative household owns the firm and receives its profits π. There is a unique produced good that can be used for consumption c and investment x. Its price is p. Money is the numéraire good. It is supplied in quantity $\overline{m}$ by the Central Bank and is distributed to the housholds.

The representative household utility is of the UV form, and is given by

$$U\left(c, \ell, \frac{m}{p}\right) + V(x, \theta) = \ln c - B\ell + \nu \ln\left(\frac{m}{p}\right) + \theta \ln x,$$

where m is the money demand of the household, ℓ is labour. Agents demand money because $\frac{m}{p}$ brings some utility.

The household budget constraint is

$$pc + px + m \leq w\ell + \overline{m} + \pi$$

where w is the nominal wage.

The representative firm produces a single good according to the technology $y = A\ell^{\alpha}$ with $0 < \alpha < 1$. y is split into consumption c and investment x.

Finally, all markets are competitive.

Assume first that prices are flexible.

1 – Write the profit maximization problem of the firm and the first order conditions to this maximization. Derive good supply y, labour demand ℓ^d and profit π as a function of w, p and parameters.

2 – Write the utility maximization problem of the household and the four first order conditions to this maximization, with respect to c, x, ℓ and m. Derive consumption demand c, investment demand x, money demand m and labour supply ℓ^s as a function of w, p, π, $\overline{m}$ and parameters.

3 – Define the general equilibrium of this economy. Find the equilibrium levels of quantities c^E, ℓ^E, x^E, y^E and m^E, wage w^E and price p^E.

4 – Can a technology shock ($\partial A > 0$), an investment demand shock ($\partial\theta > 0$) or a monetary policy shock ($\partial\overline{m} > 0$) create business cycles comovements? Discuss.

Assume now that the price of the good is fixed at a level $p^F > p^E$.

Let's consider a fix-price "equilibrium" in which the money market clears, but the good and labour market may not. We assume *voluntary exchange*, meaning that households are not forced to work or spend more than what is optimal for them, neither are firms forced to hire or produce more that what is optimal for them.

Because $p^F > p^E$, the structure of the equilibrium will be as follows. Money market will be in equilibrium, which will determine equilibrium wage w^F. Output y^F will be determined by good demand $c^F + x^F$. To that level of output will correspond a demand for labour ℓ^F. That level of labour demand will be the equilibrium employment.

5 – Find the fix-price equilibrium values $c^F, \ell^F, x^F, y^F, m^F$ and w^F.

6 – Can a technology shock ($\partial A > 0$), an investment demand shock ($\partial\theta > 0$) or a monetary policy shock ($\partial\overline{m} > 0$) create business cycles comovements in that fix-price economy? Discuss.

IV – Discussing Document 1 and 2 - "Of shocks and horrors, The causes of booms and busts", Sep 28th 2002, The Economist - and - "Reconciling Hayek's and Keynes' views of recessions", Paul Beaudry, Dana Galizia and Franck Portier, Vox.eu, May 2014

1 – To which theory do refer the two models we have seen in Lecture 4?

2 – Describe the Austrian theory of Business Cycles.

3 – What are the evidence for an "overaccumulation" theory of recessions?

4 – Why Austrian and Keynesian views of recessions might not be completely opposed?

V – Discussing Document 3 - “NBER Business Cycle Dating Committee Announcement July 19, 2021”

1 – Explain how did the paid furlough scheme out in place during the pandemics affected the measurement used by the NBER dating committee

2 – Why dating a recession that is only two months long?

VI – Discussing Document 4 - “The Challenges of Estimating Potential Output in Real Time”, Robert W. Arnold, Federal Reserve Bank of St. Louis Review, July/August 2009, 91(4), pp. 271-90.

You can skip the last section of the text that is entitled“*Challenges Associated with Projecting Potential Output*”

1 – What is the definition of potential output?

2 – Why is the concept of potential output useful to evaluate the causes of inflationary pressures?

3 – What is the Okun’s law?

4 – What is the NAIRU?

5 – In Table 1, which variable is the main contributor to potential output growth?

6 – What are the main challenges in estimating the output gap?

VII – Discussing Document 5 - “Business Cycles: real Facts and a Monetary Myth”, Finn E. Kydland and Edward C. Prescott, Federal Reserve Bank of Minneapolis Quarterly Review, Spring 1990, 14(2), pp. 3-18.

1 – What is the Koopman’s “measurement without theory” critique to the Burns $ Mitchell NBER approach to the measure of Business Cycles?

2 – What are the key growth facts highlighted by Solow?

3 – What is Lucas’s definition of Business Cycles?

4 – What is the “monetary myths” according to Kydland & Prescott?

5 – What is the Frisch’s impulsion/propagation approach?

6 – What is the Slutsky effect?

VIII – Discussing Document 6 - “The History of Econometric Ideas, Part I: Business Cycles”, Mary S. Morgan, Cambridge University Press, 1990

1 – This is a long extract of a book written by Mary Morgan, that proposes an archeology of the concept and measure of the Business Cycle in the economic literature. Read it only if you find it are interesting. You might want to identify in particular the ideas and work of the following economists: William Stanley Jevons, Henry Ludwell Moore, Clément Juglar, Wesley Clair Mitchell, Warren M. Persons, Nikolai Dmitriyevich Kondratiev, George Udny Yule, Eugen Slutsky, Ragnar Frish, Jan Tinbergen, John Maynard Keynes.

Appendix

Document 1 - "Of shocks and horrors, The causes of booms and busts", Sep 28th 2002, The Economist

In the bible, Joseph tells the pharaoh to expect seven years of plenty followed by seven lean years. That was, perhaps, the first documented business cycle – and the first (and probably last) example of an accurate economic forecast. Recorded data on business cycles go back to the early 19th century, but economists still cannot agree about what causes contractions and expansions in economic activity.

Economists have come up with all sorts of explanations, from the effect of sunspot activity on climate to the alignment of planets and their magnetic forces. The current preference is to look for more down-to- earth causes, which come in two main varieties: those that explain the business cycle as a self-perpetuating process, and those that blame recessions on shocks or policy mistakes. The main theories are:

Exogenous shocks. Recessions, it is argued, are caused by unexpected events, such as the rise in oil prices in the mid-1970s or, as some (incorrectly) tried to argue, the terrorist attacks on September 11th last year. If so, recessions are, by definition, totally unpredictable. They cannot be prevented, but once they have arrived governments can use fiscal and monetary policies to cushion demand.

Keynesian theory. John Maynard Keynes blamed recessions on the inherent instability of investment caused by "animal spirits": swings in the mood of producers, from optimism to pessimism. As investment slumps, jobs and household incomes fall, amplifying the initial drop in demand. Unemployment rises because workers will not accept the pay cuts required to price the jobless back into work. So, to bring the economy back to full employment, the government needs to pursue expansionary policies.

Real business-cycle theory. This theory, which emerged in the early 1980s, sees productivity shocks as the cause of economic fluctuations. For example, if productivity falls, current returns decline, says the theory, so workers and firms choose to work less and take more leisure. Rather than explaining the cycle in terms of market failure, as Keynes did, real business-cycle theory views a recession as the optimal response by households and firms to a shift in productivity. If so, there is no point in governments stimulating the economy. But most economists find this theory hard to swallow. Mike Mussa, a former chief economist at the IMF, and now at the Institute for International Economics, describes it as "the theory according to which the 1930s should be known not as the Great Depression, but the Great Vacation."

Policy mistakes. The late Rudi Dornbusch, an economist at the Massachusetts Institute of Technology, once remarked: "None of the postwar expansions died of old age, they were all murdered by the Fed." Almost every recession since 1945, with the exception of last year's, was preceded by a sharp rise in inflation that forced central banks to raise interest rates. The first mistake was to allow economies to overheat; the second to slam on the brakes too hard. This theory gave rise to the popular belief that recessions could be avoided so long as governments pursued prudent monetary policies to keep inflation low and stable. Yet the recessions in Japan after the 1980s bubble and America more recently suggest that price stability does not prevent booms and busts.

Austrian business-cycle theory. This is the oldest, developed by Austrian economists such as Ludwig von Mises and Friedrich Hayek in the early 20th century. Unlike Keynes, who thought recessions were caused by insufficient demand, these economists put them down to excess supply brought about by overinvestment. As a result of mutually reinforcing movements in credit, investment and profits, each boom contains the seeds of the subsequent recession and each recession the seeds of the subsequent boom. According to Hayek, output fluctuates because the short-term interest rate for loans diverges from the "natural" or equilibrium interest rate – the rate at which the supply of saving from households equals the demand for investment funds by firms. If central banks hold interest rates below this rate, credit and investment will rise too rapidly, and consumers will not save enough. This creates a mismatch between future output (which will increase as a result of higher investment) and future spending (which will fall as a result of lower saving today). Cheap credit and inflated profit expectations cause both overinvestment and "malinvestment" in the wrong kind of capital. The mismatch between saving and investment will eventually push up interest rates, making some previous investments unprofitable. Too much capacity will also reduce profits. Investment collapses, ushering in a recession. As excess capacity is cut, profits rise and investment eventually recovers. According to this theory, central banks would not be able to avoid a downturn by heading off a rise in interest rates. The only way to prevent the cycle from turning is to inject ever more credit, which becomes unsustainable. A recession is inevitable, and indeed necessary to correct the imbalance between saving and investment.

A tour of the Austrian Alps

In the second half of the 20th century, self-perpetuating theories of the business cycle were almost completely ignored in favour of theories that stressed shocks or policy mistakes. Most recessions were caused largely by economies overheating, forcing central banks to raise interest rates. Now, however, the fall in inflation has brought the inherently cyclical behaviour of credit, investment and profits to the fore.

In the late 1990s inflation rose only slightly, leading most economists to believe that growth would continue in America. Only a few economists, such as John Makin, at the American Enterprise Institute, and Stephen King, at HSBC, recognised that this cycle was different: that it was an investment-led boom that carried the seeds of its own destruction. The recent business cycles in both America and Japan displayed many "Austrian" features. Hayek argued that the natural rate of interest could rise if faster productivity growth increased expectations about profits and hence investment opportunities. This is what happened in Japan in the 1980s and in America in the 1990s. If such a shift in investment occurs, central banks need to raise interest rates. But because inflation was low (and because Austrian economics had long gone out of fashion), the Fed and the Bank of Japan failed to do so. The cost of capital therefore fell below its expected return, fuelling a surge in credit, equity prices and investment.

Investment normally accounts for about one-sixth of America's GDP, but during the three years to 2000 the investment boom pushed up its share of GDP growth to one-third. Overinvestment caused the return on capital to decline. Anybody who looked at the profits of corporate America in 2000, as reported in the national accounts (which allow for the true cost of stock options) rather than by the companies themselves, should have seen trouble coming. Profits had been falling since 1998, as before every previous recession (see chart 2). Investment will not rebound until excess capacity has been cleared and profits have improved.

Strict Austrian-school disciples would argue that because the current downturn is due to overinvestment, the Fed's repeated easing of interest rates is wrong; it delays the correction of past excesses. The present economic and financial disruption is needed to bring saving and investment back into balance. But most economists today would accept that in the face of a severe recession central banks need to act. Even the Austrian economists recognised that a collapse in confidence could push the economy into a much deeper recession than necessary. Monetary and fiscal policy therefore still has a role – not to avoid recession, but to head off a downward economic spiral.

Many economists still do not accept that this recession has been quite different from all previous post-war cycles, and that the shape of the recovery will therefore also be different. Investment-led downturns tend to last longer because it takes much more time to eliminate financial excesses than to tame inflation. High debts and excess capacity will restrain growth. The complete rejection of Austrian-school ideas over the past half- century partly reflected a desire by economists to develop a framework that could explain all business cycles. The truth is that there is no single cause of cycles. Sometimes an oil-price shock or a policy mistake may trigger a recession, but the endogenous movements in credit, investment and profits are also always at play. Indeed, the Austrian cycle may become more common again if, as this survey will argue, financial liberalisation has made bubbles in credit and investment more likely.

Document 2 - "Reconciling Hayek's and Keynes' views of recessions", Paul Beaudry, Dana Galizia and Franck Portier, Vox.eu, May 2014

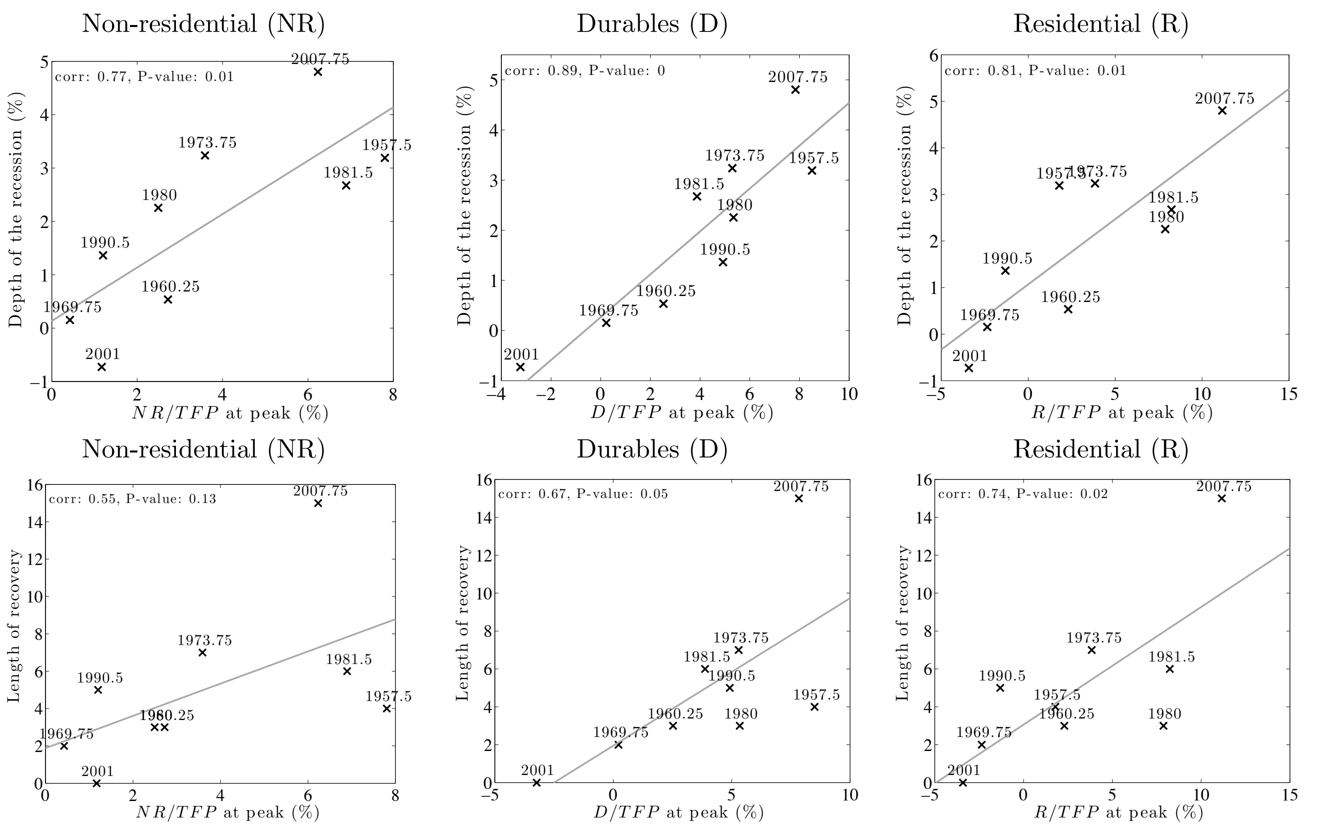
There remains considerable debate regarding the causes and consequences of recessions. Two views that are often presented as opposing, and which created controversy in the recent recession and its aftermath, are those associated with the ideas of Hayek and Keynes[1].

The Hayekian perspective is generally associated with viewing recessions as a necessary evil. According to this view, recessions mainly reflect periods of liquidation resulting from past over- (or mal-) accumulation of capital goods; periods of hangover as Paul Krugman wrote in a 1998 article in Slate. A situation where the economy needs to liquidate an excess of capital can quite naturally give rise to a recession, but government spending aimed at stimulating activity, it is argued, is not warranted since it would mainly delay the needed adjustment process and thereby postpone the recovery. This was the view of U.S. Secretary of the Treasury Mellon during the great depression[2], but it has also been considered as an accurate description of the Asian crisis of 1997 and the current great recession.

In contrast, the Keynesian view suggests that recessions reflect periods of deficient aggregate demand where the economy is not effectively exploiting the gains from trade between individuals. According to this view, policy interventions aimed at increasing investment and consumption are generally desirable, as they favour the resumption of mutually beneficial trade between individuals.

In a recent research (CEPR Discussion Paper #9966), we show that those two views are not contradictory. First, it is a fact that over the last 70 years, U.S. recessions are longer and more severe when they have been preceded by periods of particularly high accumulation of capital goods, durable goods and houses, as illustrated on Figure 1.

Depth of recession and length of recovery vs. cumulated investment



Note: Horizontal axis is capital "over-accumulation", defined as cumulated investment over past 10 years, divided by TFP and detrended using a cubic trend. Vertical axis is either depth of recession, measured as percentage difference in real GDP from

[1] See Wapshott (2012) for a popular account of the Hayek-Keynes controversy.

[2] Andrew W. Mellon is famous for his policy advice to President Hoover: "liquidate labor, liquidate stocks, liquidate the farmers, liquidate real estate. It will purge the rottenness out of the system."

peak to subsequent trough, or length of recovery, measured as the number of quarters it takes for real GDP to reach again the peak level. Data are US postwar National Accounts, business cycle datation is the NBER one.

As can be seen in this Figure, there is a very strong positive correlation between our measure of capital over-accumulation prior to a recession and either our measure of the subsequent severity of the recession or our measure of the length of the recovery. While this evidence is at best only suggestive, it does support the notion that severe recessions have generally been preceded by periods of very high investment relative to TFP growth. An interesting aspect of Figure 1 is that we see that severe recessions were generally preceded by high accumulation of all three classes of capital: housing, durables, and physical capital. This pattern would be consistent, for example, with over-accumulation caused by periods of excessively lax credit.

But, second, those "Hayekian" facts do not mean that liquidation without public intervention is desirable and painless. We indeed show that excessive capital accumulation puts the economy in a "Keynesian" regime in which it suffers from unemployment low aggregate demand.

We develop a theory that describes how the economy adjusts when it inherits from the past an excessive amount of capital goods, which could be in the form of houses, durable goods or productive capital. The building blocks of our theory are decentralized markets and search frictions on the labour market. We do not address the question of why the economy may have over-accumulated in the past[3] but ask how it reacts to such an over-accumulation once it is realized. As suggested by Hayek, such a situation can readily lead to a recession as less economic activity is generally warranted when agents want to deplete past over-accumulation. However, because of the endogenous emergence of unemployment risk in our set-up, the size and duration of the recession induced by the need for liquidation is not socially optimal. In effect, the reduced gains from trade induced by the need for liquidation creates a multiplier process that leads to an excessive reduction in activity. Although prices are free to adjust, the liquidation creates a period of deficient aggregate demand where economic activity is too low because people spend too cautiously due to increased unemployment risk. Individual spending is low because economic agents need to liquidate their stock of capital, durable goods or houses. Because spending is low, firms post less vacancies, which increases the risk of being unemployed, boosts precautionary savings and further reduces demand. In this sense, we argue that liquidation and deficient aggregate demand should not be viewed as alternative theories of recessions but instead should be seen as complements, where past over-accumulation may be a key driver of periods of deficient aggregate demand.

This perspective also makes salient the trade-offs faced by policy. In particular, a policy-maker in our environment faces an unpleasant trade-off between the prescriptions emphasized by Keynes and Hayek. On the one hand, a policy-maker would want to stimulate economic activity during a liquidation-induced recession because precautionary savings is excessively high. On the other hand, the policy-maker also needs to recognize that intervention will likely postpone recovery, since it slows down the needed depletion of excess capital. In the above-mentioned research, we provide a simple framework where both of these forces are present and can be compared. Although there are no obvious theoretical reasons why one force would always dominate, we find that during liquidation periods, aggregate demand policies are socially optimal in our framework. Casual observation also suggests that liquidations are socially painful periods that we believe should be accompanied by some aggregate demand policy.

This research also shed some light on the recent debate on secular stagnation, as put forward by Lawrence Summers. The type of decentralized market economy that we consider functions quite efficiently in growth periods when it is far below its balanced growth path, while simultaneously functioning particularly inefficiently when it is near its balanced growth path. When the economy is far below its balanced growth path level of capital, demand for capital is very strong and unemployment risk is therefore minimal. In contrast, when the economy is close to its balance growth path, it is always in an unemployment zone because the capital stock is relatively large. The associated unemployment risk then causes households to increase precautionary savings, which sustains in turn the existence of unemployment, giving rise to a permanent paradox of thrift situation in which abundance creates scarcity.

While the main mechanism in our theory has many precursors in the literature[4], we believe that our setup illustrates most clearly (1) how unemployment risk gives rise to a multiplier process for demand shocks even in the absence of price stickiness or increasing returns, (2) how this multiplier process can be ignited by periods of liquidation, and (3) how fiscal policy can and cannot be used to counter the process.

References

[3]There are several reason why an economy may over-accumulate capital. For example, agents may have had overly optimistic expectations about future expected economic growth that did not materialize, as in Beaudry and Portier [2004], or it could have been the case that credit supply was unduly subsidized either through explicit policy, as argued in Mian and Sufi [2010] and Mian, Sufi, and Trebbi [2010], or as a by-product of monetary policy, as studied by Bordo and Landon-Lane [2013].

[4]Our model structure is closely related to that presented in Guerrieri and Lorenzoni [2009], and also shares many features with Heathcote and Perri [2012].

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Cambridge, July 19, 2021 - The Business Cycle Dating Committee of the National Bureau of Economic Research maintains a chronology of the peaks and troughs of US business cycles. The committee has determined that a trough in monthly economic activity occurred in the US economy in April 2020. The previous peak in economic activity occurred in February 2020. The recession lasted two months, which makes it the shortest US recession on record.

The NBER chronology does not identify the precise moment that the economy entered a recession or expansion. In the NBER's convention for measuring the duration of a recession, the first month of the recession is the month following the peak and the last month is the month of the trough. Because the most recent trough was in April 2020, the last month of the recession was April 2020, and May 2020 was the first month of the subsequent expansion.

In determining that a trough occurred in April 2020, the committee did not conclude that the economy has returned to operating at normal capacity. An expansion is a period of rising economic activity spread across the economy, normally visible in real GDP, real income, employment, industrial production, and wholesale-retail sales. Economic activity is typically below normal in the early stages of an expansion, and it sometimes remains so well into the expansion.

The committee decided that any future downturn of the economy would be a new recession and not a continuation of the recession associated with the February 2020 peak. The basis for this decision was the length and strength of the recovery to date.

The Month of the Trough

In determining the date of a monthly peak or trough, the committee considers a number of indicators of employment and production. In the current case, all of those indicators point clearly to April 2020 as the month of the trough. On the employment side, the committee normally views the payroll employment measure produced by the Bureau of Labor Statistics (BLS), which is based on a large survey of employers, as the most reliable comprehensive estimate of employment. This series reached a clear trough in April before rebounding strongly the next few months and then settling into a more gradual rise. However, the committee recognized that this survey was affected by special circumstances associated with the COVID-19 pandemic in early 2020. In the survey, individuals who are paid but not at work are counted as employed, even though they are not in fact working or producing. Workers on paid furlough, who became more numerous during the pandemic, thus resulted in an overcount of people working. Accordingly, the committee also considered the employment measure from the BLS household survey, which excludes individuals who are paid but on furlough. This series also shows a clear trough in April. The committee concluded that both employment series were thus consistent with a business cycle trough in April.

The committee believes that the two most reliable comprehensive estimates of aggregate production are the quarterly estimates of real Gross Domestic Product (GDP) and of real Gross Domestic Income (GDI), both produced by the Bureau of Economic Analysis (BEA). Both series attempt to measure the same underlying concept, but GDP does so using data on expenditure while GDI does so using data on income. These measures estimate production that occurred over an entire quarter and are not available monthly. The most comprehensive income-based monthly measure of aggregate production is real personal income less transfers, from the BEA. The deduction of transfers is necessary because transfers are included in personal income but do not arise from production. This measure reached a clear trough in April 2020. The most comprehensive expenditure-based monthly measure of aggregate production is monthly real personal consumption expenditures (PCE), published by the BEA. This series also reached a clear trough in April 2020.

The Quarter of the Trough

In determining the date of a quarterly peak or trough, the committee relies on real GDP and real GDI as published by the BEA, and on quarterly averages of key monthly indicators. As with the monthly trough, there is strong agreement across the indicators about the timing of the quarterly trough: the indicators all point strongly to 2020Q2. Quarterly real GDP and real GDI both reached clear troughs in 2020Q2, as did quarterly averages of both the monthly payroll and household employment series.

Further Comments

The NBER's traditional definition of a recession involves a decline in economic activity that lasts more than a few months. For example, the previous shortest recession occurred in the first half of 1980 and lasted six months. However, in deciding whether to identify a recession, the committee weighs the depth of the contraction, its duration, and whether economic activity declined broadly across the economy (the diffusion of the downturn). The recent downturn had different characteristics and dynamics than prior recessions. Nonetheless, the committee concluded that the unprecedented magnitude of the decline in employment and production, and its broad reach across the entire economy, warranted the designation of this episode as a recession, even though the downturn was briefer than earlier contractions.

Committee members participating in the decision were: Robert Hall, Stanford University (chair); Robert Gordon, Northwestern University; James Poterba, MIT and NBER President; Valerie Ramey, University of California, San Diego; Christina Romer, University of California, Berkeley; David Romer, University of California, Berkeley; James Stock, Harvard University; Mark Watson, Princeton University.

Document 4 - “The Challenges of Estimating Potential Output in Real Time”, Robert W. Arnold, Federal Reserve Bank of St. Louis Review, July/August 2009, 91(4), pp. 271-90.

The Challenges of Estimating Potential Output in Real Time

Robert W. Arnold

Potential output is an estimate of the level of gross domestic product attainable when the economy is operating at a high rate of resource use. A summary measure of the economy's productive capacity, potential output plays an important role in the Congressional Budget Office (CBO)'s economic forecast and projection. The author briefly describes the method the CBO uses to estimate and project potential output, outlines some of the advantages and disadvantages of that approach, and describes some of the challenges associated with estimating and projecting potential output. Chief among these is the difficulty of estimating the underlying trends in economic data series that are volatile, subject to structural change, and frequently revised. Those challenges are illustrated using examples based on recent experience with labor force growth, the Phillips curve, and labor productivity growth. (JEL E17, E32, E62)

Federal Reserve Bank of St. Louis *Review*, July/August 2009, *91*(4), pp. 271-90.

Assessing current economic conditions, gauging inflationary pressures, and projecting long-term economic growth are central aspects of the Congressional Budget Office (CBO)'s economic forecasts and baseline projections. Those tasks require a summary measure of the economy's productive capacity. That measure, known as potential output, is an estimate of "full-employment" gross domestic product (GDP)—the level of GDP attainable when the economy is operating at a high rate of resource use. Although it is a measure of the productive capacity of the economy, potential output is not a technical ceiling on output that cannot be exceeded. Rather, it is a measure of *sustainable* output, where the intensity of resource use is neither adding to nor subtracting from short-run inflationary pressure. If actual output exceeds its potential level, then constraints on capacity begin to bind, restraining further growth and contributing to inflationary pressure. If output falls below potential, then resources are lying idle and inflation tends to fall.

In addition to being a measure of aggregate supply in the economy, potential output is also an estimate of trend GDP. The long-term trend in real GDP is generally upward as more resources—primarily labor and capital—become available and technological change allows more productive use of existing resources. Real GDP also displays short-term variation around that long-run trend, influenced primarily by the business cycle but also by random shocks whose sources are difficult to pinpoint. Analysts often want to estimate the underlying trend, or general momentum, in GDP by removing short-term variation from it. A distinct, but related, objective is to remove the fluctuations that arise solely from the effects of the business cycle.

Potential output plays a role in several areas associated with the CBO's economic forecast. In particular, we use potential output to set the level of real GDP in its medium-term (or 10-year)

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projections. In doing so, we assume that the gap between GDP and potential GDP will equal zero on average in the medium term. Therefore, the CBO projects that any gap that remains at the end of the short-term (or two-year) forecast will close during the following eight years. We also use the level of potential output as one gauge of inflationary pressures in the near term. For example, an increase in inflation that occurs when real GDP is below potential (and monetary growth is moderate) can probably be attributed to temporary factors and is unlikely to persist. Finally, potential output is an important input in computing the standardized-budget surplus or deficit, which the CBO uses to evaluate the stance of fiscal policy and reports regularly as part of its mandate.

THE CBO METHOD FOR ESTIMATING POTENTIAL OUTPUT

The CBO model for estimating potential output is based on the framework of a neoclassical, or Solow, growth model. The model includes a Cobb-Douglas production function for the nonfarm business (NFB) sector with two factor inputs, labor (measured as hours worked) and capital (measured as an index of capital services provided by the physical capital stock), and total factor productivity (TFP), which is calculated as a residual. NFB is by far the largest sector in the economy, accounting for 76 percent of GDP in 2007, compared with less than 10 percent for each of the other sectors. For smaller sectors of the economy, including farms, federal government, state and local government, households, and nonprofit institutions, simpler equations are used to model output. Those equations generally relate the growth of output in a sector to the growth of the factor input—either capital or labor—that is more important for production in that sector.[1]

To compute historical values for potential output, we cyclically adjust the factor inputs and then combine them using the production function. Cyclical adjustment removes the variation in a series that is attributable solely to business cycle fluctuations. Ideally, the resulting series will reflect not only the trend in the series, but also will be benchmarked to some measure of capacity in the economy and, therefore, can be interpreted as the potential level of the series.

For most variables in the model, we use a cyclic-adjustment equation that combines a relationship based on Okun's law with linear time trends to produce potential values for the factor inputs. Okun (1970) postulated an inverse relationship between the size of the output gap (the percentage difference between GDP and potential GDP) and the size of the unemployment gap (the difference between the unemployment rate and the "natural" rate of unemployment). According to that relationship, actual output exceeds its potential level when the unemployment rate is below the natural rate of unemployment and falls short of potential output when the unemployment rate is above its natural rate (Figure 1).

For the natural rate of unemployment, we use the CBO estimate of the non-accelerating inflation rate of unemployment (NAIRU). That rate corresponds to a particular notion of full employment—the rate of unemployment that is consistent with a stable rate of inflation. The historical estimate of the NAIRU derives from an estimated relationship known as a Phillips curve, which connects the change in inflation to the unemployment rate and other variables, including changes in productivity trends, oil price shocks, and wage and price controls. The historical relationship between the unemployment gap and the change in the rate of inflation appears to have weakened since the mid-1980s.[2] However, a negative correlation still exists; when the unemployment rate is below the NAIRU, inflation tends to rise, and when it exceeds the NAIRU, inflation tends to fall. Consequently, the NAIRU, while it is less useful for inflation forecasts, is still useful as a benchmark for potential output.

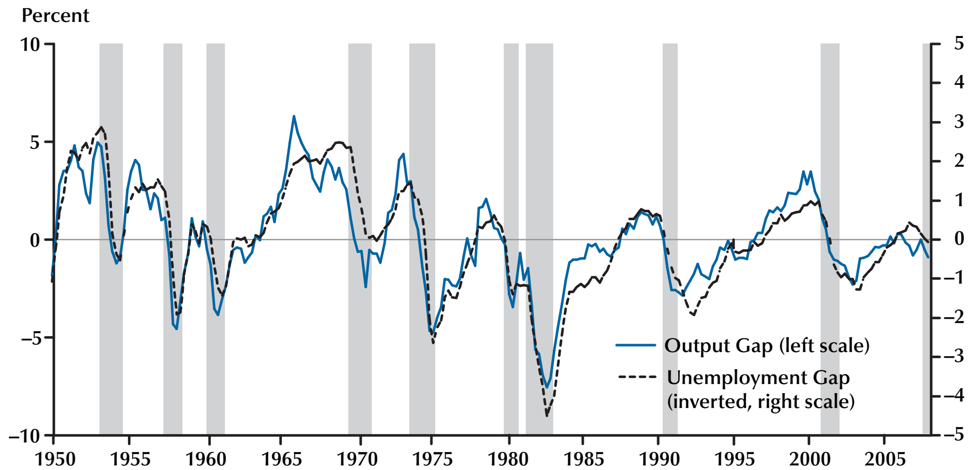
The assumption of linear time trends in the cyclic-adjustment equation implies that the poten-

[1] This section gives an overview of the CBO method. For a more complete description, see CBO (2001).

[2] For a description of the procedure used to estimate the NAIRU, see Arnold (2008).

Figure 1

Okun's Law: The Output Gap and the Unemployment Gap



NOTE: Gray bars in Figures 1, 2, 3, 5, 6, and 8 indicate recession as determined by the National Bureau of Economic Research.

tial version of each variable grows at a constant rate during each historical business cycle. Rather than constraining the potential series to follow a single time trend throughout the entire sample, the model allows for several time trends, each beginning at the peak of a business cycle. Defining the intervals of the time trends using full business cycles helps to ensure that the trends are estimated consistently throughout the historical sample. Most economic variables have distinct cyclical patterns—they behave differently at different points in the business cycle. Specifying breakpoints for the trends that occur at different stages of different business cycles (say, from trough to peak) would likely provide a misleading view of the underlying trend.

The cyclic-adjustment equation has the following form:

$$\log(X) = \text{Constant} + \alpha\left(U - U^*\right) + \beta_1 T_{1953} + \beta_2 T_{1957} + \ldots + \beta_8 T_{1990} + \varepsilon, \quad (1)$$

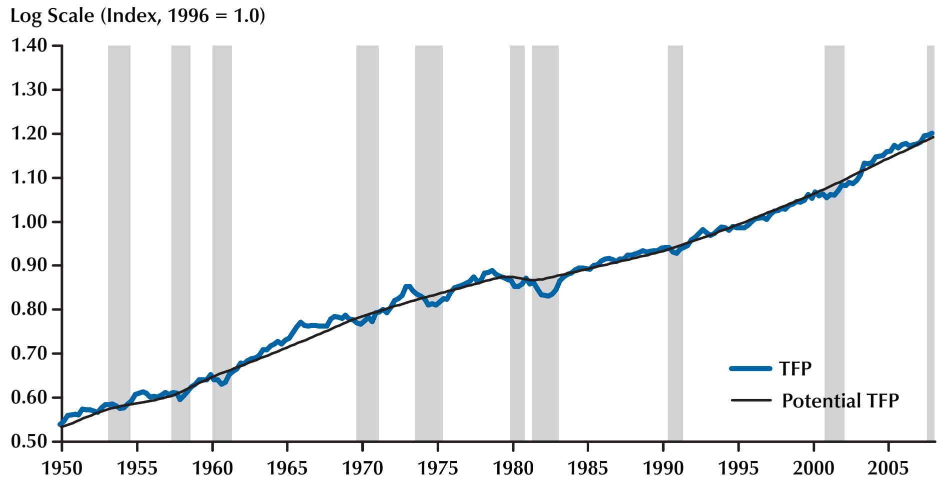
where X = the series to be cyclically adjusted,
U = unemployment rate,
U^* = NAIRU, and
T_i = zero until the business-cycle peak occurring in year i, after which it equals the number of quarters elapsed since that peak.

Equation (1), a piecewise linear regression, is estimated using quarterly data and ordinary least squares. Potential values for the series being adjusted are calculated as the fitted values from the regression, with U constrained to equal U^*. Setting the unemployment rate to equal the NAIRU removes the estimated effects of fluctuations in the business cycle; the resulting estimate gives the equation's prediction of what the dependent variable (X) would be if the unemployment rate never deviated from the NAIRU. An example of the results of using the cyclic-adjustment equation is illustrated in Figure 2, which shows TFP and potential TFP.

One question that arises is when to add a new trend break to the equation. Typically, we do not add a new breakpoint immediately after a business cycle peak because doing so would create, at least initially, a very short trend segment for the period after the peak. Such a segment would be subject to large swings as new data points were added

Figure 2

TFP and Potential TFP



to the sample because it was so short, at least initially. Because the final segment of the trend is carried forward into the projection, those swings would create instability to our medium-term projections. Consequently, we typically wait until a full business cycle has concluded before adding a new break to the trend. For example, the model does not yet include a break in 2001, though the addition of one appears to be increasingly likely.

Equation (1) is used for most, but not all, inputs in the model. One important exception is the capital input, which does not need to be cyclically adjusted to create a "potential" level because the unadjusted capital input already represents its potential contribution to output. Although use of the capital stock varies greatly during the business cycle, the potential flow of capital services is always related to the total size of the capital stock, not to the amount currently being used. Other exceptions include several variables of lesser importance that do not vary with the business cycle. Those series are smoothed using the Hodrick-Prescott filter.

As noted earlier, the method for computing historical values of potential output in the other sectors of the economy differs slightly from that used for the NFB sector. In general, the approach is to express real GDP in each of the other sectors as a function of the primary factor input (either labor or capital) in that sector and the productivity of that input. The potential levels of the primary input and its productivity are cyclically adjusted using an analog to equation (1) and then combined to estimate potential output in that sector. The list below describes how each sector is modeled.

- Farm sector: Potential GDP in this sector is modeled as a function of potential farm employment and potential output per employee.
- Government sector: Potential GDP in this sector is the sum of potential GDP in the federal government and state and local governments. Potential GDP at each level of government equals the sum of the compensation of general government employees (adjusted to potential) and government depreciation. Compensation is modeled as a function of total employment, and compensation per employee and depreciation is modeled as a function of the government capital stock.
- Nonprofit sector: Potential GDP in this sector is modeled as a function of potential

nonprofit employment and potential output per employee.

- Household sector: Although some of the GDP in the household sector consists of the compensation of domestic workers, the majority is composed of imputed rent on owner-occupied housing. As such, output in this sector is composed of a stream of housing services provided almost entirely from the capital stock. Potential GDP in this sector is modeled as a function of the stock of owner-occupied housing and an estimate of the productivity of that stock. Similar to the capital input in the NFB sector, the housing capital stock is not adjusted to potential because the unadjusted stock reflects the potential contribution to output.

For projections of potential output, the same framework is used for these sectors as is used for the NFB sector. Given projections of several exogenous variables—of which potential labor force, potential TFP growth, and the national saving rate are the most important—the growth model computes the capital stock endogenously and combines the factor inputs into an estimate of potential output. In most cases, projecting the exogenous variables is straightforward: The CBO generally extrapolates the trend growth rate from recent history through the 10-year projection period. However, the projections for some exogenous variables, most notably the saving rate, are taken from the CBO economic forecast.

Advantages and Disadvantages of the CBO Method

The CBO method for estimating and projecting potential output has several key advantages. First, it looks explicitly at the supply side of the economy. Potential output is a measure of productive capacity, so any estimate of it is likely to benefit from explicit dependence on factors of production. For example, if growth in the available pool of labor increases, then this method will show an acceleration in potential output (all other things being equal). With our approach, an increase in investment spending would also be reflected in faster growth in productive capacity.

Another advantage of a growth model is that it allows for a transparent accounting of the sources of growth. Such a growth-accounting exercise, which divides the growth of potential GDP into the contributions from each of the factor inputs, is especially useful when explaining the factors that caused a change to CBO projections. A growth-accounting exercise for our current projection is shown in Table 1.[3] The table displays the growth rates of potential output and its components for the overall economy and the NFB sector. Note that the growth rates of the factor inputs (top and middle panels of the table) are not weighted; they do not sum to the growth in potential output.

A third advantage of using a growth model to calculate potential output is that it supplies a projection for potential output that is consistent with the CBO projection for the federal budget. That consistency allows the CBO to incorporate the effects of changes in fiscal policy into its medium-term (10-year) economic and budget projections. Fiscal policy has obvious effects on aggregate demand in the short run, effects that are reflected in our short-term forecast. However, fiscal policy will also influence the growth in potential output over the medium term through its effect on national saving and capital accumulation. Because the growth model explicitly includes capital as a factor of production, it captures that effect.

Table 1 also shows the contribution of each factor input to the growth of potential output in the NFB sector by weighting each input's growth rate by its coefficient in the production function. The sum of the contributions equals the growth of potential output in the NFB sector. Computing the contributions to growth highlights the sources of any quickening or slowdown in growth. For example, the CBO estimates that potential output in the NFB sector grew at an average annual rate of 3.3 percent during the 1982-90 period and 3.5 percent during the 1991-2001 period. That acceleration can be attributed to faster growth in the capital input (which contributed 1.2 percentage points to the growth of potential output during the first period and 1.4 percentage points in the

[3] See CBO (2008).

Table 1

Key Assumptions in the CBO's Projection of Potential Output

	Average annual growth (%)						Projected average annual growth (%)		
	1950-73	1974-81	1982-90	1991-2001	2002-2007*	Total, 1950-2007*	2008-2013	2014-2018	Total, 2008-2018
Overall economy									
Potential output	3.9	3.2	3.1	3.1	2.7	3.4	2.5	2.4	2.4
Potential labor force	1.6	2.5	1.6	1.2	1.1	1.6	0.8	0.5	0.7
Potential labor force productivity†	2.3	0.7	1.4	1.9	1.6	1.8	1.6	1.9	1.7
Nonfarm business sector									
Potential output	4.0	3.6	3.3	3.5	3.0	3.6	2.8	2.8	2.8
Potential hours worked	1.4	2.3	1.7	1.1	1.0	1.5	0.7	0.4	0.6
Capital input	3.8	4.2	4.1	4.6	2.5	3.9	2.9	3.5	3.2
Potential TFP	1.9	0.7	0.9	1.3	1.5	1.4	1.4	1.4	1.4
Potential TFP excluding adjustments	1.9	0.7	0.9	1.3	1.3	1.4	1.3	1.3	1.3
TFP adjustments	0.0	0.0	0.0	0.1	0.2	‡	0.1	0.1	0.1
Price measurement§	0.0	0.0	0.0	0.1	0.1	‡	0.1	0.1	0.1
Temporary adjustment¶	0.0	0.0	0.0	‡	‡	‡	0.0	0.0	0.0
Contributions to the growth of potential									
Output (percentage points)									
Potential hours worked	1.0	1.6	1.2	0.8	0.7	1.0	0.5	0.3	0.4
Capital input	1.1	1.3	1.2	1.4	0.8	1.2	0.9	1.0	1.0
Potential TFP	1.9	0.7	0.9	1.3	1.5	1.4	1.4	1.4	1.4
Total contributions	4.0	3.6	3.3	3.5	2.9	3.6	2.8	2.8	2.8
Potential labor productivity in the NFB sector#	2.6	1.3	1.6	2.4	1.9	2.1	2.1	2.3	2.2

NOTE: Data are for calendar years. Numbers in the table may not add up to totals because of rounding.
*Values as of August 22, 2008.
†The ratio of potential output to the potential labor force.
‡Between zero and 0.05 percent.
§An adjustment for a conceptual change in the official measure of the GDP chained price index.
¶An adjustment for the unusually rapid growth of TFP between 2001 and 2003.
#The estimated trend in the ratio of output to hours worked in the NFB sector.

SOURCE: CBO.

second) and faster growth of potential TFP (which contributed 0.9 and 1.3 percentage points to the growth of potential output during the two periods, respectively). Faster growth in those two factors more than offset a slowdown in potential hours worked between the two periods. This point is addressed later.

Fourth, by using a disaggregated approach, the CBO method can reveal more insights about the economy than a more-aggregated model would. For example, the model calculates the capital input to the production function as a weighted average of the services provided by seven types of capital. Those data indicate a shift over the past few decades to capital goods with shorter service lives: A larger share of total fixed investment is going to producers' durable equipment (PDE) relative to structures, and a larger share of PDE is going to computers and other information technology (IT) capital. Because shorter-lived capital goods depreciate more rapidly, the shift toward PDE and IT capital increases the share of investment dollars used to replace worn-out capital and tends to lower net investment and the capital input. Shorter-lived capital goods are also more productive per year of service life than those that last longer and are therefore weighted more heavily in the growth model's capital input. A model that ignores the capital input or that does not disaggregate capital is likely to miss both of those effects.

On the negative side, the simplicity of our model could be perceived as a drawback. The model uses some parameters—most notably, the coefficients on labor and capital in the production function—that are imposed rather than econometrically estimated. Although that approach is standard practice in the growth-accounting literature (in part because it has empirical support), it is tantamount to assuming the magnitude of the contribution that each factor input makes to growth. With such an approach, the magnitude of that contribution will not change from year to year as the economy evolves, as it would in an econometrically estimated model. Moreover, it requires some strong assumptions that may not be consistent with the data.

A second disadvantage of using a growth model to estimate potential output is that including the capital stock introduces measurement error. Most economic variables are subject to measurement error, but the problem is particularly acute for capital, for two basic reasons. First, measuring the stock of any particular type of capital is difficult because depreciation is hard to define or measure. Purchases of plant and equipment can be tallied to produce a historical series for investment, but no corresponding source of data exists for depreciation. Second, even if accurate estimates of individual stocks were available, aggregating them into a single index would be difficult because capital is heterogeneous, differing with respect to characteristics such as durability and productivity.[4]

A third point of contention regarding the CBO approach is the use of deterministic time trends to cyclically adjust many variables in the model. Some analysts assert that relying on fixed time trends provides a misleading view of the cyclical behavior of some economic time series. They argue, on the basis of empirical studies of the business cycle, that using variable rather than fixed time trends is more appropriate for most data series.[5] However, the evidence on this point is mixed—it is very difficult to determine whether the trend in a data series is deterministic or stochastic using existing econometric techniques—and the methods used to estimate stochastic trends often yield results that are not useful for estimating potential output. That is, stochastic trends tend to produce estimates of the output gap that are not consistent with other indicators of the business cycle.

Fourth, the CBO growth model is based on an estimate of the amount of slack in the labor market, which in turn requires an estimate of the natural rate of unemployment or the NAIRU. Such

[4] The CBO capital input uses capital stock estimates (and the associated assumptions about depreciation) from the Bureau of Economic Analysis and uses an aggregation equation that is based on the approach used by the Bureau of Labor Statistics (BLS) to construct the capital input that underlies the multifactor productivity series. The CBO estimate of the capital input is quite similar to that calculated by the BLS.

[5] See, for example, Stock and Watson (1988).

estimates are highly uncertain. Few economists would claim that they can confidently identify the current NAIRU to within a percentage point. Our method is not very sensitive to possible errors in the average level of the estimated NAIRU, but it is sensitive to errors in identifying how that level changes from year to year.

Finally, the CBO model does not contain explicit channels of influence for all major effects of government policy on potential output. For example, it does not include an explicit link between tax rates and labor supply, productivity, or the personal saving rate; nor does it include any link between changes in regulatory policy and those variables. However, that does not mean that the model precludes a relationship between policy changes and any of those variables. If a given policy change is estimated to be large enough to affect the incentives governing work effort, productivity, or saving, then those effects can be included in our projection or in a policy simulation by adjusting the relevant variable in the model. For example, changes in marginal tax rates have the potential to affect labor supply. Because the Solow model does not explicitly model the factors that affect the labor input, our model includes a separate adjustment to incorporate such effects. Indeed, for the past several years, such an adjustment has been included in our model to account for the effects on the labor supply of the scheduled expiration in 2011 of the tax laws passed in 2001 and 2003. The structure of our model makes it easier to isolate (and incorporate) the effects of such policy changes than would be the case with a time-series–based model.

CHALLENGES ASSOCIATED WITH ESTIMATING AND PROJECTING POTENTIAL OUTPUT

Potential output plays a key role in the CBO economic forecast and projection. Perhaps the two most important are estimating the output gap (percentage difference between GDP and potential GDP) and providing a target for the 10-year projection of GDP. Important challenges are associated with both roles.

Challenges Associated with Estimating the Output Gap

Any method used to estimate the trend in a series, including potential output, is subject to an “end-of-sample” problem, which means that estimating the trend is especially difficult near the end of a data sample. In the case of the output gap, this is usually the period of greatest interest. Three examples from the period since 2000 illustrate the difficulties associated with estimating the level of potential output at the end of the sample period.

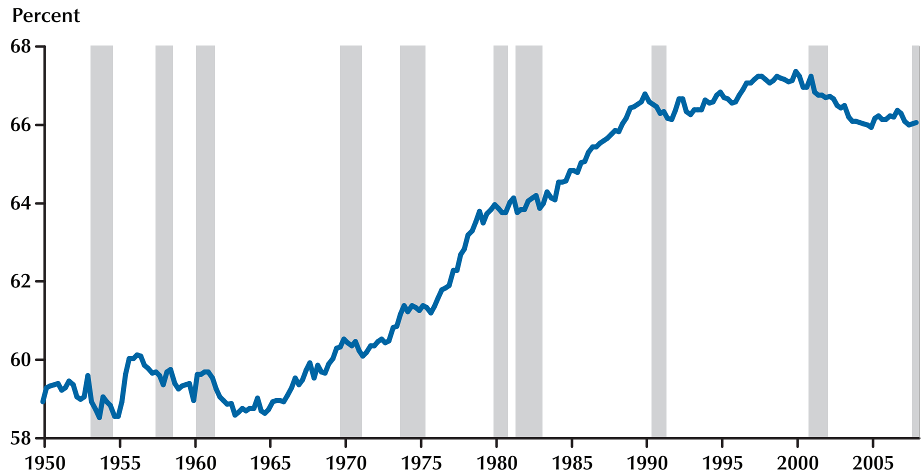
Potential Labor Force. Fundamentally, the amount of hours worked in the economy is determined by the size of the labor force, which, in turn, is largely influenced by two factors: growth in the population and the rate of labor force participation. Neither of those series is especially sensitive to business cycle fluctuations, but both are subject to considerable low-frequency variation. The discussion here focuses on how the rate of labor force participation has changed during recent years and how we have modified the CBO labor force projections as a result.

After a long-running rise that started in the early 1960s, the labor force participation rate plateaued at about 67 percent of the civilian population during the late 1990s, declined sharply between 2000 and 2003, and varied in a narrow range near 66 percent between 2003 and 2008 (Figure 3). Had that decline in the participation rate not occurred, the labor force would have had approximately 2.3 million more workers in 2008 than it actually did.

To assess its impact on potential output, the challenge during the early 2000s was to determine whether the decline in the participation rate was cyclical (i.e., workers had dropped out of the labor force because their prospects of getting a job were dim) or structural (i.e., prospective labor force participants had weighed the alternatives and found that options such as education, retirement, or child-rearing were more attractive). If the decline were due to cyclical reasons, then the dip in participation should not be reflected in the estimate of potential labor force. If the decline were due to structural reasons, however, then the

Figure 3

Labor Force Participation Rate



estimates of potential labor force and potential output should be lowered to reflect the decreased size of the potential workforce.

The drop in the participation rate also complicated the interpretation of movements in the unemployment rate, which peaked at 6.1 percent in mid-2003 and declined thereafter. During 2006 and 2007, the unemployment rate was below 5 percent, which suggested considerable tightness in the labor market. However, the decline in the participation rate implied that there existed a pool of untapped labor that could have been drawn into the workforce had there been a significant speedup in the pace of job creation. Consequently, at that time, the unemployment rate probably understated the degree of slack that existed in the labor market. Indeed, in the early stages of the expansion following the 2001 recession, we projected that the participation rate would recover as job creation picked up. It never did though, and the CBO has since concluded that the decline in the participation rate was more structural than cyclical.[6]

[6] That conclusion was based on an analysis of the factors affecting the participation rates of various demographic subgroups in the population; see CBO (2004).

Potential NFB Employment. The second challenge is associated with the behavior of employment since the end of the 2001 recession and its implications for the estimate of potential hours worked in the NFB sector. One striking feature of the economic landscape since the 2001 business cycle trough is very weak growth in employment, especially for measures derived from the Bureau of Labor Statistics' establishment survey. For example, since the trough in the fourth quarter of 2001, growth in nonfarm payroll employment averaged 0.8 percent at an annual rate, which means that payrolls were roughly 5 percent higher in the second quarter of 2008 than they were at the end of the 2001 recession. However, based on patterns in past cycles, one would have expected much faster growth in payroll employment—2.4 percent on average—and a much higher level of employment—17 percent higher than its trough value—by the second quarter of 2008 (Figure 4).

A similar pattern holds for employment in the NFB sector (which differs from the headline payroll number by excluding employees in private households and nonprofit institutions and including proprietors). In the second quarter of 2008, NFB employment was about 4 percent above

Figure 4

Payroll Employment in the Current Expansion Compared with an "Average" Cycle

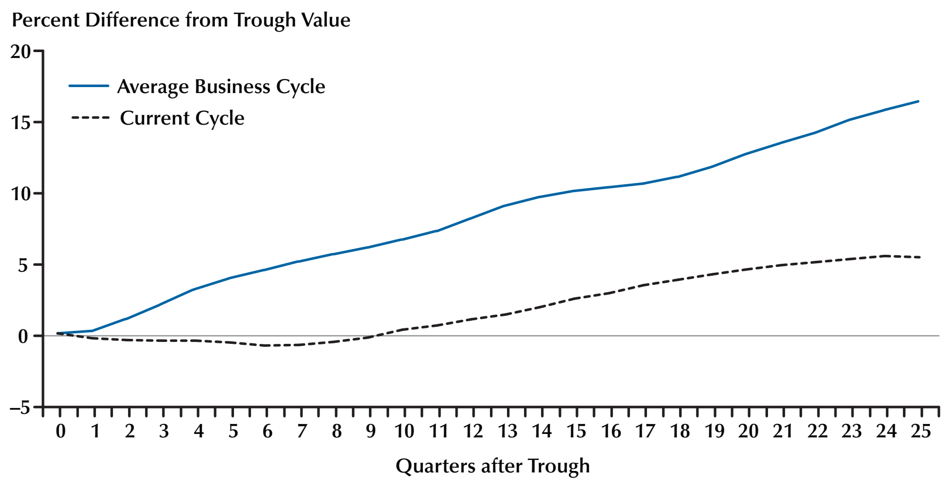


Figure 5

NFB Sector Employment as a Percent of the Civilian Labor Force

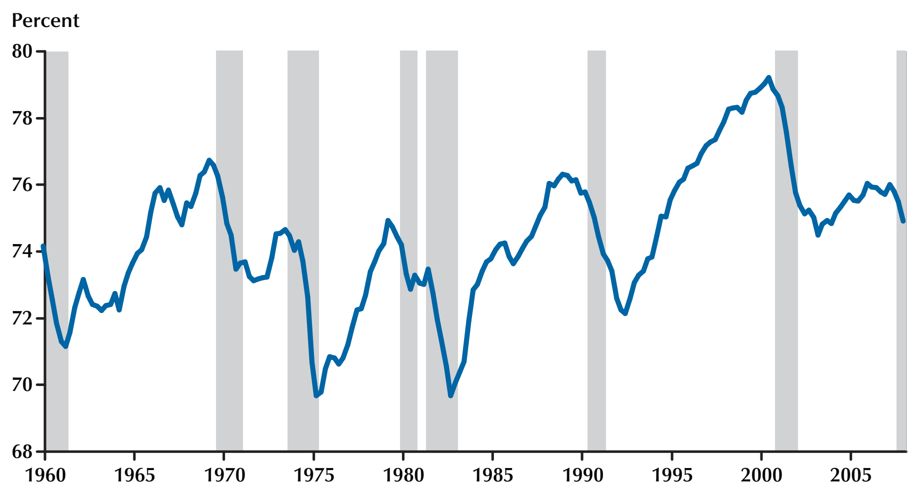
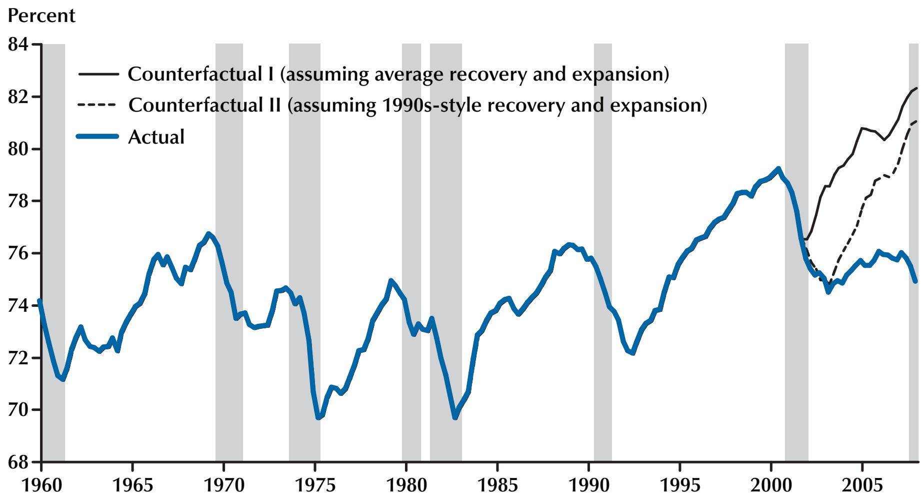


Figure 6

NFB Employment as a Percent of the Civilian Labor Force and Two Counterfactual Paths



its level at the end of the 2001 recession. Had it grown according to the pattern seen in a typical business cycle expansion, it would have been about 15 percent higher than its level at the trough of the recession.

The behavior of NFB employment since the business cycle peak in 2001 also looks very unusual when viewed from another perspective. When measured as a share of the labor force (which controls for the decline in the rate of labor force participation), NFB employment barely grew during the expansion that followed the 2001 recession (Figure 5). This is extremely unusual on two counts. First, it departs from the very strong procyclical pattern seen in most recovery and expansion periods. Typically, NFB sector employment grows much faster than the labor force during business cycle expansions, which causes a rapid increase in its share. Second, the recent behavior breaks with the long-standing upward trend in the NFB share of the labor force. Since roughly the mid-1970s, trend growth in NFB employment has exceeded trend growth in the labor force on average, leading to a steady increase in the share. Examining the peaks is a rough-and-ready way to control for business cycle variation: The share increased from about 74 percent in 1973, to just under 75 percent in 1979, to just over 76 percent in 1989, to just under 80 percent in 1999.

After the trough in 2001, the share of NFB employment declined for another two years and then increased somewhat but not anything like a normal cyclical rebound. The reasons for this behavior are not fully clear—shifts of employment to other sectors, including government and non-profit institutions, can explain only part of the shortfall—but it has important implications for the estimate of potential employment and hours worked. Specifically, the estimate of potential employment in the NFB sector is much lower than it would have been had actual NFB employment followed a more typical cyclical pattern since 2001.

To illustrate this point, consider what the NFB employment share would have looked like had it followed a more typical cyclical pattern. Figure 6 shows NFB employment as a share of the labor force along with two counterfactual paths for the share. The thin solid line shows the evolution of the NFB employment share had it grown since 2001 at the same rate as an "average" historical expansion. That path embodies much stronger employment growth than does the actual path and would imply a much higher level of potential employment as well. Arguably, that path is too strong, given that employment growth has been sluggish in the recoveries that followed the past two recessions. So the figure includes a second counterfactual path (dotted line) showing the evolution of the NFB employment share had it grown as it did during the expansion of the 1990s. It too implies much stronger employment growth than actually occurred.

For the first few years of the current business cycle, it was reasonable to expect a typical rebound in the NFB employment share, even if it was delayed relative to past expansions. If so, then the period of sluggish growth in NFB employment could be interpreted as a cyclical pattern and would not necessarily imply that the level of potential NFB employment was lower. However, as the period of sluggish growth grew longer and in light of the possibility of a business cycle peak in early 2008, the position that NFB employment would eventually rebound became increasingly untenable.

Instead, it seems increasingly likely that NFB employment will merely match the growth in the labor force in the future, rather than grow at a faster pace. One implication of that interpretation is that the experience of the late 1990s, when the NFB employment share of the labor force was very high, was unusual and is unlikely to be repeated.

Changes in the Phillips Curve and NAIRU. As noted previously, the natural rate of unemployment is an important input in the CBO model. It serves as the benchmark used to estimate the potential values of the factor inputs and, consequently, potential output. Any uncertainties associated with the size of the unemployment gap, or difference between the unemployment rate and the natural rate, will translate directly into uncertainty about the size of the output gap.

Our estimate of the natural rate, known as the NAIRU, is based on a standard Phillips curve, which relates changes in inflation to the unemployment rate, expected inflation, and various supply shock variables. In particular, the NAIRU estimate relies on the existence of a negative correlation between inflation and unemployment: If inflation tends to rise when the unemployment rate is low and tends to fall when the unemployment rate is high, then there must be an unemployment rate at which there is no tendency for inflation to rise or fall. This does not mean that the rate is stable or that it is precisely estimated, just that it must exist.

However, during the past 20 or so years, significant changes in how the economy functions have affected the relationship between inflation and unemployment and, consequently, estimates of the Phillips curve and the NAIRU. Most notably, the rate of inflation has been lower and much less volatile since the mid-1980s, a phenomenon often referred to as the Great Moderation. At the same time, the unemployment rate has trended downward, which suggests that the natural rate of unemployment has declined also. Researchers had identified several factors that would be expected to lower the natural rate, including the changing demographic composition of the workforce, changes in disability policies, and improved efficiency of the labor market's matching process. Based on internal evaluation of those factors, the CBO began to lower its estimate of the NAIRU for the period since 1990, overriding the econometric estimate at that time.[7]

More recent Phillips curve estimates are consistent with the hypothesis that a change occurred sometime during the past 20 or so years. In a recent working paper, I presented regression results from estimates of several Phillips curve specifications that suggested the presence of significant structural change since the mid-1980s.[8] Using the full data sample, from 1955 through 2007, the equations' performance appeared to be satisfactory. They fit the data well and their estimated coefficients had the correct sign, were of reasonable magnitude, and were statistically significant. However, the full-sample regressions masked evidence of a breakdown in performance that began during the mid-1980s. Estimation results from equations that allowed for structural change indicated that the fit of the equations dete-

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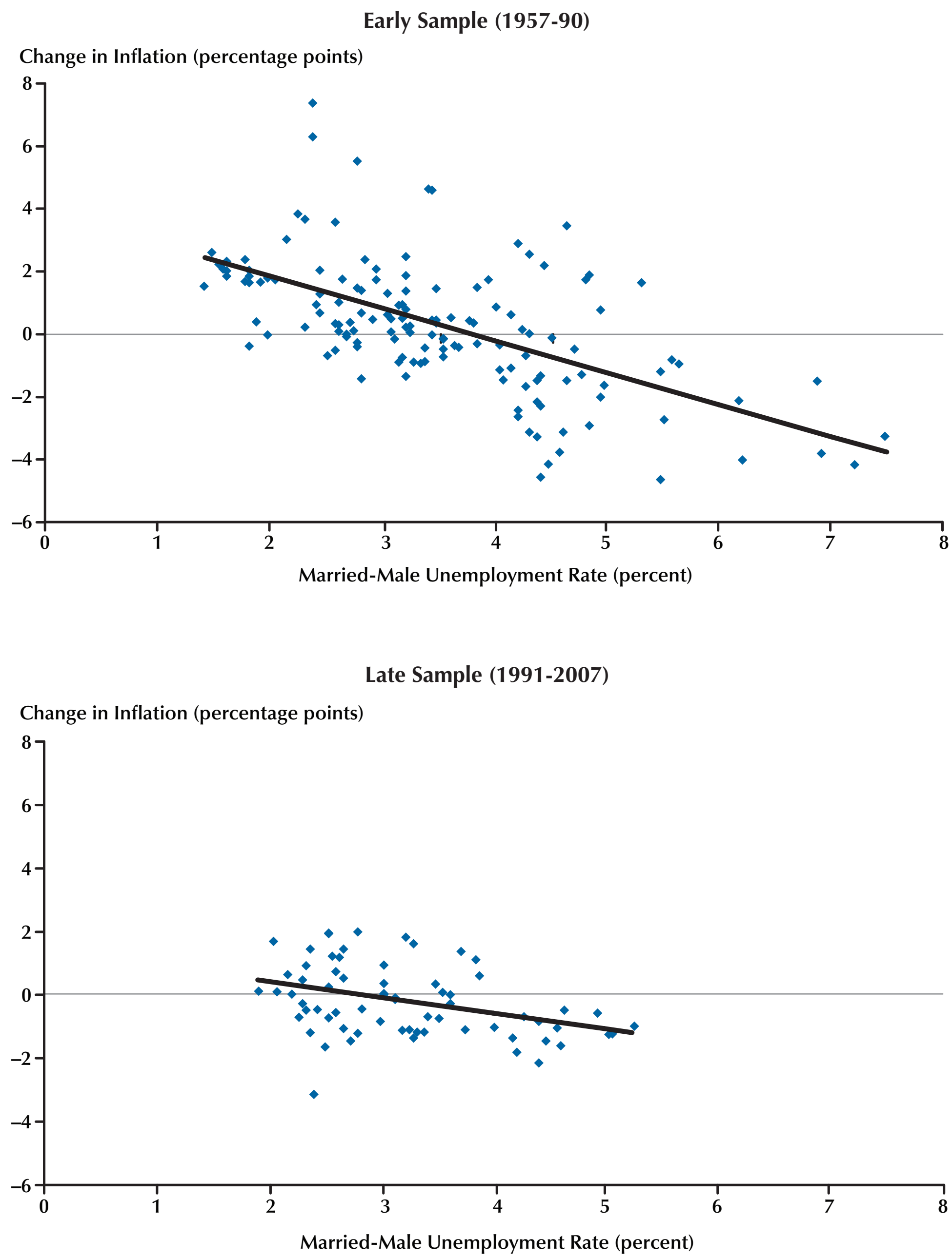
7 That analysis was later summarized in a CBO paper; see Brauer (2007).

8 See Arnold (2008).

Figure 7

Married-Male Unemployment and the Change in Inflation



NOTE: The change in inflation is defined as the difference between the quarterly rate of inflation in the personal consumption expenditure (PCE) price index and a 24-quarter moving average of PCE inflation.

riorated and that the coefficients were smaller and less statistically significant during the latter part of the data sample than they were during the earlier part. In general, the results suggest that the NAIRU is lower now than it had been during the period from 1955 through the mid-1980s, a conclusion consistent with evidence from the labor market suggesting a decline in the natural rate. The results also indicate that the Phillips curve has become less useful for predicting inflation.

However, the relationship between inflation and unemployment, though not as strong as it once was, has not collapsed completely. Consider Figure 7, which plots changes in a measure of unanticipated inflation against the married-male unemployment rate. The top panel shows data from the 1957-90 period, while the bottom panel shows data from 1991 through 2007.[9] Comparing the two panels reveals four features of the latter period. First, both graphs show a negative correlation between the two series, so there still appears to be a tradeoff between inflation and unemployment. Second, the point at which the regression line intersects the horizontal-axis intercept has moved to the left in the second panel, which is consistent with the idea that the NAIRU is lower now than it had been earlier. Third, the slope of the trend line is lower during the second part of the sample, which suggests that the inflation-unemployment tradeoff is somewhat flatter during the second period (i.e., inflation is less responsive to changes in the unemployment rate). Fourth, much less variation has occurred in both inflation and unemployment during the past 20 or so years than previously.

What do these observations imply for the estimate of potential output? The first observation—that a negative correlation still exists—means that the unemployment rate is still consistent with a stable rate of inflation. The second observation—that the NAIRU has declined—implies that the level of potential output is higher than it would have been had the NAIRU been constant. This observation also serves as a reminder that structural change in macro equations is a fact of life. It is important to monitor such equations continually to identify how economic events will affect their conclusions. The final two observations imply that Phillips curves, and by extension the NAIRU and potential output, are less useful indicators of inflationary pressure than they once were.

Challenges Associated with Projecting Potential Output

Potential output is used for more than gauging the state of the business cycle. It is also used to set the path for real GDP in the 10-year forecast that underlies the CBO budget projections. A separate set of challenges is associated with projecting the variables that underlie our estimate of potential output.

Projecting Labor Productivity I: The Late-1990s' Acceleration. Labor productivity growth during the late 1990s provides an important example of the challenges associated with projecting potential GDP.[10] The broad outlines of the story are familiar: After a long period of sluggish growth, labor productivity accelerated sharply during the second half of the 1990s and continued to grow rapidly during the 2000s. Moreover, the upswing was substantial. Trend growth in labor productivity averaged 2.7 percent between the end of 1995 and the middle of 2008, considerably faster than the 1.4 percent pace from 1974 to 1995 (Figure 8). Had it followed that pre-1996 trend of 1.4 percent instead of growing as it did, labor productivity would be 15 percent lower than it is today. Furthermore, if the 2.7 percent trend is sustained over the next decade, then the level of real GDP will be nearly 30 percent higher in 2018 than the level that would have resulted from the pre-1996 rate of growth.

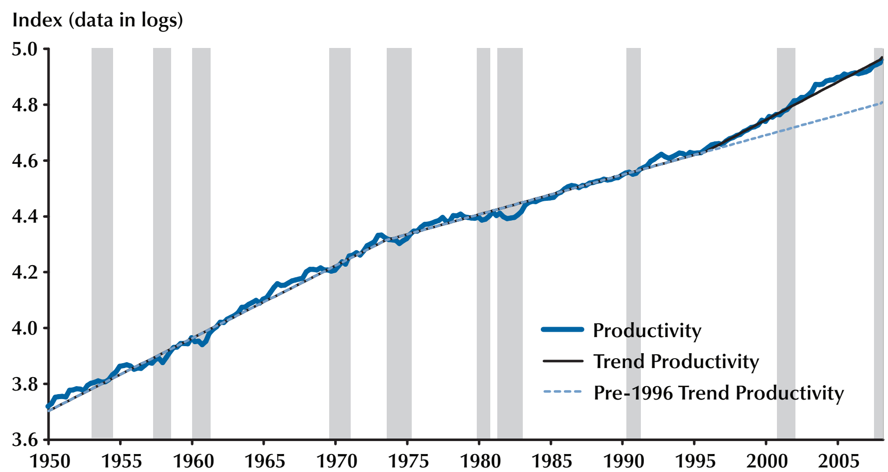
One problem for forecasters was that the productivity acceleration was largely unexpected. In the mid-1990s, few analysts anticipated such a dramatic increase in the trend rate of growth.

[9] The working paper estimated Phillips curve equations using different price indices and used Chow tests to determine when the structural break occurred in each equation. For the personal consumption expenditures price index, the break was found in 1991. The married-male unemployment rate was used in the estimation because it is better insulated from demographic shifts than the overall unemployment rate.

[10] In our model, we actually project potential TFP—the projection for potential labor productivity is implied by the projections for potential TFP and capital accumulation.

Figure 8

Labor Productivity Growth and Trend (1950-2008)



In January 1995, for example, the CBO projected that labor productivity growth would average 1.3 percent annually for the 1995-2000 period, a pace similar to the average for the prior 20 years. The Clinton administration and the Blue Chip Consensus of private forecasters projected similar rates of growth.

Another problem for forecasters was that the productivity surge in the late 1990s went unrecognized until very late in the decade for two basic reasons. First, labor productivity is fairly volatile, with growth rates that can swing widely from quarter to quarter. As a result, a period of two or three years is a short window within which to discern a new trend. Moreover, the acceleration followed a period of subpar growth (productivity growth averaged only 0.22 percent annually between the end of 1992 and 1995:Q3); so, initially, the faster growth appeared to be just making up lost ground rather than establishing a new, higher trend growth rate. The postwar data sample includes several episodes of faster- or slower-than-trend productivity growth that were later reversed.

Second, early vintages of productivity data for the late 1990s proved to be understated and, therefore, painted a misleading picture of the productivity trend. Only after several revisions did a stronger pattern emerge. Using real-time data culled from our forecast databases, Table 2 shows that data available in 1996, 1997, and 1998 showed only a small rise in productivity growth starting in late 1995. For example, data available in early 1997 showed labor productivity growing by only 0.3 percent on average from 1995:Q4 through 1996:Q3. The story changes markedly using currently available data: Labor productivity growth for that period was actually 3 percent.

A similar case holds for 1998 and 1999. Data available in January 1998 showed labor productivity growth averaging 1.8 percent between 1995:Q4 and 1997:Q3. That rate has since been revised upward by 0.6 percentage points, to 2.4 percent. The growth rate for the three-year period ending in 1998:Q3 also has been revised from 2.0 percent (using data from early 1999) to 2.5 percent (using currently available data).

The information in Table 2 highlights an important point. Productivity data are revised frequently, and the revisions can be large enough to alter analyses of trends in productivity growth. Indeed, after being revised upward several times during the late 1990s, productivity data have been revised downward somewhat during recent years.

Table 2

Changes in Estimates of Average Annual Growth Rate for Labor Productivity

		Average annual rate of growth (%)		
Date of forecast	Period	Initial estimate (using original data)	Current estimate (using current data)	Revision (percentage points)
January 1997	1995:Q4–1996:Q3	0.3	3.0	2.7
January 1998	1995:Q4–1997:Q3	1.8	2.4	0.6
January 1999	1995:Q4–1998:Q3	2.0	2.5	0.5
January 2000	1995:Q4–1999:Q3	2.7	2.4	–0.3

NOTE: Each forecast is based on productivity data that extend through the third quarter of the previous year. Numbers in the table may not add up to totals because of rounding.

SOURCE: CBO based on data from the BLS.

In January 2000, labor productivity growth for 1995:Q4 to 1999:Q3 was estimated at 2.7 percent; that estimate has since been revised to 2.4 percent.

The revisions to productivity data highlight the difficulty in recognizing a change in the underlying trend growth rate and suggest that we should be circumspect about data series until they have undergone revision. This is especially true if the data show a shift in trend (as in the late 1990s) or if they are not consistent with other economic indicators.

Projecting Labor Productivity II: Shifting Sources of Growth. Another aspect of labor productivity growth during the past decade—a shift in its sources—has complicated the analysis of trends and made projections difficult. With our model we can easily divide the growth in labor productivity into two components: capital deepening (increases in the amount of capital available per worker) and TFP. Capital per worker can rise over time not only because investment provides more capital goods for workers to use, but also because the quality of those goods improves over time and investment can shift from assets with relatively low levels of productivity (e.g., factories) to those with higher productivity levels (e.g., computers). Because TFP is calculated as a residual—the growth contributions of labor and capital are subtracted from the growth in output—any growth in labor productivity that is not attributed to capital deepening will be assigned to TFP.

With this in mind, the contributions of capital deepening and TFP to the growth in labor productivity since 1995 can be calculated. The results of such a growth-accounting exercise are shown in Table 3. Those results show that capital deepening was the primary source of the surge in labor productivity growth in the late 1990s and that faster TFP growth was the primary source of productivity growth during the period after the business cycle peak in 2001. Between the early (1991-95) and the late (1996-2001) part of the past decade, labor productivity growth stepped up from about 1.5 percent, on average, to 2.5 percent per year. Growth in capital per worker accounted for 80 percent (0.8 percentage points) of that 1-percentage-point increase, according to our estimates. Faster TFP growth was responsible for the rest of the step-up in productivity growth, or about 0.2 percentage points.

Since the 2001 recession, however, the sources of labor productivity growth have completely reversed. Business investment fell substantially in 2001 and 2002 and remained weak in 2003, thus slowing the growth in capital deepening relative to that in the late 1990s. Consequently, the contribution of capital per worker to labor productivity growth fell by 0.7 percentage points between 2001 and 2005 relative to the 1996-2001 period. At the same time, however, TFP growth was accelerating sharply, especially in 2003. The CBO estimates that TFP was responsible for all

Table 3

Contributions of Capital Deepening and TFP to Labor Productivity Growth (1990-2006)

	Average annual growth rate			Change	
	1991-95	1996-2001	2002-06	1991-95 to 1996-2001	1996-2001 to 2002-06
Contribution of capital deepening (percentage points)	0.50	1.33	0.62	0.83	–0.72
Contribution of TFP growth (percentage points)	1.04	1.21	2.07	0.18	0.86
Labor productivity (%)	1.54	2.54	2.65	1.00	0.11

NOTE: Numbers in the table may not add up to totals because of rounding.

SOURCE: CBO using data from the BLS and Bureau of Economic Analysis.

of the acceleration in labor productivity in the 2001-06 period.

A natural question is whether labor productivity will grow as rapidly over the next 10 years as during the past decade. But the experience since 1995 illustrates why that question is so hard to answer. Labor productivity growth is volatile, its measurement is subject to large revisions, and the reasons for changes in its rate of growth are not well understood. Consequently, it is a difficult variable to forecast; past patterns and recent data provide only a rough guide to future labor productivity. Explanations for the recent acceleration help to determine whether any of the changes to growth since 1995 will reverse or recur in the next 10 years.

Projecting Labor Productivity III: Explaining the Acceleration. Although it is hard to say conclusively that one factor is the sole cause of the post-1995 acceleration in productivity growth, most economists point to IT as the primary source. This case is easiest for the late 1990s and more difficult for the period since 2001. As noted previously, the majority of the productivity acceleration for 1996-2000 can be attributed to capital deepening, which was one result of a huge increase in business investment. During the late 1990s, not only did investment boom, but it was heavily tilted toward IT capital (Figure 9). The CBO estimates suggest that faster capital deepening accounted for 80 percent of the upswing in labor productivity growth during the late 1990s and that IT capital accounted for 75 percent of the contribution from capital deepening.

In addition, it appears that rapid technological change in IT industries (including computers, software, and telecommunications) caused faster TFP growth in those industries. It also appears that the pace of technological change was fast enough, and those industries were large enough, for faster TFP growth in that sector of the economy to support overall TFP growth during the late 1990s.[11] However, because TFP growth did not accelerate during the late 1990s, it appears that faster TFP growth in the IT sectors merely offset slower TFP growth elsewhere.

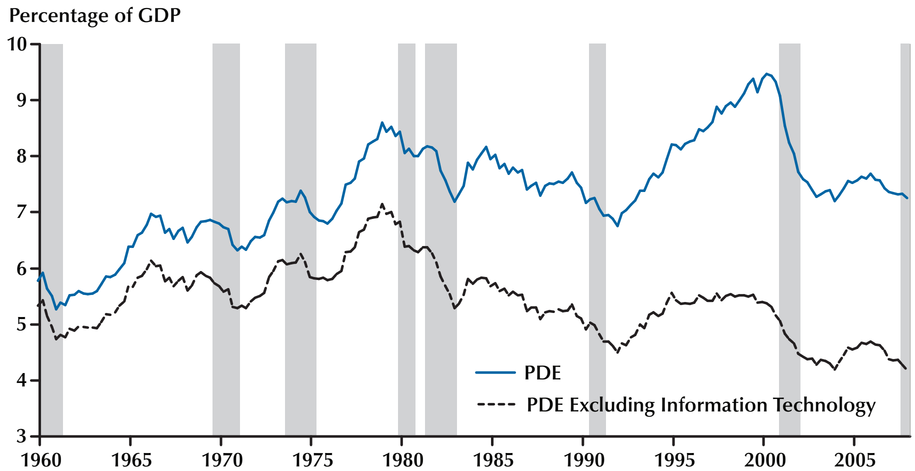
It is somewhat harder to make the case that IT spending was the primary source of the continued rapid growth in labor productivity since the business cycle peak in 2001. One obvious problem with this explanation is that spending on IT capital collapsed after 2000, which strongly suggests that IT capital was not the reason for the continued surge. According to our estimates, nearly 80 percent of the post-2001 growth in labor productivity can be attributed to TFP, with only 20 percent accounted for by capital deepening.

Despite those estimates, the continued growth in labor productivity could still be the result of

[11] According to estimates by Oliner and Sichel (2000), for example, the computer and semiconductor industries accounted for about half of TFP growth from 1996 through 1999, even though those industries composed only about 2.5 percent of GDP in the NFB sector during those years.

Figure 9

Investment in Producers' Durable Equipment



IT spending if a lag exists between the time when the capital is installed and when businesses achieve productivity gains. Several theories, not necessarily mutually exclusive, have been proposed to explain why such a delay could occur. They include the possibility that there are adjustment costs associated with large changes in the capital stock; the possibility that computers are an example of a general-purpose technology, like dynamos and electric motors, that fundamentally change the way businesses operate but take time to produce results; and the possibility that there is a link between IT spending and investment in intangible capital, which refers to spending that is intended to increase future output more than current production but does not result in ownership of a physical asset. As computing power becomes cheaper and more pervasive, managers can invent new business processes, work practices, and organizational structures, which in turn allow companies to produce entirely new goods and services or to improve their existing products' convenience, quality, or variety.

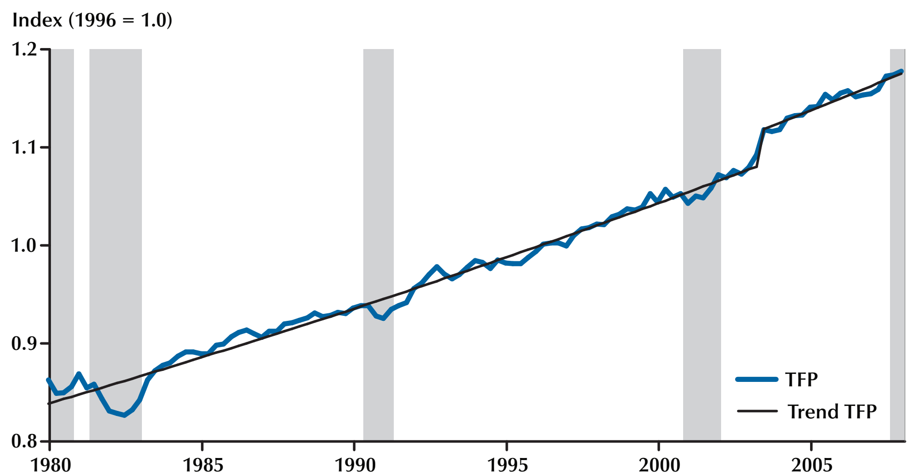
All of these theories could explain the increase in TFP growth. However, all would be expected to have a gradual effect on TFP, raising the growth rate by a small amount over an extended period. In fact, the TFP data display a very steady trend during the 1980s, 1990s, and early 2000s; then a very abrupt increase, occurring entirely in 2003; and then a return to the previous growth trend thereafter (Figure 10). This behavior is somewhat puzzling and hard to reconcile with explanations that rely on a lagged impact of IT spending during the late 1990s. We interpret the abrupt increase as a one-time boost to productivity engendered by the IT revolution—the burst of investment in IT capital allowed firms to raise their efficiency to a higher level but not to permanently increase the rate of productivity growth. Our estimate of potential TFP includes an adjustment that temporarily raises its growth rate to include a level shift similar to that shown in Figure 10.

CONCLUSION

Potential output is a difficult variable to estimate largely because it is an unobservable concept. There are many ways to compute the economy's productive potential. Some methods rely on purely statistical techniques. Others, including the CBO method, rely on statistical procedures

Figure 10

TFP and Trend (1980-2008)



NOTE: Data are adjusted to exclude the effects of methodological changes in the measurement of prices.

grounded in economic theory. However, all of the methods have difficulty estimating the trend in GDP near the end of the data sample, which is usually the period of greatest interest. Because the trend at the end of the data sample is the trend that is projected into the future, any errors in estimating the end-of-sample trend will be carried forward into the projection. The process is further complicated by factors that alter the interpretation of recent economic events, including data revisions and structural change.

In addition to describing the CBO method and highlighting the pros and cons of our approach, this paper describes how we dealt with some developments during the past several years that complicated estimation of potential output. As a general principle, we try to make our estimate of potential output as objective as possible, but as this review of recent problems indicates, estimating potential GDP in real time often involves weighing contradictory evidence. Deciding whether or not, or how much, to change a trend growth rate for TFP, for example, often has a large effect on the estimate of potential for the medium term.

This review demonstrates that the economic landscape is continually changing and that estimates of the trend in any variable, including potential GDP, are affected by those changes. Oftentimes, what looks like a new trend in a series disappears after successive revisions. This factor argues for a conservative approach to estimating such trends and being judicious about changes in those trends.

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Business Cycles: Real Facts and a Monetary Myth*

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Ever since Koopmans (1947) criticized Burns and Mitchell's (1946) book on *Measuring Business Cycles* as being "measurement without theory," the reporting of business cycle facts has been taboo in economics. In his essay, Koopmans presents two basic criticisms of Burns and Mitchell's study. The first is that it provides no systematic discussion of the theoretical reasons for including particular variables over others in their empirical investigation. Before variables can be selected, Koopmans argues, some notion is needed of the theory that generates the economic fluctuations. With this first criticism we completely agree: Theory is crucial in selecting which facts to report.

Koopmans' second criticism is that Burns and Mitchell's study lacks explicit assumptions about the probability distribution of the variables. That is, their study lacks "assumptions expressing and specifying how random disturbances operate on the economy through the economic relationships between the variables" (Koopmans 1947, p. 172). What Koopmans has in mind as such relationships is clear when he concludes an overview of Burns and Mitchell's so-called measures with this sentence: "Not a single demand or supply schedule or other equation expressing the behavior of men [i.e., people] or the technical laws of production is employed explicitly in the book, and the cases of implicit use are few and far between" (p. 163). Koopmans repeatedly stresses this need for using a structural system of equations as an organizing principle (pp. 169–70). Economists, he argues, should first hypothesize that the aggregate time series under consideration are generated by some probability model, which the economists must then estimate and test. Koopmans convinced the economics profession that to do otherwise is unscientific. On this point we strongly disagree with Koopmans: We think he did economics a grave disservice, because the reporting of facts—without assuming the data are generated by some probability model—is an important scientific activity. We see no reason for economics to be an exception.

As a spectacular example of facts influencing the development of economic theory, we refer to the growth facts that came out of the empirical work of Kuznets and others. According to Solow (1970, p. 2), these facts were instrumental in the development of his own neoclassical growth model, which has since become the most important organizing structure in macroeconomics, whether the issue is one of growth or fluctuations or public finance. Loosely paraphrased, the key growth facts that Solow lists (on pp. 2–3) are

- Real output per worker (or per worker-hour) grows at a roughly constant rate over extended time periods.
- The stock of real capital, crudely measured, grows at a roughly constant rate which exceeds the growth rate of labor input.

*The authors thank the National Science Foundation for financial support.

- The growth rates of real output and the stock of capital goods tend to be similar, so the capital-to-output ratio shows no systematic trend.
- The rate of profit on capital has a horizontal trend.

These facts are neither estimates nor measures of anything; they are obtained without first hypothesizing that the time series are generated by a probability model belonging to some class. From this example, no one can deny that the reporting of growth facts has scientific value: Why else would Kuznets have received a Nobel Prize for this work? Or Solow, as well, for developing a parsimonious theory that rationalizes these facts—namely, his neoclassical growth model?

The growth facts are not the only interesting features of these aggregate time series. Also of interest are the more volatile changes that occur in these and other aggregates—that is, the cyclical behavior of the time series. These observations are interesting because they apparently conflict with basic competitive theory, in which outcomes reflect people's ability and willingness to substitute between consumption and leisure at a given point in time and between consumption at different points in time.

The purpose of this article is to present the business cycle facts in light of established neoclassical growth theory, which we use as the organizing framework for our presentation of business cycle facts. We emphasize that the statistics reported here are not measures of anything; rather, they are statistics that display interesting patterns, given the established neoclassical growth theory. In discussions of business cycle models, a natural question is, Do the corresponding statistics for the model economy display these patterns? We find these features interesting because the patterns they seem to display are inconsistent with the theory.

The study of business cycles flourished from the 1920s through the 1940s. But in the 1950s and 1960s, with the development of the structural system-of-equations approach that Koopmans advocated, business cycles ceased to be an active area of economic research. Now, once again, the study of business cycles, in the form of recurrent fluctuations, is alive. At the leading research centers, economists are again concerned with the question of why, in market economies, aggregate output and employment undergo repeated fluctuations about trend.[1]

Instrumental in bringing business cycles back into the mainstream of economic research is the important paper by Lucas (1977), "Understanding Business Cycles." We follow Lucas in defining *business cycles* as the deviations of aggregate real output from trend. We complete his definition by providing an explicit procedure for calculating a time series trend that successfully mimics the smooth curves most business cycle researchers would draw through plots of the data. We also follow Lucas in viewing the business cycle facts as the statistical properties of the comovements of deviations from trend of various economic aggregates with those of real output.

Lucas' definition differs importantly from that of Mitchell (1913, 1927), whose definition had guided students of business cycles up until World War II. Mitchell represents business cycles as sequences of expansions and contractions, particularly emphasizing turning points and phases of the cycle. We think the developments in economic theory that followed Mitchell's work dictate Lucas' representation of cycles.

Equipped with our operational definition of cyclical deviations, we present what we see as the key business cycle facts for the United States economy in the post–Korean War period (1954–1989). Some of these facts are fairly well known; others, however, are likely to come as a surprise because they are counter to beliefs often stated in the literature.

An important example of one of these commonly held beliefs is that the price level always has been procyclical and that, in this regard, the postwar period is no exception. Even Lucas (1977, p. 9) lists procyclical price levels among business cycle regularities. This perceived fact strongly influenced business cycle research in the 1970s. A more recent example of this misbelief is when Bernanke (1986, p. 76) discusses a study by King and Plosser (1984): "Although some points of their analysis could be criticized (for example, there is no tight explanation of the relation between transaction services and the level of demand deposits, and the model does not yield a strong prediction of price procyclicality), the overall framework is not implausible." Even more recently, Mankiw (1989, p. 88), in discussing the same paper, points out that "while the story of King and Plosser can explain the procyclical behavior of money, it cannot explain the procyclical behavior of prices." We shall see that, in fact, these criticisms are based on what is a myth. We show that during the 35 years since the Korean War, the price level has displayed a clear *countercyclical* pattern.

Other misperceptions we expose are the beliefs that

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[1] The view of Hayek (1933, p. 33) in the 1930s and Lucas (1977, p. 7) in the 1970s is that answering this question is one of the outstanding challenges to economic research.

the real wage is either countercyclical or essentially uncorrelated with the cycle and that the money stock, whether measured by the monetary base or by M1, leads the cycle.

The real facts documented in this paper are that major output components tend to move together over the cycle, with investment in consumer and producer durables having several times larger percentage deviations than does spending on nondurable consumption.

Alternative Views of Business Cycles

To many, when we talk about cycles, the picture that comes to mind is a sine wave with its regular and recurrent pattern. In economics and other sciences, however, the term *cycle* refers to a more general concept. One of the best-known examples of cycles is the sunspot cycle, which varies in length from under 10 years to nearly 20 years. The significant fact about cycles is the recurrent nature of the events.

In 1922, at a Conference on Cycles, representatives from several sciences discussed the cyclical phenomena in their fields. The participants agreed on the following definition (quoted in Mitchell 1927, p. 377) as being reasonable for all the sciences: "In general scientific use . . . the word (cycle) denotes a recurrence of different phases of plus and minus departures, which are often susceptible of exact measurement." Our definition of business cycles is consistent with this general definition, but we refer to departures as *deviations.*

Mitchell's Four Phases

In 1913, Wesley C. Mitchell published a major work on business cycles, in it reviewing the research that had preceded his own. The book presents his largely descriptive approach, which consists of decomposing a large number of time series into sequences of cycles and then dividing each cycle into four distinct phases. This work was continued by Mitchell (1927) and by Burns and Mitchell (1946), who defined business cycles as

> . . . a type of fluctuation found in the aggregate economic activity of nations that organize their work mainly in business enterprises: a cycle consists of expansions occurring at about the same time in many economic activities, followed by similarly general recessions, contractions, and revivals which merge into the expansion phase of the next cycle; this sequence of changes is recurrent but not periodic; in duration business cycles vary from more than one year to ten or twelve years; they are not divisible into shorter cycles of similar character with amplitudes approximating their own (p. 3).

From the discussion in their books, it is clear the authors view business cycles as consisting of four phases that inevitably evolve from one into another: prosperity, crisis, depression, and revival. This view is expressed perhaps most clearly by Mitchell ([1923] 1951, p. 46), who writes: "Then in order will come a discussion of how prosperity produces conditions which lead to crises, how crises run into depressions, and finally how depressions after a time produce conditions which lead to new revivals." Mitchell clearly had in mind a theoretical framework consistent with that view. In defending the use of the framework of four distinct cyclical phases, Mitchell later wrote that "most current theories explain crises by what happens in prosperity and revivals by what happens in depression" (Mitchell 1927, p. 472). (For an extensive overview of these theories and their relationship to Mitchell's descriptive work, see Haberler 1937.)

We now know how to construct model economies whose equilibria display business cycles like those envisioned by Mitchell. For example, a line of research that gained attention in the 1980s demonstrates that cyclical patterns of this form result as equilibrium behavior for economic environments with appropriate preferences and technologies. (See, for example, Benhabib and Nishimura 1985 and Boldrin 1989.) Burns and Mitchell would have been much more influential if business cycle theory had evolved in this way. Koopmans (1957, pp. 215–16) makes this point in his largely unnoticed "second thought" on Burns and Mitchell's work on business cycles.

In retrospect, it is now clear that the field of business cycles has moved in a completely different direction from the one Mitchell envisioned. Theories with deterministic cyclical laws of motion may a priori have had considerable potential for accounting for business cycles; but in fact, they have failed to do so. They have failed because cyclical laws of motion do not arise as equilibrium behavior for economies with empirically reasonable preferences and technologies—that is, for economies with reasonable statements of people's ability and willingness to substitute.

Frisch's Pendulum

As early as the 1930s, some economists were developing business cycle models that gave rise to difference equations with random shocks. An important example appears in a paper by Ragnar Frisch ([1933] 1965). Frisch was careful to distinguish between impulses in the form of random shocks, on the one hand, and their propagation over time, on the other. In contrast with proponents of modern business cycle theory, he emphasized damped oscillatory behavior. The concept of equilibrium was interpreted as a system at rest (as it is, for instance, in the science of mechanics).

The analogy of a pendulum is sometimes used to describe this view of cycles. Shocks are needed to provide "energy in maintaining oscillations" in damped cyclical systems. Frisch reports that he was influenced by Knut Wicksell, to whom he attributes the following: "If you hit a wooden rocking horse with a club, the movement of the horse will be very different to that of the club" (quoted in Frisch [1933] 1965, p. 178). The use of the rocking horse and pendulum analogies underscores their emphasis on cycles in the form of damped oscillations.

The research of Frisch and Wicksell received considerable attention in the 1930s, but no one built on their work. Construction stopped primarily because the neoclassical growth model and the necessary conceptual tools (particularly the Arrow–Debreu general equilibrium theory) had not yet been developed. Since the tools to do quantitative dynamic general equilibrium analysis weren't available, whereas statistical time series techniques were advancing rapidly, it's not surprising that quantitative system-of-equation models—especially the Keynesian income-expenditure models—received virtually all the attention.

Slutzky's Random Shocks

An entirely different way of generating cycles is suggested by the statistical work of Eugen Slutzky (1937). Slutzky shows that cycles resembling business fluctuations can be generated as the sum of random causes—that is, by a stable, low-order, stochastic difference equation with large positive real roots.

The following exercise illustrates how Slutzky's method can generate cycles. Let the random variable e_t take the value 0.5 if a coin flip shows heads and −0.5 if tails. Assume that

$$(1) \qquad y_{t+1} = 0.95y_t + e_{t+1}.$$

By repeated substitution, y_t is related to current and past shocks in the following way:

$$(2) \qquad y_t = e_t + 0.95e_{t-1} + 0.95^2e_{t-2} + \cdots + 0.95^{t-1}e_1 + 0.95^t y_0.$$

The y_t are geometrically declining sums of past shocks. Given an initial value y_0 and a fair coin, this stochastic difference equation can be used to generate a random path for the variable y.

Chart 1

Cycles Generated by Slutzky's Mechanism

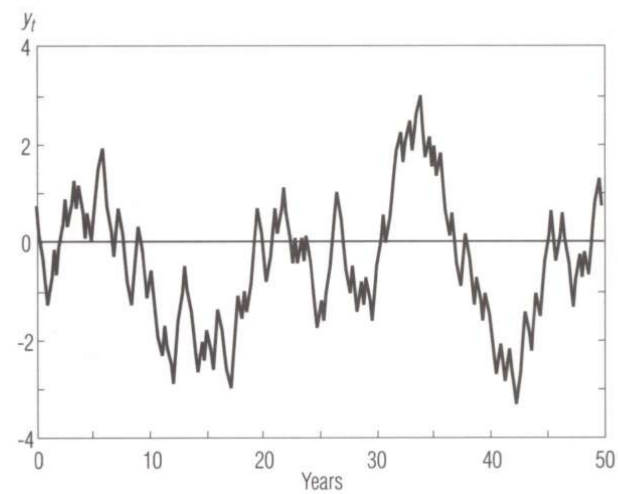


Chart 1 plots a time series generated in this way. The time series displays patterns that Burns and Mitchell (1946) would characterize as business cycles.[2] The amplitudes and duration of cycles are variable, with the duration varying from 1 to 12 years and averaging about 3½ years. The time series seems to display cycles in Mitchell's sense of expansions containing the seed for recessions and vice versa. But, by construction, recessions do not contain the seed of subsequent expansions. At each point in time, the expected future path is monotonic and converges to the zero mean, with 5 percent of the distance being closed in each quarterly time period.

Another demonstration of the role that random shocks can play appears in a paper by Adelman and Adelman (1959). Using the Klein–Goldberger model, they show that by adding random shocks to the model, it produces aggregate time series that look remarkably like those of the post–World War II economy of the United States. The deterministic version of this model converges almost monotonically to a point. This exercise forcefully demonstrates that a stochastic process can generate recurrent cycles while its deterministic version can converge monotonically.

Advancing to Lucas' Deviations

In the 1940s and the 1950s, while macroeconometric

[2]King and Plosser (1989) describe a well-defined, judgment-free scheme that successfully mimics the Burns and Mitchell procedure. The description of the cycles in the time series plotted in Chart 1 is based on this procedure.

system-of-equations models were being developed, important theoretical advances were being made along entirely different fronts. By the early 1960s, economists' understanding of the way economic environments work in general equilibrium had advanced by leaps and bounds. The application of general equilibrium theory in dynamic environments led to theoretical insights on the growth of economies; it also led to important measurements of the parameters of the aggregate production function that formed the foundation for neoclassical growth theory. Thus, by the late 1960s, there were two established theories competing for dominance in aggregate economics. One was the behavioral-empirical approach reflected in the Keynesian system-of-equations models. The other was the neoclassical approach, which modeled environments with rational, maximizing individuals and firms. The neoclassical approach dominated public finance, growth theory, and international trade. As neoclassical theory progressed, an unresolvable conflict developed between the two approaches. The impasse developed because dynamic maximizing behavior is inconsistent with the assumption of invariant behavioral equations, an assumption that underlies the system-of-equations approach.

Not until the 1970s did business cycles again receive attention, spurred on by Lucas' (1977) article, "Understanding Business Cycles." There, Lucas viewed business cycle regularities as "comovements of the deviations from trend in different aggregative time series." He defined the business cycle itself as the "movements about trend in gross national product." Two types of considerations led Lucas to this definition: the previously discussed findings of Slutzky and the Adelmans, and the important advances in economic theory, especially neoclassical growth theory. We interpret Lucas as viewing business cycle fluctuations as being of interest because they are at variance with established neoclassical growth theory.

Another important theoretical advance of the 1960s and 1970s was the development of recursive competitive equilibrium theory. This theory made it possible to study abstractions of the aggregate economy in which optimizing economic behavior produces behavioral relations in the form of low-order stochastic difference equations. The role these advances played for Lucas' thinking is clear, as evident from one of his later articles discussing methods and problems in business cycle theory (see Lucas 1980).

In contrast with Mitchell's view of business cycles, Lucas does not think in terms of sequences of cycles as inevitable waves in economic activity, nor does he see a need to distinguish among different phases of the cycle. To Lucas, the comovements over time of the cyclical components of economic aggregates are of primary interest, and he gives several examples of what he views as the business cycle regularities. We make explicit and operational what we mean by these terms and present a systematic account of the regularities. When that step is implemented quantitatively, some regularities emerge that, in the 1970s, would have come as a surprise—even to Lucas.

Modern Business Cycle Theory

In the 1980s and now in the early 1990s, business cycles (in the sense of recurrent fluctuations) increasingly have become a focus of study in aggregate economics. Such studies are generally guided by perceived business cycle regularities. But if these perceptions are not in fact the regularities, then certain lines of research are misguided.

For example, the myth that the price level is procyclical largely accounts for the prevalence in the 1970s of studies that use equilibrium models with monetary policy or price surprises as the main source of fluctuations. At the time, monetary disturbances appeared to be the only plausible source of fluctuations that could not be ruled out as being too small, so they were the leading candidate. The work of Friedman and Schwartz (1963) also contributed to the view that monetary disturbances are the main source of business cycle fluctuations. Their work marshaled extensive empirical evidence to support the position that monetary policy is an important factor in determining aggregate output, employment, and other key aggregates.

Since the early studies of Burns and Mitchell, the emphasis in business cycle theory has shifted from essentially pure theoretical work to quantitative theoretical analysis. This quantitative research has had difficulty finding an important role for monetary changes as a source of fluctuations in real aggregates. As a result, attention has shifted to the role of other factors—technological changes, tax changes, and terms-of-trade shocks. This research has been strongly guided by business cycle facts and regularities such as those to be presented here.

Along with the shift in focus to investigating the sources and nature of business cycles, aggregate analysis underwent a methodological revolution. Previously, empirical knowledge had been organized in the form of equations, as was also the case for the early rational expectations models. Muth (1960), in his pioneering work on rational expectations, did not break with this system-of-equations tradition. For that reason, his econometric program did not come to dominate. Instead, the program which has prevailed is the one that organizes empirical knowledge around preferences, technology, information structure, and policy rules or arrangements. Sargent (1981) has led the development of tools for inferring values of parameters characterizing these elements, given the behavior of the aggregate time series. As a result, aggregate economics is no longer a separate and entirely different field from the rest of economics; it now uses the same tools and empirical knowledge as other branches of economics, such as finance, growth theory, public finance, and international economics. With this development, measurements and quantitative findings in those other fields can be used to restrict models of business cycles and make our knowledge about the quantitative importance of cyclical disturbances more precise.

Business Cycle Deviations Redefined

Because economic activity in industrial market economies is characterized by sustained growth, Lucas defines business cycles as deviations of real gross national product (GNP) from trend rather than from some constant or average value. But Lucas does not define *trend*, so his definition of business cycle deviations is incomplete. What guides our, and we think his, concept of trend is steady state growth theory. With this theory there is exogenous labor-augmenting technological change that occurs at a constant rate; that is, the effectiveness of labor grows at some constant rate. Steady state growth is characterized by per capita output, consumption, investment, capital stock, and the real wage all growing at the same rate as does technology. The part of productive time allocated to market activity and the real return on capital remain constant.

If the rate of technological change were constant, then the trend of the logarithm of real GNP would be a linear function of time. But the rate of technological change varies both over time and across countries. (*Why* it varies is the central problem in economic development or maybe in all of economics.) The rate of change clearly is related to the arrangements and institutions that a society uses and, more important, to the arrangements and institutions that people expect will be used in the future. Even in a relatively stable society like the United States since the Second World War, there have been significant changes in institutions. And when a society's institutions change, there are changes in the productivity growth of that society's labor and capital. In the United States, the rate of technological change in the 1950s and 1960s was significantly above the U.S. historical average rate over the past 100 years. In the 1970s, the rate was significantly below average. In the 1980s, the rate was near the historical average. Because the underlying rate of technological change has not been constant in the period we examine (1954–1989), detrending using a linear function of time is inappropriate. The scheme used must let the average rate of technological change vary over time, but not too rapidly.

Any definition of the trend and cycle components, and for that matter the seasonal component, is necessarily statistical. A *decomposition* is a representation of the data. A representation is useful if, in light of theory, it reveals some interesting patterns in the data. We think our representation is successful in this regard. Our selection of a trend definition was guided by the following criteria:

- The trend component for real GNP should be approximately the curve that students of business cycles and growth would draw through a time plot of this time series.
- The trend of a given time series should be a linear transformation of that time series, and this transformation should be the same for all series.[3]
- Lengthening the sample period should not significantly alter the value of the deviations at a given date, except possibly near the end of the original sample.
- The scheme should be well defined, judgment free, and cheaply reproducible.

These criteria led us to the following scheme. Let y_t, for $t = 1, 2, \ldots, T$, denote a time series. We deal with logarithms of a variable, unless the variable is a share, because the percentage deviations are what display the interesting patterns. Moreover, when an exponentially growing time series is so transformed, it becomes linear in time. Our trend component, denoted τ_t, for $t = 1, 2, \ldots, T$, is the one that minimizes

$$(3) \quad \textstyle\sum_{t=1}^{T} (y_t - \tau_t)^2 + \lambda \sum_{t=2}^{T-1} [(\tau_{t+1} - \tau_t) - (\tau_t - \tau_{t-1})]^2$$

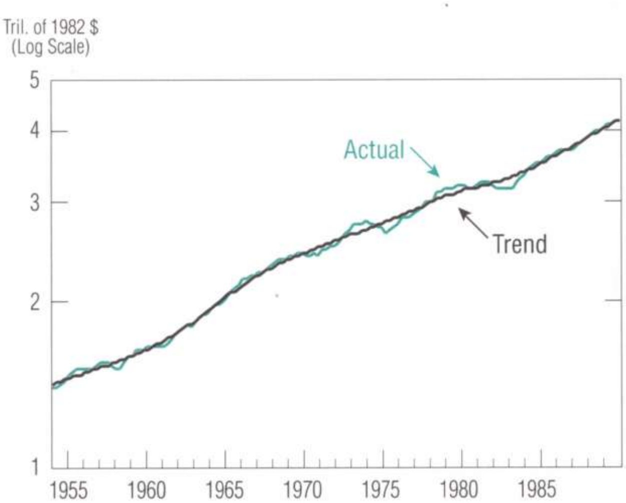
[3]The reason for linearity is that the first two moments of the transformed data are functions of the first two moments, and not the higher moments, of the original data. The principal rationale for the same transformation being applied to all time series is that it makes little sense to carry out the analogue of growth accounting with the inputs to the production function subject to one transformation and the outputs subject to another.

for an appropriately chosen positive λ. (The value of λ will be specified momentarily.) The first term is the sum of the squared deviations $d_t = y_t - \tau_t$. The second term is multiple λ of the sum of the squares of the trend component's second differences. This second term penalizes variations in the growth rate of the trend component, with the penalty being correspondingly larger if λ is larger.

The first-order conditions for this convex minimization problem are linear and can be solved for the τ_t.[4] We found that if the time series are quarterly, a value of $\lambda = 1600$ is reasonable. With this value, the implied trend path for the logarithm of real GNP is close to the one that students of business cycles and growth would draw through a time plot of this series, as shown in Chart 2. The remaining criteria guiding our selection of a detrending procedure are satisfied as well.

We have learned that this procedure for constructing a smooth curve through the data has a long history in both the actuarial and the natural sciences. Stigler (1978) reports that actuarial scientists used this method in the 1920s. He also notes that John von Neumann, who undoubtedly reinvented it, used it in the ballistics literature in the early 1940s. That others facing similar problems developed this simple scheme attests to its reasonableness. What is surprising is that economists took so long to exploit this scheme and that so many of them were so hostile to the idea when it was finally introduced into economics.[5]

Chart 2
Actual and Trend of U.S. Real Gross National Product
Quarterly, 1954–1989



Source of basic data: Citicorp's Citibase data bank

Business Cycle Facts and Regularities

We emphasize that our selection of the facts to report is guided by neoclassical growth theory. This theory, currently the established one in aggregate economics, is being used not only to study growth and development but also to address public finance issues and, more recently, to study business cycles. The facts we present here are the values of well-defined statistics for the U.S. economy since the Korean War (1954–1989). We refer to consistent patterns in these numbers as business cycle *regularities*.

The statistics presented in Tables 1–4 provide information on three basic aspects of the cyclical behavior of aggregates:

- The amplitude of fluctuations
- The degree of comovement with real GNP (our measure of pro- or countercyclicality)
- The phase shift of a variable relative to the overall business cycle, as defined by the behavior of cyclical real GNP.

We emphasize that, except for the share variables shown in Table 2, these statistics are percentage, not absolute, deviations. For instance, the percentage deviation of investment expenditures is more than three times that of total real GNP. Since this component averages less than one-fifth of real GNP, its absolute volatility is somewhat less than that of total output.

In the tables, the degree of contemporaneous comovement with real GNP is indicated in the $x(t)$ column. The statistics in that column are the correlation coefficients of the cyclical deviations of each series with the cyclical deviations of real GNP. A number close to one indicates that a series is highly *procyclical;* a number close to one but of the opposite sign indicates that a series is *countercyclical.* A number close to zero means that a series does not vary contemporaneously with the cycle in any systematic way, in which case we say the series is *uncorrelated* with the cycle.

The remaining columns of the tables also display

[4]A short FORTRAN subroutine that efficiently computes the trend and deviations components is available on request to the Research Department, Federal Reserve Bank of Minneapolis. The computation time required by this algorithm increases linearly with the length of the sample period, as do storage requirements.

[5]This approach was introduced in an unpublished paper by Hodrick and Prescott (1980).

correlation coefficients, except the series have been shifted forward or backward, relative to real GNP, by from one to five quarters. To some extent these numbers indicate the degree of comovement with GNP. Their main purpose, however, is to indicate whether, typically, there is a phase shift in the movement of a time series relative to real GNP. For example, if for some series the numbers in the middle of each table are positive but largest in column $x(t-i)$, where $i > 0$, then the numbers indicate that the series is procyclical but tends to peak about *i* quarters before real GNP. In this case we say the series *leads* the cycle. Correspondingly, a series that *lags* the cycle by $j > 0$ quarters would have the largest correlation coefficient in the column headed by $x(t+j)$. For example, productivity is a series that leads the cycle, whereas the stock of inventories is one that lags the cycle.

We let the neoclassical growth model dictate which facts to examine and how to organize them. The aggregate economy can be divided broadly into three sectors: businesses, households, and government. In the business sector, the model emphasizes production inputs as well as output components. Households allocate income earned in the business sector to consumption and saving. In the aggregate, there is an accounting relation between household saving and business investment. Households allocate a fraction of their discretionary time to income-earning activities in the business sector. The remaining fraction goes to nonmarket activities, usually referred to as leisure but sometimes (perhaps more appropriately) as input to household production. This time-allocation decision has received little attention in growth theory, but it is crucial to business cycle theory. The government sector, which is at the heart of public finance theory, also could play a significant role for business cycles.

Table 1

Cyclical Behavior of U.S. Production Inputs

Deviations From Trend of Input Variables
Quarterly, 1954–1989

		Cross Correlation of Real GNP With										
Variable *x*	Volatility (% Std. Dev.)	*x*(*t*−5)	*x*(*t*−4)	*x*(*t*−3)	*x*(*t*−2)	*x*(*t*−1)	*x*(*t*)	*x*(*t*+1)	*x*(*t*+2)	*x*(*t*+3)	*x*(*t*+4)	*x*(*t*+5)
Real Gross National Product	1.71	−0.03	0.15	0.38	0.63	0.85	1.00	0.85	0.63	0.38	0.15	−0.03
Labor Input												
Hours (Household Survey)	1.47	−0.10	0.05	0.23	0.44	0.69	0.86	0.86	0.75	0.59	0.38	0.18
Employment	1.06	−0.18	−0.04	0.14	0.36	0.61	0.82	0.89	0.82	0.67	0.47	0.25
Hours per Worker	0.54	0.08	0.21	0.35	0.49	0.66	0.71	0.59	0.43	0.29	0.11	−0.02
Hours (Establishment Survey)	1.65	−0.23	−0.07	0.14	0.39	0.66	0.88	0.92	0.81	0.64	0.42	0.21
GNP/Hours (Household Survey)	0.88	0.11	0.21	0.34	0.48	0.50	0.51	0.21	−0.02	−0.25	−0.34	−0.36
GNP/Hours (Establishment Survey)	0.83	0.40	0.46	0.49	0.53	0.43	0.31	−0.07	−0.31	−0.49	−0.52	−0.50
Average Hourly Real Compensation (Business Sector)	0.91	0.30	0.37	0.40	0.42	0.40	0.35	0.26	0.17	0.05	−0.08	−0.20
Capital Input												
Nonresidental Capital Stock*	0.62	−0.58	−0.61	−0.51	−0.48	−0.31	−0.08	0.16	0.39	0.56	0.66	0.70
Structures	0.37	−0.45	−0.51	−0.55	−0.53	−0.44	−0.29	−0.10	0.09	0.25	0.38	0.45
Producers' Durable Equipment	0.99	−0.57	−0.58	−0.53	−0.41	−0.22	0.02	0.26	0.47	0.62	0.70	0.71
Inventory Stock (Nonfarm)	1.65	−0.37	−0.33	−0.23	−0.05	0.19	0.50	0.72	0.83	0.81	0.71	0.53

*Based on quarterly data, 1954:1–1984:2.
Source of basic data: Citicorp's Citibase data bank

The standard version of the neoclassical growth model abstracts from money and therefore provides little guidance about which of the nominal variables to examine. Given the prominence that monetary shocks have held for many years as the main candidate for the impulse to business cycles, it seems appropriate that we also examine the cyclical behavior of monetary aggregates and nominal prices.

Real Facts

□ Production Inputs

We first examine real (nonmonetary) series related to the inputs in aggregate production. The cyclical facts

Table 2

Cyclical Behavior of U.S. Output and Income Components

Deviations From Trend of Product and Income Variables
Quarterly, 1954–1989

		Cross Correlation of Real GNP With										
Variable *x*	Volatility (% Std. Dev.)	*x*(*t*−5)	*x*(*t*−4)	*x*(*t*−3)	*x*(*t*−2)	*x*(*t*−1)	*x*(*t*)	*x*(*t*+1)	*x*(*t*+2)	*x*(*t*+3)	*x*(*t*+4)	*x*(*t*+5)
Real Gross National Product	1.71	−0.03	0.15	0.38	0.63	0.85	1.00	0.85	0.63	0.38	0.15	−0.03
Consumption Expenditures	1.25	0.25	0.41	0.56	0.71	0.81	0.82	0.66	0.45	0.21	−0.02	−0.21
Nondurables & Services	0.84	0.20	0.38	0.53	0.67	0.77	0.76	0.63	0.46	0.27	0.06	−0.12
Nondurables	1.23	0.29	0.42	0.52	0.62	0.69	0.69	0.57	0.38	0.16	−0.05	−0.22
Services	0.63	0.03	0.25	0.46	0.63	0.73	0.71	0.60	0.49	0.39	0.23	0.07
Durables	4.99	0.25	0.38	0.50	0.64	0.74	0.77	0.60	0.37	0.10	−0.14	−0.32
Investment Expenditures	8.30	0.04	0.19	0.39	0.60	0.79	0.91	0.75	0.50	0.21	−0.05	−0.26
Fixed Investment	5.38	0.09	0.25	0.44	0.64	0.83	0.90	0.81	0.60	0.35	0.08	−0.14
Nonresidential	5.18	−0.26	−0.13	0.05	0.31	0.57	0.80	0.88	0.83	0.68	0.46	0.23
Structures	4.75	−0.40	−0.31	−0.17	0.03	0.29	0.52	0.65	0.69	0.63	0.50	0.34
Equipment	6.21	−0.18	−0.04	0.14	0.39	0.65	0.85	0.90	0.81	0.62	0.38	0.15
Residential	10.89	0.42	0.56	0.66	0.73	0.73	0.62	0.37	0.10	−0.15	−0.34	−0.45
Government Purchases	2.07	0.00	−0.03	−0.03	−0.01	−0.01	0.05	0.09	0.12	0.17	0.27	0.34
Federal	3.68	0.00	−0.05	−0.08	−0.09	−0.09	−0.02	0.03	0.06	0.10	0.19	0.24
State & Local	1.19	0.06	0.10	0.17	0.25	0.26	0.25	0.20	0.16	0.19	0.27	0.36
Exports	5.53	−0.50	−0.46	−0.34	−0.14	0.11	0.34	0.48	0.53	0.53	0.53	0.45
Imports	4.92	0.11	0.18	0.30	0.45	0.61	0.71	0.71	0.51	0.28	0.03	−0.19
Real Net National Income												
Labor Income*	1.58	−0.18	−0.02	0.18	0.42	0.68	0.88	0.90	0.80	0.62	0.40	0.19
Capital Income**	2.93	0.10	0.24	0.44	0.63	0.79	0.84	0.60	0.30	0.02	−0.19	−0.29
Proprietors' Income & Misc.†	2.70	0.11	0.24	0.38	0.55	0.62	0.68	0.46	0.29	0.11	0.02	−0.10

*Employee compensation is deflated by the implicit GNP price deflator.

**This variable includes corporate profits with inventory valuation and capital consumption adjustments, plus rental income of persons with capital consumption adjustment, plus net interest, plus capital consumption allowances with capital consumption adjustment, all deflated by the implicit GNP price deflator.

†Proprietors' income with inventory valuation and capital consumption adjustments, plus indirect business tax and nontax liability, plus business transfer payments, plus current surplus of government enterprises, less subsidies, plus statistical discrepancy.

Source of basic data: Citicorp's Citibase data bank

are summarized in Table 1. Since it is not unreasonable to think of the inventory stock as providing productive services, we include this series with the labor and capital inputs.

The two most common measures of the labor input are aggregate hours-worked according to the household survey and, alternatively, the payroll or establishment survey. We see in Table 1 that total hours with either measure is strongly procyclical and has cyclical variation which, in percentage terms, is almost as large as that of real GNP. (For a visual representation of this behavior, see Chart 3.) The capital stock, in contrast, varies smoothly over the cycle and is essentially uncorrelated with contemporaneous real GNP. The correlation is large, however, if the capital stock is shifted back by about a year. In other words, business capital lags the cycle by at least a year. The inventory stock also lags the cycle, but only by about half a year. In percentage terms, the inventory stock is nearly as volatile as quarterly real GNP.

The hours-worked series from the household survey can be decomposed into employment fluctuations on the one hand and variations in hours per worker on the other. Employment lags the cycle, while hours per worker is nearly contemporaneous with it, with only a slight lead. Much more of the volatility in total hours worked is caused by employment volatility than by changes in hours per worker. If these two subseries were perfectly correlated, their standard deviations would add up to the standard deviation of total hours. Although not perfectly correlated, their correlation is quite high, at 0.86. Therefore, employment accounts for roughly two-thirds of the standard deviation in total hours while hours per worker accounts for about one-third.

As a measure of the aggregate labor input, aggregate hours has a problem: it does not account for differences across workers in their relative contributions to aggregate output. That is, the hours of a brain surgeon are given the same weight as those of an orderly. This

Table 3

Cyclical Behavior of U.S. Output and Income Component Shares

Deviations From Trend of Product and Income Variables
Quarterly, 1954–1989

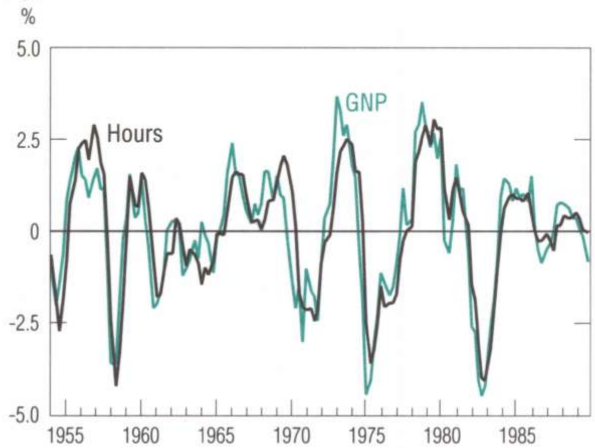
Variable x	Mean % of GNP	Volatility (% Std. Dev.)	Cross Correlation of Real GNP With										
			$x(t-5)$	$x(t-4)$	$x(t-3)$	$x(t-2)$	$x(t-1)$	$x(t)$	$x(t+1)$	$x(t+2)$	$x(t+3)$	$x(t+4)$	$x(t+5)$
Gross National Product													
Consumption Expenditures	63.55	0.58	0.29	0.15	−0.06	−0.32	−0.56	−0.78	−0.68	−0.52	−0.33	−0.17	−0.03
Nondurables & Services	54.79	0.70	0.06	−0.08	−0.27	−0.51	−0.72	−0.89	−0.73	−0.50	−0.23	0.01	0.18
Durables	8.76	0.33	0.36	0.43	0.48	0.54	0.56	0.53	0.36	0.15	−0.10	−0.31	−0.44
Investment Expenditures	15.85	1.07	0.03	0.18	0.36	0.56	0.75	0.87	0.71	0.47	0.18	−0.09	−0.30
Fixed Investment	15.16	0.56	0.11	0.25	0.40	0.57	0.74	0.81	0.77	0.61	0.40	0.14	−0.08
Change in Business Inventories	0.69	0.69	0.04	0.07	0.24	0.40	0.56	0.69	0.48	0.22	−0.05	−0.25	−0.40
Government Purchases	20.13	0.57	0.04	−0.09	−0.25	−0.40	−0.55	−0.61	−0.52	−0.36	−0.15	0.09	0.28
Net Exports	0.47	0.45	−0.51	−0.51	−0.48	−0.43	−0.37	−0.28	−0.17	0.00	0.17	0.30	0.38
Net National Income*													
Labor Income	58.57	0.47	−0.29	−0.36	−0.45	−0.52	−0.47	−0.39	−0.03	0.23	0.42	0.48	0.46
Capital Income	24.38	0.42	0.19	0.25	0.36	0.43	0.48	0.43	0.17	−0.13	−0.35	−0.48	−0.46
Proprietors' Income & Misc.	17.04	0.34	0.18	0.19	0.17	0.17	0.06	0.00	−0.16	−0.19	−0.20	−0.11	−0.11

*For explanations of the national income components, see notes to Table 2.
Source of basic data: Citicorp's Citibase data bank

Chart 3

Deviations From Trend of U.S. Real Gross National Product and Hours Worked*

Quarterly, 1954–1989



*The estimate of hours worked uses the establishment survey.
Source of basic data: Citicorp's Citibase data bank

disparity would not be problematic if the cyclical volatility of highly skilled workers resembled that of the workers who are less skilled. But it doesn't. The hours of the less-skilled group are much more variable, as established in one of our recent studies (Kydland and Prescott 1989). Using data for nearly 5,000 people from all major demographic groups over the period 1969–82, we found that, cyclically, aggregate hours is a poor measure of the labor input. When people were weighted by their relative human capital, the labor input for this sample and period varied only about two-thirds as much as did aggregate hours. We therefore recommend that the cyclical behavior of labor productivity (as reported by GNP/hours in Table 1) be interpreted with caution.

Since the human-capital-weighted cyclical measure of labor input fluctuates less than does aggregate hours, the implicit real wage (the ratio of total real labor compensation to labor input) is even more procyclical than average hourly real compensation. (For the latter series, see Table 1.) This finding that the real wage behaves in a reasonably strong procyclical manner is

Table 4

Cyclical Behavior of U.S. Monetary Aggregates and the Price Level

Deviations From Trend of Money Stock, Velocity, and Price Level
Quarterly, 1954–1989

Variable *x*	Volatility (% Std. Dev.)	Cross Correlation of Real GNP With										
		x(*t*−5)	*x*(*t*−4)	*x*(*t*−3)	*x*(*t*−2)	*x*(*t*−1)	*x*(*t*)	*x*(*t*+1)	*x*(*t*+2)	*x*(*t*+3)	*x*(*t*+4)	*x*(*t*+5)
Nominal Money Stock*												
Monetary Base	0.88	−0.12	0.02	0.14	0.25	0.36	0.41	0.40	0.37	0.32	0.28	0.26
M1	1.68	0.01	0.12	0.23	0.33	0.35	0.31	0.22	0.15	0.09	0.07	0.07
M2	1.51	0.48	0.60	0.67	0.68	0.61	0.46	0.26	0.05	−0.15	−0.33	−0.46
M2 - M1	1.91	0.53	0.63	0.67	0.65	0.56	0.40	0.20	−0.01	−0.21	−0.39	−0.53
Velocity*												
Monetary Base	1.33	−0.26	−0.15	0.00	0.22	0.40	0.59	0.50	0.37	0.22	0.08	−0.08
M1	2.02	−0.24	−0.19	−0.12	−0.01	0.14	0.31	0.32	0.27	0.20	0.10	0.00
M2	1.84	−0.63	−0.59	−0.48	−0.29	−0.05	0.24	0.34	0.40	0.43	0.44	0.43
Price Level												
Implicit GNP Deflator	0.89	−0.50	−0.61	−0.68	−0.69	−0.64	−0.55	−0.43	−0.31	−0.17	−0.04	0.09
Consumer Price Index	1.41	−0.52	−0.63	−0.70	−0.72	−0.68	−0.57	−0.41	−0.24	−0.05	0.14	0.30

*Based on quarterly data, 1959:1–1989:4.
Source of basic data: Citicorp's Citibase data bank

counter to a widely held belief in the literature. [For a fairly recent expression of this belief, see the article by Lawrence Summers (1986, p. 25), which states that there is "no apparent procyclicality of real wages."]

□ *Output Components*

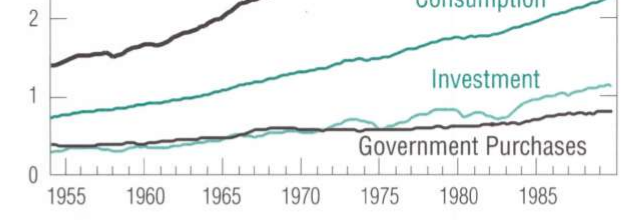
Real GNP is displayed in Chart 4, along with its three major components: consumption, investment, and government purchases. These three components do not quite add up to real GNP, the difference being accounted for by net exports and change in business inventories. Because household investment in consumer durables behaves similarly to fixed investment in the business sector, we have added those two series. By far the largest component (nearly two-thirds) of total output is consumption of nondurable goods and services. This component, moreover, has relatively little volatility. The chart shows that the bulk of the volatility in aggregate output is due to investment expenditures.

The cyclical components (relative to cyclical real GNP) of consumer nondurables and services, consumer durable investment, fixed investment, and government purchases are reported in Table 2 and plotted in Charts 5–8. From the table and charts, we can see that all but government purchases are highly procyclical. Household and business investment in durables have similar amplitudes of percentage fluctuations. Expenditures for consumer durables leads slightly while nonresidential fixed investment lags the cycle, especially investment in structures. Consumer nondurables and services is a relatively smooth series.

Chart 4

U.S. Real Gross National Product and Its Components*

Quarterly, 1954–1989



*Consumption includes nondurable goods and services. Investment is the sum of consumer durable investment and business fixed investment. Components do not add to total because net exports and the change in business inventories are excluded.

Source of basic data: Citicorp's Citibase data bank

Some of the interesting features of the other components are that government purchases has no consistent pro- or countercyclical pattern, that imports is procyclical with no phase shift, and that exports is procyclical but lags the cycle by from six months to a year.

The cyclical behavior of the major output components, measured as shares of real GNP, is reported in Table 3. Using fractions rather than the logarithms of the series permits us to include some series that could not be used in Table 2 because they are negative during some quarters. We see that the change in business inventories is procyclical. Net exports is a countercyclical variable, with the association being strongest for exports shifted back by about a year.

□ *Factor Incomes*

Tables 2 and 3 also provide information about factor incomes, which are the components of national income. The cyclical behavior of factor incomes is described in terms of their levels (Table 2) and their shares of GNP (Table 3). Since proprietors' income includes labor and capital income, we treat this component (plus some small miscellaneous components) separately. We find that proprietors' income, as a share of national income, is uncorrelated with the cycle.

Table 2 shows that both labor income and capital income are strongly procyclical and that capital income is highly volatile. Table 3 shows that, measured as shares of total income, labor income is countercyclical while capital income is procyclical.

Nominal Facts

The statistical properties of the cyclical components of various nominal aggregates are summarized in Table 4, and four of these series along with cyclical real GNP are plotted on Charts 9–12.

□ *Monetary Aggregates*

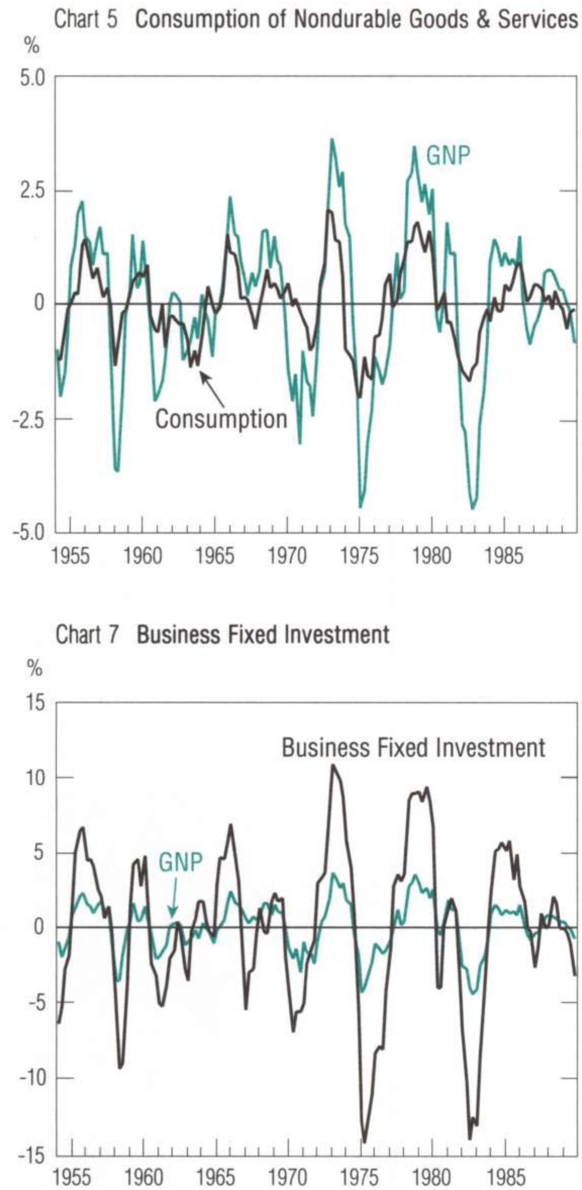
There is no evidence that either the monetary base or M1 leads the cycle, although some economists still believe this monetary myth. Both the monetary base and M1 series are generally procyclical and, if anything, the monetary base lags the cycle slightly.

An exception to this rule occurred during the expansion of the 1980s. This expansion, so long and steady, has even led some economists and journalists to

Charts 5–8

Deviations From Trend of U.S. Real Gross National Product and Its Components

Quarterly, 1954–1989



Source of basic data: Citicorp's Citibase data bank

Charts 9–12

Deviations From Trend of U.S. Real Gross National Product and Selected Nominal Aggregates

Quarterly, 1959–1989*

Chart 9 **Monetary Base**

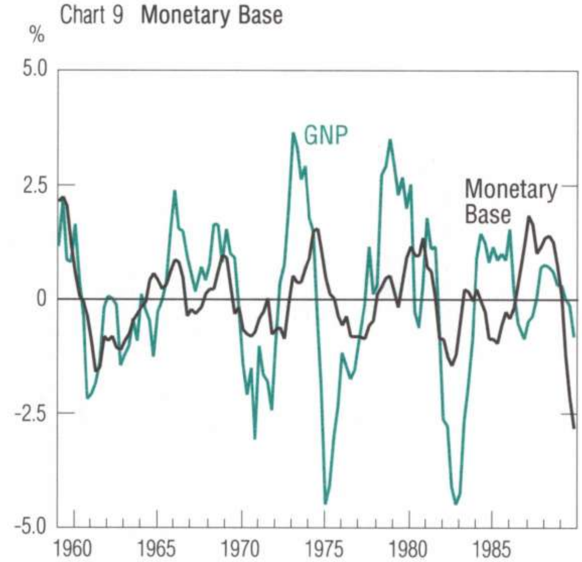


Chart 10 **M1**

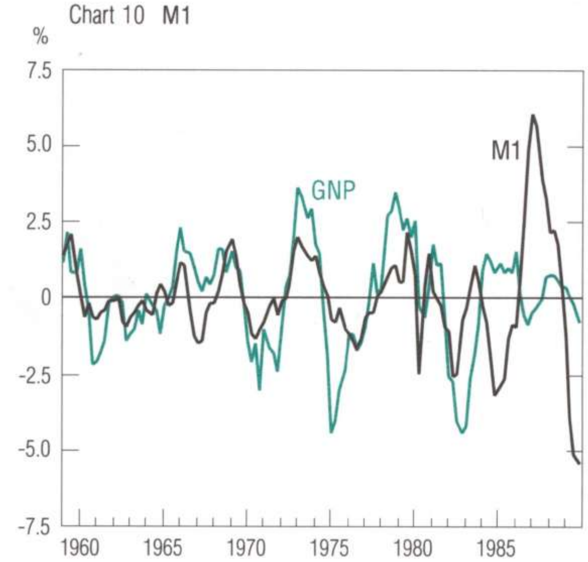


Chart 11 **M2**

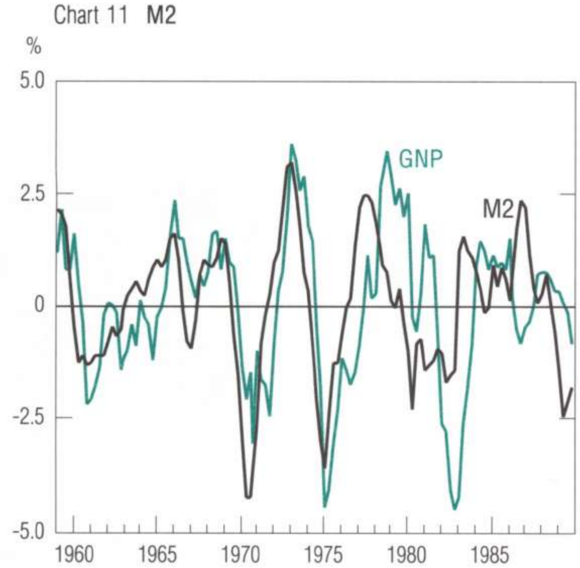
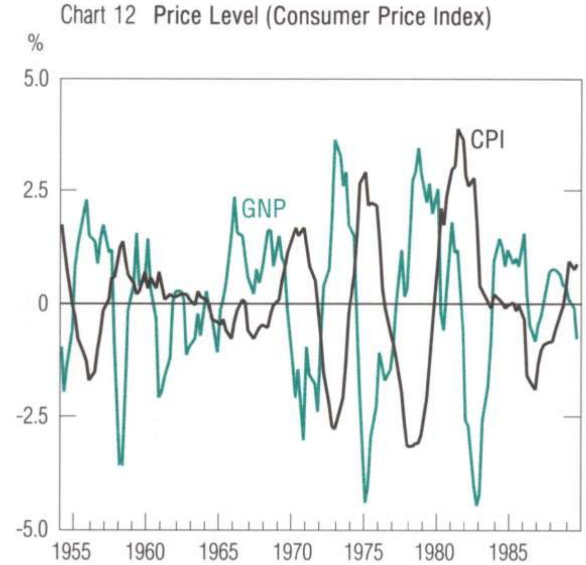


Chart 12 **Price Level (Consumer Price Index)**



*For the price level, 1954–1989.

Source of basic data: Citicorp's Citibase data bank

speculate that the business cycle is dead (Zarnowitz 1989 and *The Economist* 1989). During the expansion, M1 was uncommonly volatile, and M2, the more comprehensive measure of the money stock, showed some evidence that it leads the cycle by a couple quarters.

The difference in the behavior of M1 and M2 suggests that the difference of these aggregates (M2 minus M1) should be considered. This component mainly consists of interest-bearing time deposits, including certificates of deposit under $100,000. It is approximately one-half of annual GNP, whereas M1 is about one-sixth. The difference of M2 − M1 leads the cycle by even more than M2, with the lead being about three quarters.

From Table 4 it is also apparent that money velocities are procyclical and quite volatile.

□ *Price Level*

Earlier in this paper, we documented the view that the price level is always procyclical. This myth originated from the fact that, during the period between the world wars, the price level *was* procyclical. But because of the Koopmans taboo against reporting business cycle facts, no one bothered to ascertain the cyclical behavior of the price level since World War II. Instead, economists just carried on, trying to develop business cycle theories in which the price level plays a central role and behaves procyclically. The fact is, however, that whether measured by the implicit GNP deflator or by the consumer price index, the U.S. price level clearly has been countercyclical in the post-Korean War period.

Concluding Remarks

Let us reemphasize that, unlike Burns and Mitchell, we are not claiming to measure business cycles. We also think it inadvisable to start our economics from some statistical definition of trend and deviation from trend, with growth theory being concerned with trend and business cycle theory with deviations. Growth theory deals with both trend and deviations.

The statistics we report are of interest, given neoclassical growth theory, because they are—or maybe were—in apparent conflict with that theory. Documenting real or apparent systematic deviations from theory is a legitimate activity in the natural sciences and should be so in economics as well.

We hope that the facts reported here will help guide the selection of model economies to study. We caution that any theory in which procyclical prices figure crucially in accounting for postwar business cycle fluctuations is doomed to failure. The facts we report indicate that the price level since the Korean War moves countercyclically.

The fact that the transaction component of real cash balances (M1) moves contemporaneously with the cycle while the much larger nontransaction component (M2) leads the cycle suggests that credit arrangements could play a significant role in future business cycle theory. Introducing money and credit into growth theory in a way that accounts for the cyclical behavior of monetary as well as real aggregates is an important open problem in economics.[6]

[6] Two interesting attempts to introduce money into growth theory are the work of Cooley and Hansen (1989) and Hodrick, Kocherlakota, and D. Lucas (1988). Their approach focuses on the transaction role of money.

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Introduction to business cycles

The nineteenth-century economists did not, for the most part, recognise the idea or existence of regular cycles in economic activity. Instead, they thought in terms of 'crises', a word implying an abnormal level of activity and used in several different ways: it could mean a financial panic (that is the peak turning point in a commercial or financial market) or it could mean the period of deepest depression when factories closed. A financial crisis was not necessarily followed by an economic depression, but might be a phenomenon solely of the financial markets.[1] There were a few exceptions amongst economists and industrial commentators who recognised a cyclical pattern in economic activity, including, for example, Marx.[2] In addition, there was no agreed theory about what caused a crisis, indeed it sometimes seemed that not only each economist but each person had their own pet theory of the cause of crises in the economy. One telling example, 'The First Annual Report of the Commissioner of Labor' in the USA in 1886 listed all the causes of depressions reported to the various committees of Congress. This list ran to four pages, from

> Administration, changes in the policies of
> Agitators, undue influence of

through various economic and institutional reasons to

> War, absorption of capital by destruction of
> property during
> Work, piece.[3]

[1] Kindleberger's recent analysis (1978) of 'crises' rather than cycles shows how varied the meaning of the term was in the nineteenth century and earlier.

[2] Schumpeter (1954) discusses the emergence of the business cycle as a unit of analysis in the nineteenth century and suggests that apart from Marx and Jevons, economists were not much interested in this field.

[3] The full list is quoted by Hull (1926), pp. 262–6.

There was not much chance of making sense of so many causes using the standard ways of gathering evidence in support of theories in nineteenth-century economics. Casual empiricism: the use of odd facts, a few numbers, or individual cases could not penetrate very deeply into the causes of complex macroeconomic behaviour over time. Full-blown empiricism, such as that of the German historical school, suffered from the inability to synthesise its masses of evidence into coherent theories. The third source of evidence for economists, introspection, was more suited to investigating microeconomic behaviour. It was against this background that serious quantitative study of business cycles began in the late nineteenth century. The development of a standard approach was not likely to be straightforward, for there was no agreed theory or even definition of the phenomena. Instead, there was a considerable amount of blundering about in the dark looking for the business cycle. Did it exist or was it just a tendency; was it a fixed recurring cycle; was each cycle different; was it predictable? Econometricians tried to discover the answers to their questions directly from the evidence; but there were many different ways of choosing data, of analysing them and of exploring the relationships of interest. As for the evidence itself, although standard national income accounts did not exist, there was a considerable body of data available on prices, output, financial markets and so on, which was suitable for business cycle analysis.

The development of the econometric analysis of business cycles is described in four chapters. Chapters 1 and 2 discuss the histories of two different approaches to statistical study of the business cycle. These approaches were associated with different notions of the business cycle and employed different methodologies, which together implied different analytical techniques. The first chapter deals with the early attempts to confirm clearly defined theories in which economic cycles were assumed to be regular standard events subject to statistical analysis. In contrast, Chapter 2 covers the work of those who rejected a strong theoretical input in their quantitative study and who regarded each observed cycle as a different individual event. These economists used statistical evidence and techniques to enrich a descriptive approach which was in some cases theory-seeking and in others blatantly empirical in outlook; questions of measurement and definition dominated their work. The methods used by both groups did not go uncriticised. Chapter 3 shows how statistical discussions on time-series issues led to an understanding of the design features required to make economic theories of the business cycle amenable to econometric study. Chapter 4 describes how a fully-fledged econometric approach to business cycle measurement and testing emerged, dependent on both mathematically expressed theory and statistical understanding.

CHAPTER 1

Sunspot and Venus theories of the business cycle

Wiliam Stanley Jevons was one of the first economists to break away from the casual tradition of applied work which prevailed in the nineteenth century and combine theory with statistical data on many events to produce a general account of the business cycle. Then, in the early twentieth century when statistical work was still uncommon, Henry Ludwell Moore adopted more sophisticated techniques to develop an alternative theory of the cycle. This chapter relates how Jevons and Moore set about building their theories. Both of them relied heavily on statistical regularities in the formation of their hypotheses, which featured periodic economic cycles caused by exogenous changes in heavenly bodies. The development of the econometric approach to business cycle analysis between the 1870s and the 1920s is shown in their work, and in its changing reception from reviewers. Jevons' and Moore's cycle work deserves to be taken seriously as pioneering econometrics, yet the periodic cycle programme they initiated did not prove to be very influential. The last section of this chapter explores why this was so.

1.1 Jevons' sunspot theory

Jevons' sunspot theory of the cycle has always been the object of mirth to his fellow economists, despite the fact that by the time he began to work on the subject in the 1870s he was already well known and distinguished for his contributions to mainstream economics.[1] He was also renowned for statistical analyses of various problems, particularly the problem of index numbers. This statistical work included some fine studies of economic fluctuations, for example, on the seasonal

[1] William Stanley Jevons (1835–82) was best known during his life for his writings on practical economic and policy issues. His theoretical contributions and his role in developing marginal analysis were not recognised till later. He was a keen advocate and user of statistics and mathematics in economics, on which see S. M. Stigler (1982) and Schabas (1984).

variations in money markets. In contrast, his empirical work in a series of papers from 1875 to 1882 on solar periods and economic cycles won him little praise.[2]

The initial hypothesis of Jevons' first paper on trade cycles was that the sunspot cycle led to a weather cycle which in turn caused a harvest cycle and thence a price cycle (Jevons, 1875: see (1884), Paper VI). The problem was that there was little evidence to back up his hypothesis: there was no obvious cycle in grain prices to match the 11.1-year sunspot cycle. He thought that this was probably because there were many other influences on the harvest besides the sun, and that grain prices were also influenced by social, political and economic factors. In other words, the sun's effect on the harvest cycle, and thus the price cycle, was overshadowed by these other disturbing factors for which his methods could not control.

Encouraged by the fact that Schuster had already found a cycle in German wine vintages that matched the sunspot cycle in length, Jevons decided to use agricultural data from the thirteenth and fourteenth centuries (collected by Rogers (1866)) to test his hypothesis. He felt that using data from these earlier centuries would lessen the problem of interference from other factors and consequently that agricultural price data would show clear cycles. Jevons did not even have sunspot data for this early period; he simply assumed that the length of the sunspot cycle had been the same in the thirteenth and fourteenth centuries as it was in the nineteenth century.

Jevons' analysis of the price data was interesting. His method was to lay out the data, for a number of price series for different crops over a 140-year period, on a grid representing 11 years. Analysis of the grid seemed to show a similar pattern of variation in the prices of each of the crops. In addition the variation in the aggregate figures seemed to Jevons to be greater than that which would be expected by the impact of purely accidental causes such as wars, plagues and so forth (even after omitting outlying observations). He also checked the number of maxima and minima of crop prices occurring in any given year of the 11-year grid and found that their distribution looked distinctly

[2] Jevons' interest in time-series fluctuations may have grown out of his earlier work on meteorology, in which area he first published. His work on sunspot cycles was made known in papers given at meetings of the British Association (Section F) and in contributions to *The Times* and to *Nature*. These appear, together with much of his other statistical work, in the collection *Investigations in Currency and Finance* published posthumously in 1884. Other items and letters can be found in Jevons' *Papers*, published recently under the editorship of R. D. Collison Black (see Vols. IV and V, 1977, and Vol. VII, 1981). These include a previously unpublished but very interesting unfinished piece on cycles which Jevons wrote for the *Princeton Review*.

non-uniform.[3] The evidence of the variation in prices revealed by the grid convinced Jevons of the existence of an 11-year cycle in crop prices. He admitted that his results would be more persuasive if the price series on different crops were independent sets of observations, which of course they were not. But Jevons did not assert positive proof based on his analysis, which was just as well because he soon discovered that if he analysed the prices on a grid of 3, 5, 7, 9 or 13 years the results were just as good![4]

Returning to the nineteenth century in his search for evidence, Jevons began to look at cycles in commercial credit instead of in agricultural prices. His analysis of the dates of nineteenth-century financial crises produced an average cycle length of 10.8 years, a little less than the sunspot cycle of 11.1 years. He tried to get around this slight difference by suggesting that his sunspot theory, combined with the theory of the credit cycle as a mental or psychological phenomenon, would produce such an observed average cycle length. He wrote:

> It may be that the commercial classes of the English nation, as at present constituted, form a body, suited by mental and other conditions, to go through a complete oscillation in a period nearly corresponding to that of the sun-spots. In such conditions a comparatively slight variation of the prices of food, repeated in a similar manner, at corresponding points of the oscillation, would suffice to produce violent effects. A ship rolls badly at sea, when its period of vibration corresponds nearly with that of the waves which strike it, so that similar impulses are received at similar positions. A glass is sometimes broken by a musical sound of the same tone as that which the glass produces when struck. A child's swing is set and kept in motion by a very small push, given each time that it returns to the same place. If, then, the English money market is naturally fitted to swing or roll in periods of ten or eleven years, comparatively slight variations in the goodness of harvests repeated at like intervals would suffice to produce those alternations of depression, activity, excitement, and collapse which undoubtedly recur in well-marked succession ... I am aware that speculations of this kind may seem somewhat far-fetched and finely-wrought; but financial collapses have recurred with such approach to regularity in the last fifty years, that either this or some other explanation is needed. (Jevons, 1875: see (1884), pp. 184–5)

So in this combined theory, the sunspot cycle caused a harvest cycle which in turn maintained or intensified the natural cyclical motion of

[3] Probability held an important place in Jevons' philosophy of science, but he rarely, as in this example, used probability ideas in conjunction with statistical evidence (but see Aldrich (1987)).

[4] Jevons did not immediately publish his 1875 paper because of this discovery, which he noted in a letter to J. Mills in 1877 (1977, IV, letter 482). The paper was prepared by Jevons for publication in the *Investigations* (1884).

the economy. This idea of a combination of endogenous and exogenous causes turned out to be an important element in econometric models of the business cycle in the 1930s.

Convinced that sunspot cycles and business cycles were of the same length, Jevons resolved early in 1878

> to prove the matter empirically, by actual history of last century occurrences. (Jevons (1977), IV, letter 511)

This work was mainly concerned with dating the various commercial crises of the previous 150 years or so going back to the South Sea Bubble. Aware that his commitment to the theory clouded his judgement, Jevons wrote at this point:

> I am free to confess that in this search I have been thoroughly biased in favour of a theory, and that the evidence which I have so far found would have no weight, if standing by itself. (Jevons, 1878: see (1884), p. 208)

He used the cycle dates of other commentators to buttress his case wherever possible. He searched avidly for evidence of crises where there appeared to be ones missing in the sequence and even for evidence which would make the cycles come at exactly equal intervals by suggesting reasons why crises had been either accelerated or retarded by a few months from their due date.

The evidence of regular credit cycles did not, according to Jevons, stand by itself: there was also the coincidence of cycle lengths. The extended series of commercial crises gave him an average cycle length of somewhere between 10.3 and 10.46 years. Luckily, this corresponded with a new estimate of the sunspot cycle length, which was now put at 10.45 years. Jevons argued:

> Judging this close coincidence of results according to the theories of probabilities, it becomes highly probable that two periodic phenomena, varying so nearly in the same mean period, are connected as cause and effect. (Jevons, 1878: see (1884), p. 195)

His theory was based on a 'perfect coincidence' which

> is by itself strong evidence that the phenomena are causally connected. (Jevons, 1878: see (1884), p. 210)

The recalculation of the sunspot cycle length so close to that of the average interval between commercial crises enabled Jevons to disown his earlier 'fanciful' explanation of the difference between them. He now described the idea of a mental cycle pushed at regular intervals by a harvest cycle as merely a construct to explain that difference.

Despite the coincidence of cycle lengths, Jevons was aware that there

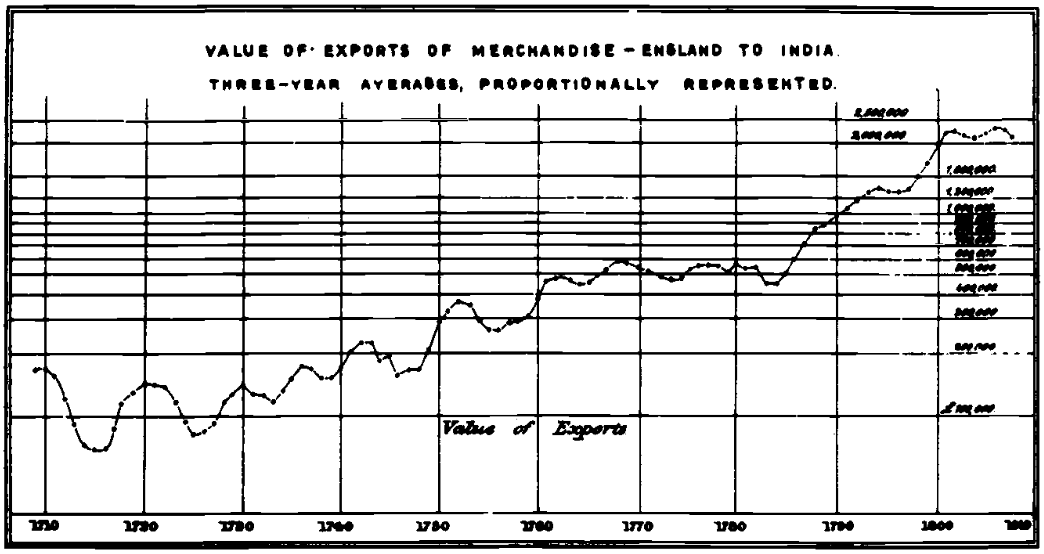


Figure 1 Jevons' trade cycle graph
Source: Jevons (1884), Plate XIV

was still no evidence of how the sunspots caused the commercial cycle since there was no obvious cycle in the intermediary of agricultural prices. He thought he could see a way round this difficulty when he observed that famines in India appeared to have a cyclical pattern similar to the sunspot cycle. Jevons believed that he would be able to observe the causal relationship between sunspots and agricultural output directly by studying events in the tropical and subtropical regions (e.g. India), where other disturbing factors affecting crop prices were likely to be less important than in the industrialised zones such as Britain. He theorised that when agricultural output was low and food prices were high in India, the Indian peasant had no money left over to buy British cotton textiles, thus Britain's exports fell and a commercial cycle was induced in the economic affairs of Britain. He did not regard this theory as being incompatible with the presence of other, more immediate, causes of crises such as bank failures (which could, as he pointed out, be connected with the Indian trade). Jevons explored this new version of his theory in 1878 and 1879 and claimed to find strong evidence of the imported cycle in trade with India. One of Jevons' graphs (of which he was particularly proud), showing his evidence of 10–year variation in trade is given here (Figure 1). This moving average chart of the value of exports to India, on a logarithmic scale, is quite an advanced representation of data for the period. But the statistical arguments in this later work were less sophisticated than those used in his first paper, for he relied solely on the comparison of dates, the comparison of average periods and the visual evidence of his time-series graphs.

Jevons had little more to add either to his theory or to the evidence in succeeding notes and letters. Further evidence was in his own mind unnecessary, since he was by now thoroughly seduced by his 'treacherous', 'entrancing' and 'beautiful coincidence', as he referred to his theory.[5] He noted that the cycles in commercial crises were more regular than the sunspots! But he did not allow this interesting fact to cast any doubt on the causal relationship running from sunspots to commercial crises. Indeed, he was so confident of this, that on several occasions he predicted the months of a future turning point in the commercial cycle and in his last paper on the subject in 1882, he even used the evidence of a cyclical peak in corn prices in Delhi to infer the presence of a preceding peak in the sunspot data (see Jevons (1981), VII, Paper XIV).

It is clear, from the accounts of his papers and from his own responses, that Jevons' sunspot theory was ridiculed, rather than criticised, by many of his scientific colleagues. They poked fun at the theory as a whole by suggesting unlikely relationships between sunspot cycles and other phenomena. Indeed, Jevons' articles and letters were part of a general literature of the time connecting sunspots with other recurring events. This literature involved a fair sprinkling of spoofs, and it is not always easy to tell which were meant seriously. Compare, for example, the titles 'Locusts and Sunspots' and 'Sunspots and the Nile' with one of Jevons' own short notes 'Sunspots and the Plague' (three articles from *Nature* in late 1878 and early 1879). Jevons felt that one anonymous contribution in the *Journal of the Royal Statistical Society* for 1879: 'University Boat Races and Sunspot Cycles' was particularly aimed at him.[6] Nevertheless, Jevons was not abashed; the more his theory was made fun of, the more serious Jevons became about it:

> I never was more in earnest ... It is no jest at all.
>
> (Jevons (1884), pp. 215 and 221)

Despite the ridicule of many of his colleagues, a few did take Jevons' work seriously. These tended to be those on the fringes of the profession, such as John Mills (a banker and frequent correspondent of Jevons on sunspots), whose prime area of influence was in commercial life. One exception was his successor at Manchester, Adamson, who

[5] Such language was sprinkled through Jevons' work on the sunspot cycle; these three come from letters 488 and 566 (1877 and 1878) in the *Papers* (1977, IV), and the 1878 *Nature* paper (1884, p. 207).

[6] See Jevons (1977, V, p. 51) for this particular episode and S. M. Stigler (1982).

wrote to Jevons towards the end of 1878 with some detailed criticism of the theory and its presentation.[7] Adamson criticised the validity of some of the 'crises' dates found by Jevons and he doubted that cycles would be exactly the same in length because he thought that there would be an increasing interval between crises as industrialisation took place. The lack of exact cycles in corn prices and output also worried him and he found Jevons' hypothesis of the link between these cycles unconvincing. On the other hand, Adamson was willing to believe that the sun would have an impact on the economic environment; but he complained that Jevons had provided no 'mode of connexion' between sunspots and agricultural output.

Adamson's criticisms were unusual in that he attacked Jevons' work at the level of evidence and the causal mechanism of his theory. It might be thought surprising that more of his contemporaries did not take this line, since it is clear that the evidence was not convincing, even by the standards of Jevons' earlier statistical work on other topics. But this was not the main thrust of most contemporary criticism. Yet nearly 60 years later, in his 'Centenary Allocution' on Jevons in 1936, J. M. Keynes took precisely this line of attack, criticising Jevons directly for the flimsy evidence on which he based his inductive argument and for the lack of attention to the time relation between cause and effect (and only indirectly for not making an economic factor the main cause of the cycle).

How did Jevons attempt to justify his views in the face of the derision of his colleagues and defend the fact that, despite the lack of convincing evidence, he gave credence to an imported periodic economic cycle? The answer lies in Jevons' use of an inductive methodology which formed an important element of his philosophy of science writings. In the context of defending his sunspot theory, Jevons explained his inductive logic as follows:

> We must proceed upon the great principle of inductive method, as laid down by Laplace and the several great mathematicians who created the theory of probability. This principle is to the effect that the most probable cause of an event which has happened is that cause which if it existed would most probably lead to that effect ... if we perceive a distinctly periodic effect, and can discover any cause which recurs at exactly equal intervals, and is the only discoverable cause recurring in that period, this is probably the cause of which we are in search. Such is the *prima facie* result, drawn simply on the ground that such a cause if existing would have effects with the required period, and there is no

[7] Letter 564, Adamson to Jevons, in *Papers* (1977, IV). Adamson took over Jevons' chair at Owens College, Manchester in 1875 when Jevons moved back to University College, London, as Professor of Political Economy (1875–81).

other cause which could be supposed with any probability to give that result. But this *prima facie* probability is immensely strengthened if we can give other reasons for believing that a cause of the nature supposed, apart from the question of its period, is likely to have effects of the kind we are attributing to it. In short, mere equality of period is a perfectly valid ground of inductive reasoning; but our results gain much in probability if we can analyse and explain the precise relation of cause and effect.[8]

(Jevons (1981), VII, pp. 93–4)

From this passage, it is clear that for Jevons, the similarity between the lengths of the sunspot cycle and the intervals between commercial crises was the primary evidence. (It has to be confessed that Jevons' imagination also played an important initial part in the induction. Once he had spotted his 'beautiful coincidence' he became hooked, alighting upon every new scrap of evidence on the timing of a crisis in a particular year to back up his coincidence.[9]) This coincidence, through the use of *probability inference*, became for Jevons a causal relationship. Evidence of the explanatory links in the causal chain between sunspots and economic cycles would, he argued, increase the probability of the theory being correct but it was not essential to his argument. The evidence of causal connection, then, was not an important consideration either for Jevons or for those contemporaries who ridiculed his work.

The unsympathetic attitude of Jevons' contemporaries is equally understandable, for both Jevons' methodology and the domain of his theory were beyond the pale of late nineteenth-century economics. The method of theorising in classical economics involved deductions from premises which were 'known to be true' partly, at least, on the basis of internal personal knowledge, while Jevons' method of theory building was an inductive process dependent on external statistical evidence. His vision of periodic economic cycles was also alien to the nineteenth-century concern with 'crises', a term which was applied to many different circumstances and involved many different causes. Even those few economists in the nineteenth century who believed in the idea of cycles in economic activity, such as Marx and Juglar, did not see them as exact cycles. Jevons' idea that spots on the sun, an entity outside the economic system, should directly cause exact cycles in the economy was considered quite bizarre.

[8] This was the methodological and intellectual side of his elaborate defence in an unfinished piece intended for the *Princeton Review*. In this paper, reproduced in Jevons (1981) *Papers*, Jevons perceived the arguments against him to be twofold: the second argument was concerned with the possible influence of the sun on economic life and was at a commonsense level.

[9] Jevons even passed on his obsession to his son, H. Stanley Jevons who continued to proselytise the sunspot cause; in 1910 he published a paper linking the sun's activity with a 42-month cycle, the so-called short business cycle.

So, while Jevons was proposing a general explanation of a general cyclical phenomenon, others were busy explaining individual 'crises' by a large number of different causes. Juglar was proposing a general cycle theory built on statistical data but, as we shall see in the next chapter, he analysed and used the evidence on each individual cycle separately. Jevons' induction was of a different and an unusual kind: the evidence on individual cycles was not important, it was the standard or average characteristics of the data which were paramount. His theory was about an underlying exact cycle not about the observed cycles of different lengths. This is why Jevons' work, despite its paucity of statistical evidence and serious analysis, marks an important step in the development of econometrics: he relied on evidence of uniformity in statistical data, from which a general theory was derived using inductive reasoning.

1.2 Moore's Venus theory

If Jevons' sunspot theory appeared to be cranky, it was nothing to the theory eventually proposed by Henry Ludwell Moore to account for economic cycles.[10] Moore seemed to have suffered from the same fatal fascination that afflicted Jevons; once persuaded of the idea of some outside periodic cause of economic cycles Moore could not give it up but strove to take the explanation back to its first cause. He developed his theory in two books. The first in 1914 found weather cycles to be the cause of business cycles. The second book in 1923 extended the causal chain back to movements to the planet Venus. Yet Moore's work represented a considerable advance over Jevons'. The main difference, which will quickly become apparent, was that for Moore, the causal chain of explanation between the weather and business cycles was the main object of study. He arrived at his explanatory theory from a mixture of prior economic theory and a serious statistical investigation into the regularities and relationships in the evidence.

It was Moore's belief that explaining the cyclical fluctuations of the economy and finding laws to fit them constituted the

> fundamental problems of economic dynamics. (Moore (1914), p. 1)

In solving this problem, he proposed to abandon the standard methodological approach (which he variously described as the deductive *a priori* method, the comparative static or the *ceteris paribus*

[10] Henry Ludwell Moore (1869–1958) led a very sheltered academic life with few professional responsibilities until his retirement in 1929 (unlike Mitchell, his contemporary and fellow American who features in Chapter 2). His mathematical and statistical knowledge were

method) on the grounds that it could never tell economists anything about the real dynamic economy which was constantly shifting and changing like the sea. The mainstream theory and method were of course interconnected: comparative statics was concerned with comparing two positions (not with explaining the path of change between them), and was perhaps appropriate for understanding endogenously caused irregular cycles. Moore believed that an exogenously determined periodic cycle theory demanded different methods of study. Method and theory had to be appropriately matched and he was very conscious that in adopting a periodic cycle theory and statistical analysis he was flouting conventional methods of analysis as well as conventional theory.[11]

Moore began his first book, *Economic Cycles – Their Law and Cause* (1914), by fixing on weather cycles as the exogenous cause of economic fluctuations and set about 'proving' this hypothesis by providing the evidence for the explanatory links in the theory. He started his statistical investigation with the harmonic analysis of rainfall from 1839 to 1910 in the Ohio Valley (for which meteorological records were the longest available). He decided that the periodogram showed two significant cycles, one with a period of 8 years and one of 33 years. The best fit to the data was obtained with these two periods plus their semi-harmonics. After this initial frequency analysis, he moved on to find the correlation between the Ohio Valley rainfall series and rainfall in Illinois (the most important state for corn production, and, luckily, one which had long kept weather records) for the period 1870–1910. He then worked out the critical growth periods for grain crops by correlating rainfall in each month with yields. By assuming that the same periodic cycles he found in the initial Ohio Valley rainfall data followed through to the months of critical growth in Illinois grain crops and thence to the crop yields in the USA as a whole, he felt able to conclude

largely self-taught, although he did attend courses on correlation from Karl Pearson. His five books – all utilising the econometric approach – spanned the years 1911 to 1929. G.J. Stigler's (1962) illuminating study of Moore gives further biographical details; Christ (1985) surveys his econometric work.

11 Surveys of business cycle theories in the 1920s and 1930s (e.g. Haberler (1937)) show that amongst professional economists, only Jevons and Moore believed in an exogenously caused cycle based on weather cycles. It should also be clear that Moore's is a broader argument about the theory, the way it was constructed, and the way data were used. For example, a contemporary mainstream study of the cycle by Robertson (1915) used comparative static arguments with statistical data series, but without any statistical analysis or explanation of the dynamic path of the economy. It is, of course, arguable whether Moore succeeded in his aims; Schumpeter (1954) believed that Moore never got beyond a 'statistically operative comparative statics'; true dynamics coming with Frisch and others in the 1930s (see Chapters 3 and 4).

that the rhythmical movement in rain and in crop yields were causally related.

In the next section of this book, Moore examined the relationship between crop yields and crop prices. He found that yield/price schedules moved up and down with the general level of prices. Moore then tried to relate crop yields to production. Using the production of pig-iron as the proxy for output, he discovered that the best correlation between output and crop yields was for a two-year lag (yields preceding pig-iron) and concluded that this was evidence of a

> positive, intimate connection and very probably a direct causal relation. (Moore (1914), p. 110)

The next link in his theory was between output and prices of pig-iron. His empirical work produced a relationship which he interpreted as a positive demand curve for pig-iron. This of course contradicted the negative relationship between the price and quantity stipulated by standard economic theory, and he used this opportunity to launch a strident attack on conventional economic method and demand theory. (In fact, contemporary reviewers claimed that Moore had found a supply curve; the whole episode is discussed in Chapters 5 and 6.)

Moore, like Jevons, was not deterred by an unusual result. He fitted the positive demand curve into his general picture of the cycle and his explanation ran as follows: crop yields rise (due to the stage of the rain cycle), so the volume of trade increases (with a lag), the demand for producers' goods rises, employment rises, demand for crops rises and so general prices rise. The process was reversed when the rain cycle caused yields to decline. This model was finally confirmed for Moore when he found that the correlation of crop yields and general prices was highest with a lag of four years. The chart, Figure 2, shows the degree of concurrence between the two series in this final relationship (note that trends and the lag have both been eliminated). He used this correlation to justify his inference that the rhythmical features of crop yields are duplicated in prices; thus completing the cycle explanation, from weather through to general prices.

Moore concluded:

> The principle contribution of this Essay is the discovery of the law and cause of Economic Cycles. The rhythm in the activity of economic life, the alternation of buoyant, purposeful expansion with aimless depression, is caused by the rhythm in the yield per acre of the crops; while the rhythm in the production of the crops is, in turn, caused by the rhythm of changing weather which is represented by the cyclical changes in the amount of rainfall. The law of the cycles of rainfall is the law of the cycles of the crops and the law of Economic Cycles. (Moore (1914), p. 135)

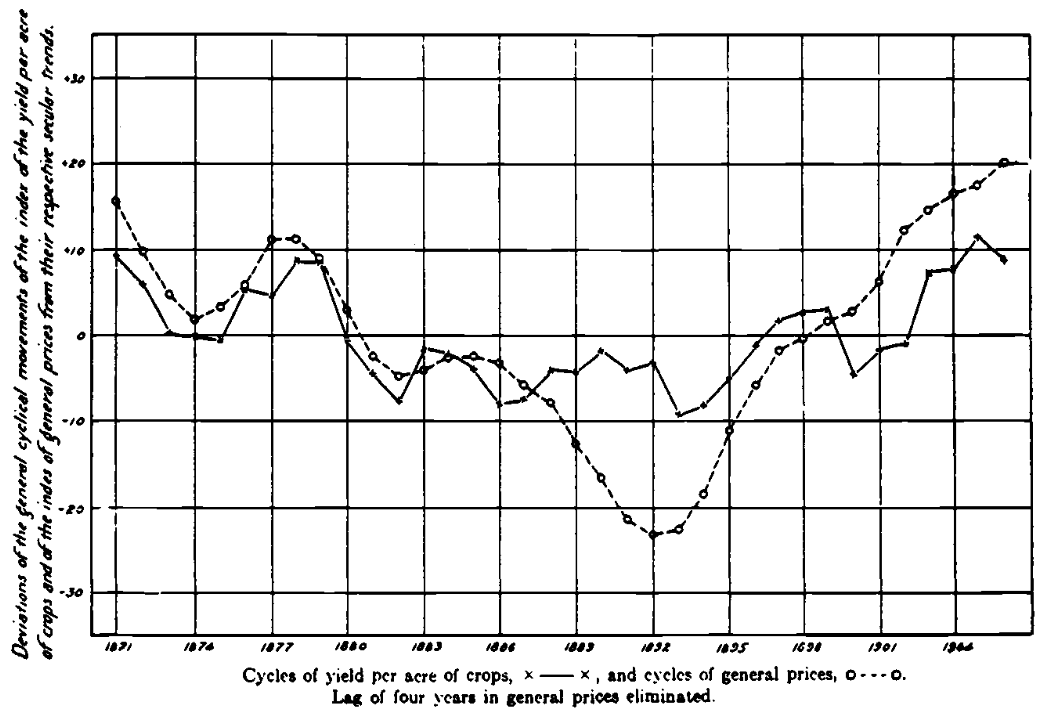


Figure 2 Moore's business cycle graph
Source: Moore (1914), p. 123, Figure 27

Moore's conviction that rainfall cycles caused economic fluctuations rivalled Jevons' belief in his sunspot theory. But, whereas for Jevons the evidence for his theory was only of secondary importance, for Moore it was primary. He worked industriously to discover and statistically verify the causal connections in the chain of evidence in order to provide a convincing explanation of the economic cycle.

Moore's handling of the evidence can be characterised as highly technological compared to those of both his predecessors and his contemporaries. His range of statistical methods included harmonic analysis, correlation, multiple regression with three variables (mainly in fitting cubic trend lines) and time-series decomposition (into three components: trend, cycle and temporary fluctuations). He worried about the goodness of fit of the lines and curves he used, particularly the sine curves, and worked out the standard error to measure the goodness of fit. Like Jevons, Moore imputed causality to certain empirical results, such as to correlations of a certain level. Unfortunately, Moore used his rather high-powered methods somewhat carelessly and uncritically. His work contrasts with Jevons who, with less concern for evidence and far less data at hand, had carried out his modest data manipulations with some care. Despite his faults, Moore's analysis of business cycles was far more sophisticated than any other statistical treatment of the period, for even those economists such as Mitchell (1913) or Robertson (1915) who did introduce statistical data into their work on economic cycles in these early years of the twentieth century did not carry out any statistical analysis of their data.

Although Moore's work was extraordinary, it did not strike all other economists of the time as outrageous. The book received a far better reception than had been awarded to Jevons' work; some reviews were critical, but none was derisory. Persons recognised, in his review, that Moore's ideas were out of harmony with those of most contemporary economists but was himself enthusiastic, believing Moore had carefully and successfully shown why economic cycles occurred. He particularly approved of Moore's methods which allowed

> the data, rather than the pre-conceived notions of the investigator to mould the conclusion. (Persons (1915), p. 645)

Yule's comments were not so enthusiastic, but he was not unfriendly. He found the correlation coefficients convincing and believed that Moore had presented a strong case that weather cycles were

> at least a very important contributory cause of economic cycles. (Yule (1915), p. 303)

But Yule criticised Moore for not using harmonic analysis directly on the economic data to reveal and analyse the economic cycles.

Lehfeldt's review, on the other hand, was one of those that made enjoyable reading for everyone except the author of the book. Lehfeldt decided that the best way to defend the application of statistical methods to the discovery of economic laws (i.e. econometrics) was to attack Moore's practice of the subject:

> In economics, hitherto, the sources of light available have been two: abstract reasoning of what should happen under certain simplified assumptions, and undigested statistics as to what does happen in the real world. Each of these is important, indeed indispensable ... But though abstract economic reasoning can be put into algebraic symbols, it leads to nothing quantitative, and, in the opinion of the reviewer, the greatest step forward in the science, possible at the present time, is to apply the modern methods of statistics to yield quantitative generalisations, under the guidance of abstract analysis, and so slowly and cautiously to build up flesh on the skeleton of theory.
>
> There are very few workers in this field as yet, which makes it all the more of a pity that one of them should lay himself open to accusations of unsoundness. Research along these lines is not easy, for it requires a thorough grasp of the mathematical theory of statistics, patience to do the

lengthy arithmetic involved, and ceaseless and acute criticism of the mathematical processes in the light of common sense and everything that is known about the subject-matter. Prof. Moore is weak in this last arm; he has, let us say, the artillery of mathematics and the plodding infantry of numbers, but not the aerial corps to save him from making attacks in the wrong directions. (Lehfeldt (1915), p. 410)

P. G. Wright's review (1915) was in rather similar vein but kindlier in manner and with some compliments to Moore's pioneering efforts. Wright was not convinced by Moore's theory and evidence and he reported some results from his own calculations with Moore's data. Moore had not analysed effective rainfall (rainfall in the critical growing months), just assumed that the 8-year cycle he found in annual rainfall was applicable. Wright carried out an harmonic analysis of effective rainfall to reveal a 4-year cycle instead of Moore's 8-year cycle. Wright believed that evidence from correlations would be consistent with the harmonic analysis but easier to understand and interpret, so he also worked out the correlogram of various data series to see if these gave evidence of cycles and of what length. To his surprise, the correlogram evidence was not consistent with Moore's, for he found a 3- to 4-year and a 7-year cycle in rainfall; a 7-year cycle in crop yields; and a 9-year cycle in general prices. No wonder Wright found Moore's 8-year cycle theory not proven! This is one of the earliest examples in econometrics where a criticism was made more pointed by reworking the author's data with a different hypothesis or method of attack.[12]

The fundamental difference in attitude in these reviews of Moore's work, compared to the contemporary response to Jevons, was that the evidence and the way it was analysed was being taken seriously, seriously enough indeed to prompt one reviewer to undertake his own data analysis. The econometric approach, though still relatively crude and untried, was treated with sympathy and accepted as having something to offer. But Moore's reviewers were part of the small group of economists involved in statistical economics and econometrics and they applauded Moore's pioneering spirit while criticising his performance. It would be wrong therefore to take their sympathy as evidence that econometrics had become in any sense a mainstream activity by 1914, for it certainly had not.

Moore's second book on the business cycles, *Generating Economic Cycles*, was published in 1923. In response to the criticisms of his earlier book, Moore carried out his statistical work on the economic cycle with

[12] These data experiments made very effective arguments and were a feature of the more theoretical investigations into time-series methods undertaken by Yule and others in the late 1920s, discussed in Chapter 3.

more diligence. For example, he used harmonic analysis directly on the economic data as suggested in Yule's review. Moore also widened the applicability of his economic theory by extending the time period he used and the coverage to three countries, namely Britain, France and the USA. The persistence of the 8-year cycles he was interested in and the fact that they occurred, according to his evidence, not only in all three countries but in synchronisation in those countries, made his case more convincing.

In terms of the content of his business cycle theory, Moore was still, like Jevons, searching

> for a single explanation of the entire rhythm and its constituent parts.
> (Moore (1923), p. 15)

The new version of Moore's theory involved a relaxation of his earlier view that there was one single generating cycle (the exogenous or propagating cycle). He now believed that the economic cycle was the result of economic and social influences combined with the main exogenous cause. But although Moore was no longer prepared to claim that 'the law of cycles of rainfall is . . . the law of Economic Cycles', his claims as to the nature of the generating cycle (the main cause) were far more extravagant. In an imaginative search for the mechanism generating the weather cycles, he fixed on Venus as the first cause, and in particular the fact that, at 8-year intervals, Venus came between the earth and the sun. The last chapter of the book went into various recent discoveries in physics which Moore believed shed light on exactly how Venus, in conjunction with the earth and the sun, might be supposed to cause rainfall cycles on earth. Sunspots were just one part of this explanatory chain which involved Venus causing interference in solar radiation to the earth, negative ionisation, weather disturbances and finally rainfall cycles.[13] He must have guessed that his Venus theory would not be well received because he opened his book with a report of Galileo's rejection of Kepler's idea that the moon caused the tides. Moore pleaded by analogy that readers should not dismiss his own ideas as ridiculous out of hand.

Moore's ideas on the cause of economic cycles had already appeared as separate articles between 1919 and 1921, so by the time of his 1923

[13] It is interesting to note that Davis (1941) suggested that if there was a connection between ionisation and health then Jevons' sunspot theory might be rehabilitated in conjunction with the psychological cycle theories. Those who have become addicted to the sunspot theory from reading this chapter might like to know that Garcia-Mata and Shaffner (1934) attempted to connect sunspots with a psychological explanation and that an ionisation/health connection has recently been made by some environmental health specialists! For those more interested in the econometric tests of the theory, see the recent debate on Granger-type causality testing of Jevons' hypotheses in the *Southern Economic Journal* (1982 and 1983).

book, the Venus theory was no longer new. Perhaps this was why his theory met with polite scepticism from the serious commentators rather than with the outrage accorded to Jevons' sunspot theory.[14] For example, Hotelling commented on Moore's Venus theory as follows:

> The trouble with all such theories is the tenuousness, in the light of physics, of the long chain of causation which they are forced to postulate. Even if a statistical test should yield a very high correlation, the odds thus established in favour of such an hypothesis would have to be heavily discounted on account of its strong *a priori* improbability. (Hotelling (1927), p. 289)

The basic theory of periodic economic cycles caused by an outside body appeared to be no more acceptable to Moore's audience than it had been in Jevons' day. It comes as no surprise then to learn that as a theory of the cycle, his work had no direct influence on other economists.[15] Moore's reviewers essentially urged the same sort of criticisms of his second work as they had of the first, in challenging his periodogram method and his evidence of 8-year cycles in the economy, rather than in seriously arguing with the Venus theory. Indeed, Moore's assumption of periodic cycles and use of frequency analysis methods on economic data were rarely copied in econometrics during the remainder of the period covered by this study.[16] The reasons for this are discussed in the next section.

Of importance for the history of econometrics is that Moore, in both his cycle books, had concentrated his energies on the statistical evidence of the economic interactions involved in the business cycle and the statistical analysis of these relationships rather than on the relation between the economic cycle and the exogenous causal factor. Moore's concern with evidence and statistical explanation compared to that of Jevons, and the matching change in contemporaries' responses, are both indicative of the development of the econometric approach by the early years of the twentieth century. Yet, it was some years before

[14] The theory gained him a certain notoriety in the popular press. Academic reviewers may have been kind to Moore because it was known that he was particularly sensitive to criticism (see G.J. Stigler (1962) for discussion of this point).

[15] For this reason G.J. Stigler (1962) judged it to be a failure. However, Moore's cycle work did prove an influential example in econometrics, particularly, as Stigler himself argued, in the field of statistical demand curves. Moore developed dynamic econometric models of the demand and supply for agricultural commodities in the 1920s and, as we shall see in later chapters, these cobweb models fed back into mainstream economic dynamics and thence into Tinbergen's econometric work on business cycles in the 1930s (see also Morgan (1988)). Thus, the correct judgement seems to be that Moore's cycle work did influence economics, but only indirectly.

[16] The time-series approach has of course been revived more recently, beginning with the Princeton Time Series Project directed by Morgenstern (see Chapter 3 n. 10, and Chapter 8.1) and Tukey in the early 1960s (see Granger (1961)).

Moore's broad econometric approach to the explanation of economic cycles, involving a large number of relationships linking different parts of the economy, was taken up by Tinbergen who produced the first macroeconometric models in the late 1930s.

1.3 The decline of periodic cycle analysis

Few economists between 1920 and 1950 followed up Moore's promising start in using periodogram analysis in economics. Those few who did use the idea of periodic cycles and its associated methods were much less ambitious in their theorising and in the scope of their studies than Moore had been, as the following two examples by William Beveridge and by E. B. Wilson show.[17]

Beveridge's studies of wheat prices, weather and export cycles were published in 1920–2, but his data and main results have been used since as a textbook example. He gathered annual data on wheat prices which covered 4 centuries and 48 different places in Western and Central Europe. These observations were then reduced to one single index of wheat prices. He noticed that the shape of the graph of this index (once it had been detrended) changed quite dramatically with the onset of industrialisation, becoming less jagged and irregular after 1800. He believed that this was because

> another disturbing influence – the credit cycle – begins to show itself, and cannot be eliminated by any simple method. The character of the curve is visibly altered in these later years; beyond 1869 it could not without correction be trusted as an indication of harvest conditions.
>
> (Beveridge (1921), p. 432)

Beveridge then applied frequency analysis to his data, calculating over 300 different periods, and made a detailed examination of the results. He selected 19 dominant cycles (noting especially that Moore's 8-year cycle was not one of these). Beveridge sought corroborating evidence for these 19 cycles in the behaviour of the weather and of heavenly bodies, but caution prevented him making the sort of leap from slight evidence to total belief in a periodic connection between variables that Jevons had made. Instead, he suggested reasons why the exact periodicity might not hold in economic cycles, arguing that, because weather

[17] W. H. Beveridge (1879–1963) was an economist and statistician who specialised in applied social questions. He was director of the London School of Economics (1919–37) but is more famous as the architect of the British welfare state of the post-war era. E. B. Wilson (1879–1964) was a Harvard mathematician associated with econometrics and on the editorial board of *Econometrica* from 1933 until the late 1940s. The other early work besides Moore's to use harmonic analysis has been reviewed by Cargill (1974).

cycles are temporary phenomena, or span more than the unit of one year, they would affect crops differently in different periods.

While Beveridge had the residues of a causal model behind his analysis of the relationship between wheat prices and weather cycles, there was no explanatory model behind Wilson's 1934 analysis of monthly data on business activity covering 140 years. His work was really a test of the hypothesis that business cycle data show definite periods: the results rather suggested that they do not. In particular, he found neither Jevons' sunspot-related cycle nor Moore's Venus-related cycle of significance. Both Wilson and Beveridge tested the cycles they found in the data. Beveridge concentrated on whether his results were compatible with other evidence. Wilson used statistical tests to see whether the cyclical patterns of business cycle data were similar to those found in chance series. (He found that his dominant periods were rejected as not significant by the test recommended by Schuster and by the tests of others who rejected Schuster's test.) Wilson and Beveridge both tested subperiods of the data to see whether the results were the same for both periods, and Wilson also tried forecasting from subperiod to subperiod with little success.

Beveridge and Wilson showed a decline in confidence in the idea of periodic economic cycles compared to Moore and a preference for using the periodogram method to describe data and analyse its characteristics, rather than letting it play a central role in theory-building, as Moore had done. There were a number of sound reasons for this decline in confidence and for the relatively sparse usage of the periodogram method in economics in the period. By virtue of his leading example in this field, Moore's work naturally provided the focus for contemporary discussion of these issues.

One obvious reason for the decline in the periodic cycle programme was that economists did not believe that the business cycle was anything more than roughly regular. Critics of Moore's work were suspicious of the periodicity claimed for the economic cycle, a double claim since it was central to his hypothesis and it was implied by his usage of the periodogram method. Ingraham, for example, believed that *a priori* it was more probable that business cycles had an average length of 8 years, rather than Moore's hypothesis of an exact periodicity of 8 years. Ingraham showed that Moore's evidence was not inconsistent with this alternative hypothesis of an average cycle length, and was thus able to conclude his review of Moore's second book:

> All [Moore's] data could be explained as well by a theory of cycles which merely assumed that in general the length of cycles is close to that of their average length, showing no true periodicity. This is to the average

> economist much the more likely hypothesis on an a priori judgement, and so it should stand until we have much more complete statistical data.
> (Ingraham (1923), p. 765)

The assumption of periodicity also implied a symmetric shape to the cycle, whereas Crum (1923) pointed out that observed economic cycles tended to be asymmetric in form (i.e. the upturn is different from the downturn).

Even if economists were willing to suspend their disbelief in periodicity, the difficulties of interpreting the periodogram results remained. On the one hand, the method seemed to produce too many cycles to allow sensible economic interpretation: for example, Beveridge's (1922) tally of 19 significant cycles was followed by a lengthy and confused discussion of the meaning of his results. Mitchell (1927) and Schumpeter (1939) refer to similar difficulties of interpretation. On the other hand, Moore's exclusive concentration on the 8-year cycle seemed to be too simplistic a characterisation of economic cycle data. P. G. Wright (1924) considered, after experimenting with the data, that it was more likely Moore had two superimposed periodicities, or different periodicities for different subsections of the data, than a single 8-year cycle. Wright believed his interpretation was compatible with an internal origin to the business cycle from psychological or economic factors, which might be expected to vary their time periods. Others also suggested that over the long term, periodicities in economic series were likely to change rather than remain constant and Frisch (1927) concluded the method was too rigid for this reason.

But apart from these rather nebulous problems of interpretation, there were a number of technical difficulties in using the periodogram method which were recognised at the time. Moore's early discussion of what he required from the method indicates some of these difficulties. In the first place, Moore said of the method:

> It must exhaust the data in the search for possible cycles; that is to say, the data must be made to yield all the truth they contain relating to the particular problem in hand. Frequently in the past, spurious periodicities have been presented as real periodicities, chiefly because the investigator started with a bias in favor of a particular period and did not pursue his researches sufficiently far to determine whether his result was not one among many spurious, chance periodicities contained in his material. In the search for real periodicities the data must be exhaustively analyzed.
> (Moore (1914), p. 137)

Despite his wise words, Moore's own work did not live up to the standards of unbiased behaviour that he had set himself. His 1914 theory had involved an 8-year cycle and his affection for this result led him astray when in 1923 he found that an harmonic analysis of British prices indicated significant cycles of 8.7 years and 7.4 years but not 8.0 years. Moore had tried to retrieve his theory by averaging the results from the two significant periods to gain a significant result for the 8-year cycle, a procedure which Ingraham (1923) pointed out was invalid in harmonic analysis.

Exhaustive and unbiased analysis was not sufficient to counteract the manifold dangers of spurious periodicities for as Moore reminded his readers there was a second technical difficulty:

> The method must render possible the discrimination between a true periodicity, having its origin in a natural cause and persisting with a change in the samples of statistics, and a spurious periodicity which is purely formal, having its origin in accidental characteristics of the statistical sample and disappearing, or radically altering its character, when different samples of statistics are made the basis of the computation.
> (Moore (1914), pp. 137–8)

Once again Moore failed to follow his own good advice, although others did so. Beveridge and Wilson were both aware of the problem and tried to eliminate spurious periodicities due to sampling by testing different subperiods of the data and Wilson applied statistical tests to try and determine real from chance periodicities.[18] Crum (1923) perceived this problem differently. He believed that non-homogeneity of the sample due to long-term trend changes, as well as irregular disturbances and even seasonal variations in the data, lead to a 'blurring' in the measurement of the maxima of the periodogram, making it difficult to determine the true periodicities. Yule (1927) went further with this argument and claimed that, if the effect of the disturbances were to carry through into later periods, the true periodicities could not be recovered by harmonic analysis.

Another way of avoiding 'chance' or 'spurious' periodicities, and thus one of the requirements for successful use of the method, was to use a long data period. Yet, economic time-series data, as Mitchell (1927) pointed out, usually covered only rather short periods and were therefore inadequate. For example, Moore's data set was only 40 years in length which, Ingraham argued, was

> insufficient to establish the existence of any period of length as great as eight years. (Ingraham (1923), p. 765)

P. G. Wright (1922) confirmed that this was so by carrying out a die throwing experiment to demonstrate that using harmonic analysis on

[18] Kendall (1945) afterwards claimed that most of Beveridge's 19 significant periods were accounted for by sampling error.

his chance series of numbers gave results similar to those Moore had obtained on his rainfall data. Data series were rarely, like Beveridge's series of wheat prices, long enough to make accurate results possible. Yet the periodogram method and the calculation it required gave a specious accuracy to results, of which economists – for example, Crum (1923) – were justly suspicious. On the other hand, use of a long data series brought a severe computing problem: periodogram analysis was very laborious to carry out by hand because it required such a large amount of computation. Contemporary evidence of the time taken to carry out periodogram analysis by hand calculator is difficult to gain, but Beveridge's work in particular must have taken many hours.[19]

Last, but by no means least, of the technical issues was that periodogram analysis could be of only very limited use to econometricians. Economists, even the most empirically minded, were primarily interested in *the relationships between variables* rather than *the behaviour of one variable*, while the harmonic analysis available at the time was applicable to one variable at a time and could not be used to investigate relationships between variables. This was certainly one of the drawbacks which struck Frisch when he started a theoretical study of time-series methods in the mid-1920s.[20]

With all the drawbacks of assumption, interpretation and technique mentioned here, it is no wonder that Crum (1923) concluded the synopsis of his critical study of the method with these words:

> The average form of the cycle is at best not very precisely determined, and its breaking up into a group of harmonic components of differing periods and amplitudes is certainly bewildering and probably furnishes no true insight into the nature of economic fluctuation. (Crum (1923), p. 17)

Yet there was one very good reason why econometricians might have persevered with harmonic analysis: the method was specifically applicable to time-related data. The alternative correlation methods, to the degree that inference from them was dependent on data observations being independent through time, were therefore unsatisfactory for prediction purposes. Indeed, it was the general view of economists analysing business cycle data in the 1920s and 1930s that methods which involved the theory of probability could not be used because the observations of business cycle data were related through time. This

[19] Some idea is given by the report that the time taken for an harmonic analysis on an early IBM computer in 1961 using 400 observations (about the size of Beveridge's data set) was three hours.

[20] Frisch (1927), p. 2. This point was entirely missed by both Cargill (1974) and Morgenstern (1961) in discussing the way frequency analysis was discarded in econometrics during the 1930s.

view was not considered inconsistent with the wide use of a whole range of statistical techniques on time series (this apparent paradox will be further discussed in Chapter 8). The fact was that for contemporaries, most other statistical techniques were preferable to periodogram analysis. The methods of correlation and decomposition of time series into component parts were more widely understood than harmonic analysis which demanded a relatively high level of mathematical comprehension. P.G. Wright (1922) also argued that the method of correlation was better than harmonic analysis because it was more sensitive to changes in cyclical features and trends, and allowed more thoughtful interpretation and critical judgement. In addition, the correlation method could be used directly to measure the association between variables.

Despite the fact that Jevons and Moore are recognised as pioneers in econometric business cycle research, the reasons for their lack of continuing influence on cycle analysis are now clearer. First, their assumption of a periodic cycle generated from outside the economy was unattractive to most economists and the frequency methods which accompanied their assumption were found to be ill-suited for econometric work. In the second place, there was a positive burgeoning of statistical work on the business cycle in the 1920s which involved alternative ideas and methods and which were more easily accessible to business cycle students of the period. Thirdly, a preference for description over explanation was a common trait of the development of statistical work on business cycles in the 1920s and 1930s regardless of the actual tools used. In the face of this strong alternative empirical statistical programme, the econometric approach advanced by Jevons and Moore, which tried to build theories of the cycle out of the statistical regularities and relationships in the data, lay dormant until revitalised by Tinbergen in the late 1930s. Meanwhile, the alternative programme deserves our attention.

CHAPTER 2

Measuring and representing business cycles

Clément Juglar, Wesley Clair Mitchell and Warren M. Persons used statistics in a different fashion from Jevons and Moore. These three economists believed that numerical evidence should be used on a large scale because it was better than qualitative evidence, but their use of data was empirical in the sense that they did not use statistical features of the data as a basis for building cycle theories. Further generalisations are difficult, because their aims differed: Juglar wanted to provide a convincing explanation for the cycle, Mitchell sought an empirical definition of the cyclical phenomenon, whilst Persons aimed to provide a representation of the cycle. The chapter begins with a brief discussion of Juglar's work in the late nineteenth century, and then deals at greater length with the work of Mitchell and Persons, whose empirical programmes dominated statistical business cycle research in the 1920s and 1930s.

The applied work discussed here grew in parallel to that described in Chapter 1, for Juglar was a contemporary of Jevons, and Mitchell and Persons overlapped with Moore. The developments that the three economists, Juglar, Mitchell and Persons, pioneered came to be labelled 'statistical' or 'quantitative' economics but never bore the tag of 'econometrics' as Moore's work has done. These statistical approaches are important to the history of econometrics, not only because they provide a comparison with the work of Chapter 1, but also because it was by no means so clear to those who worked in the early twentieth century that econometrics was a distinctly separate programme from statistical economics. The distinction was clear enough by the 1940s for the two rival parties to indulge in a methodological debate over the correct use of statistics in economic research. Comparisons between the statistical and econometric programmes are drawn at various points during the discussion and an assessment of the two approaches comes at the end of the chapter.

2.1 Juglar's credit cycle

Clément Juglar's[1] interest in business cycles was stimulated in the 1850s by his study of a table of financial statistics for France. Like his contemporary Jevons, he was immediately struck by the recurrence of 'crises' evident in such data. Unlike Jevons, Juglar set out to find all the causes of these crises and eliminate the secondary causes one by one to leave only the necessary determining cause which was constant for every cycle. The theory that Juglar proposed was more acceptable to economists of the late nineteenth century than Jevons' sunspot theory, both because it was developed from current ideas on credit crises and because his way of reasoning from statistical data was less alien to them.

In the first edition of his book on crises in 1862, *Des crises commerciales et de leur retour périodique*, Juglar studied the histories of all the crises in the nineteenth century in three different countries, France, England and the USA. He discussed each crisis in turn and argued that events such as wars, revolutions and famines were causes of the moment, which might determine the exact crisis date or the particular characteristics of the cycle but they did not determine the fact of the crisis or cycle. Such causes were only the 'last drop which caused the basin to overflow' (Juglar (1862), p. v), not determining causes in the true sense. He concluded that the one constant common cause of all these cycles was changes in the conditions of credit. Juglar's 'proof' of the constant or determining cause rested on two planks: one was an appeal to the regularity of certain sequences or patterns in the tables of data, and the other was the novel reason that a discussion of the monetary history of each crisis was repetitious and liable to cause ennui in the reader, a proof by boredom!

The second edition of Juglar's book (1889) was very much enlarged and enriched with further evidence although the method of approach was essentially the same. He presented much of the evidence in distinctive exhibits. These were in the form of tables, to be read downwards (containing only the maxima and minima figures of each cycle) with a graphic indication (not drawn to scale) of the cycle by the side of the figures. An example is presented here (Figure 3). He concluded that over the nineteenth century the credit cycle was common to, and more or less simultaneous in, three countries with

[1] Clément Juglar (1819–1905) trained as a physician but became interested in demography and statistics and thence economics in the 1840s (see entry in the *International Encyclopedia of the Social Sciences*).

Différence en plus ou moins des principaux articles des bilans de la Banque de France.

ANNÉES.	BANQUE DE FRANCE (millions de francs) CIRCULATION. Maxima.	CIRCULATION. Minima.	ENCAISSE. Maxima.	ENCAISSE. Minima.	ESCOMPTE. Maxima.	ESCOMPTE. Minima.
1840........						
1841						
1842						
1843						
1844						
1845........			+ 131			
1846........	+ 106					
1847 (crise).		— 77		— 242	+ 216	
1848						
1849						
1850						
1851........			+ 550			— 227
1852						
1853........	+ 472					
1854						
1855						
1856........				— 476		
1857 (crise).		— 178			+ 535	
1858........						— 280
1859........			+ 494			
1860						
1861						
1862........	+ 343					
1863						
1864 (crise).		— 149		— 494	+ 441	
1865						
1866						
1867						
1868........						— 404
1869						
1870........			+ 1.166			
1871........				— 920		
1872						
1873 (crise).	+ 2.345				+ 2.578	
1874						
1875						
1876						
1877........			+ 1.883			
1878						
1879........		— 970				—2.205
1880						
1881........				— 531		
1882 (crise).	+ 852		+ 408		+ 1.351	
1883						
1884........	+ 1.061					
1885						
1886........		— 504	+ 775			—1.311
1887........						

Figure 3 Juglar's table-graph of credit cycles
Source: Juglar (2nd edn, 1889), p. 154

different institutional regimes (that is, complete freedom of the banking system in the USA, freedom and monopoly in England and monopoly in France).

It was clear to Juglar, from the evidence of the first edition of his book, that cycles were variable in length and in amplitude, but that the sequence of activity was regular. He described this sequence as 'periodic', by which he meant that the sequence repeated itself, not that there was an exact periodic cycle. The terms 'constancy' and 'regularity' in Juglar's discussion referred to the necessary cause of cycles and the sequence of events not to the timing or appearance of cycles. This view was unchanged by his further research, although by the second edition he had developed a definite phase categorisation of his credit cycle sequence: prosperity (5–7 years), panic or crises (a few months to a few years) and liquidation or depression (a few years). Since his own evidence 'did not allow one to do more than observe that a crisis returned in a period of between 5 to 10 years' (Juglar (1889), p. 164), Juglar strongly rejected Jevons' theory of regular periodic cycles.[2] Juglar also found Jevons' sunspot theory untenable on other grounds. First, he believed that the changing environment of commerce and industry would cause changes in the cyclical pattern. Secondly, he believed that exogenous factors like the weather (or sunspots) could only be disturbing influences, not true causes. Lastly, he challenged those who believed in such meteorological causes to find a good economic barometer!

The novelty of both Juglar's and Jevons' work to contemporaries was that they used lots of data instead of a few odd numbers, that they were concerned with cycles not crises, and that they used statistical data to recognise regularity in the behaviour and causes of economic cycles. These general similarities hid important differences, for Juglar differed from Jevons about the nature of the regularity involved, the causes of economic cycles and, most important, they used their data in different ways. Juglar justified the usage of unusually long runs of statistics on the grounds that it added scientific rigour to his work. He argued that statistical evidence was better than verbal evidence because it uncovered the regularity in events and enabled him to see causes and links which might otherwise be hidden in the historical description. Despite his belief that statistical evidence was somehow more scientific

[2] Despite Juglar's insistent rejection of statistical regularity, cycles of 9–10 years in length are called 'Juglars'. This was due to Schumpeter (1939) who classified the various cycles by average length and named them after their main exponents (thus Kitchin, Juglar, Kuznets and Kondratieff cycles). His two-volume study of the business cycle contained no statistical analysis to speak of, but developed a complex theory of the interactions of these different cycles.

than qualitative evidence, Juglar used his statistical data in much the same way as qualitative evidence. He discussed the data on each individual cycle and then, as it were, piled up the cases to provide support for his theory by repetition. Each cycle was treated as an individual event with certain features in common; but these features were not connected to statistically measurable regularities of the data. Indeed, he made a point of their statistical variability. So Juglar's explanation and evidence for his theory was based on the aggregate of the individual occurrences, although it could have been built up on the basis of one cycle only. Jevons, on the other hand, expected the constancy of economic behaviour to be reflected in statistical regularities in the mass of the data. Not only was the pattern of economic activity to be revealed in statistically regular cycles, but these statistical regularities also played an important part in the development and design of Jevons' sunspot theory of their causation.

Historical judgement suggests that because of the way each used statistical data, Jevons' work more nearly qualifies as the first econometric theory and treatment of the business cycle, while Juglar is more appropriately seen as a pioneer in the important and parallel programme in 'quantitative economics'. Their contemporaries made no such distinction, rather they were impressed by Juglar's sensible economic reasoning about the cyclical nature of economic activity and deplored the eccentricity of Jevons' sunspot theory.[3]

2.2 The statistical approach of W. C. Mitchell

2.2.1 Mitchell's first statistical study of business cycles

Wesley Clair Mitchell, like Jevons, had a history of statistical studies to his credit when his first book on business cycles appeared in 1913, one year before that of Moore.[4] The aim and overall method of *Business Cycles and their Causes* were described in only two pages (out of over 600) and are best presented by direct quotation:

[3] For example, Schumpeter (1954) suggested that the cycle ousted the crisis in the period 1870 to 1914 and credited Juglar's work as the decisive factor in this change of perception. He conceived Mitchell's (1913) work to be in the same spirit as Juglar's and to mark the start of the next generation of business cycle study.

[4] Wesley Clair Mitchell (1878–1948) was a member of the loosely knit group of American economists known as the institutionalist school. The main influences on his work were Veblen and Dewey. Mitchell helped to found the National Bureau of Economic Research in 1920 and was its director 1920–45. He also helped found the New School for Social Research in New York. There are many sources of information about Mitchell, a contemporary starting point is Burns (1952).

> One seeking to understand the recurrent ebb and flow of economic activity characteristic of the present day finds these numerous explanations [of business cycles] both suggestive and perplexing. All are plausible, but which is valid? None necessarily excludes all the others, but which is the most important? Each may account for certain phenomena; does any one account for all the phenomena? Or can these rival explanations be combined in such a fashion as to make a consistent theory which is wholly adequate?
>
> There is slight hope of getting answers to these questions by a logical process of proving and criticizing the theories. For whatever merits of ingenuity and consistency they may possess, these theories have slight value except as they give keener insight into the phenomena of business cycles. It is by study of the facts which they purport to interpret that the theories must be tested.
>
> But the perspective of the invesigation [*sic*] would be distorted if we set out to test each theory in turn by collecting evidence to confirm or to refute it. For the point of interest is not the validity of any writer's views, but clear comprehension of the facts. To observe, analyze, and systematize the phenomena of prosperity, crisis, and depression is the chief task. And there is better prospect of rendering service if we attack this task directly, than if we take the round about way of considering the phenomena with reference to the theories.
>
> This plan of attacking the facts directly by no means precludes free use of the results achieved by others. On the contrary, their conclusions suggest certain facts to be looked for, certain analyses to be made, certain arrangements to be tried. Indeed, the whole investigation would be crude and superficial if we did not seek help from all quarters. But the help wanted is help in making a fresh examination into the facts.
>
> (Mitchell (1913), pp. 19–20)

It would be a mistake to think that Mitchell rejected any role for business cycle theories, for he recognised here quite clearly that theories determined which 'facts' should be examined. But although theories dictated which data to look at, Mitchell was not sure whether he ought to test each theory of the cycle individually (and notice that his idea of testing included both verification and refutation) or attempt to investigate all the theories together. He decided that the principal need was for more detailed information on business cycle behaviour.

Mitchell's empirical programme was to start with the facts and determine the relative importance of different causal factors in the cycle, using both quantitative and qualitative research. Quantitative evidence predominated because, as Mitchell said,

> in his efforts to make accurate measurements [of the causal factors] the economic investigator cannot devise experiments, he must do the best he can with the cruder gauges of statistics. (Mitchell (1913), p. 20)

He decided to study only a short period, 1890–1911, making up a 'sufficient number of cases' from which to generalise by looking at the evidence of four countries. His study of the facts consisted of an exhaustive 350-page presentation of the statistics of business cycles for the four countries, accompanied by a commentary and arranged by topic. These topics included the usual ones of prices and profits, but also the less usual aspects of cycles such as the behaviour of migration. His analysis of the data was severely limited, consisting of taking averages for each of the two decades and comparing the behaviour of different countries using graphs.

Following his statistical enquiry Mitchell attempted to redefine what an adequate business cycle theory should look like. His study suggested that

> A theory of business cycles must therefore be a descriptive analysis of the cumulative changes by which one set of business conditions transforms itself into another set.
>
> The deepest-seated difficulty in the way of framing such a theory arises from the fact that while business cycles recur decade after decade each new cycle presents points of novelty. Business history repeats itself, but always with a difference. This is precisely what is implied by saying that the process of economic activity within which business cycles occur is a process of *cumulative* change. (Mitchell (1913), p. 449)

In this definition, no general theory is completely adequate because each cycle is unique and demands a unique explanation. Since each cycle grows out of, and is dependent on, the previous events, it is impossible completely to explain any cycle because the analysis can never be carried back far enough. Mitchell realised these difficulties and proposed that instead of seeking a general theory, attention should be focussed on the recurring phases of the cycle, from revival to depression.

Having presented the data, Mitchell set himself to study the interactions of the elements of the cycle as the cycle passed through its various phases. There was no natural starting place for the process and arbitrarily he chose the point following on from a depression. This investigation did not involve quantitative analysis, but the knowledge of the statistical evidence was used to evaluate the various cycle theories. He suggested that

> none of the theories of business cycles ... seems to be demonstrably wrong, but neither does any one seem to be wholly adequate.
> (Mitchell (1913), p. 579)

He concluded that cycles are not uniform but irregular; that they do not always follow the same pattern of phases and their intensities and elements differ. He claimed these differences were often due to 'extraneous factors' such as weather, war, government policy changes, etc. In addition, Mitchell thought that, with improved business barometers (already widely used in the US and commercially successful), future cycles could be predicted and might be controlled to some extent.

Lastly, Mitchell believed that longer term changes in economic organisation (being cumulative) had an evolutionary effect on cycles. To illustrate this point he discussed the changes that had occurred in the nature of cycles from agricultural crises (in the pre-industrial period) to credit crises and thence to business cycles (as industrialisation had taken place). These long-term developments he claimed were gradual and continuous (unlike the earlier stage theories of development held by some nineteenth-century German economists), and were associated with institutional or organisational change in the economy.

Mitchell's book was very well received and was regarded by contemporaries as an important contribution to statistical economics. Pigou was particularly convinced by Mitchell's description of the complexity of the cycle:

> For the great value of this lies in its realism and concreteness – in the fact that the skeleton does not appear as a skeleton, but as a being of flesh who lives and moves. (Pigou (1914), p. 81)

Persons, who was also very taken with the book, wrote:

> It is comprehensive; it contains the most complete and careful statistical study of the phenomena connected with business cycles; ... it is a scientific study by an author who, obviously, is not seeking to establish a pet theory. (Persons (1914), p. 795)

In a later judgement, Dorfman (1949) suggested that the theoretical contributions in Mitchell's book were not appreciated by contemporary readers because the language was commonplace and the method descriptive.

There was some truth in this latter point for Mitchell had presented the statistical evidence and other evidence before discussing theories; he had then drawn his conclusions about the nature of the cycle from an investigation in which theoretical ideas and statistical evidence were discussed in conjunction. Despite admitting (though only in a footnote) the difficulties of studying such a complex phenomenon in this way,

> the intellectual instruments of analysis are unequal to the complex problem of handling simultaneous variations among a large number of inter-related functions. (Mitchell (1913), p. 450)

Mitchell's descriptive analysis remained wedded to conventional methods based on the unaided intellect; he spurned the aid of both mathematical and statistical tools.[5] It is not clear that these tools would have helped him, for his investigations into the evidence had convinced him that not only was each cycle a unique event, but that cycles did not even possess unifying characteristics such as the same pattern of phases or the same elements in their changes. These beliefs coupled with his empirical approach made it impossible for Mitchell to produce a simple but general model of the cycle as Juglar had done from his observation that the cycle had a standard sequence of events.

The fact that Mitchell believed each cycle was different also precluded him from adopting a data reduction approach involving either the summary statistics such as means and variances, like Jevons' reliance on average cycle length, or using regression and correlation as Moore was to do in 1914. Moore's book provides the obvious contemporary comparison. Their divergent views on the nature of the business cycle were naturally reflected in the way that each used statistical evidence. For Mitchell, the raw data gave evidence for the lack of uniformity, and this effectively excluded both statistical analysis and any general explanation or theory of the business cycle. For Moore, the raw data hid the uniformity and constancy of the cycles; he therefore depended on statistical analysis in his search for regularity and relationships in the data and a general theory of the cycle.

Despite their differences, both Mitchell and Moore were pioneers in their time. Few other economists had followed the pointers laid by Juglar and Jevons. Like these predecessors, both Mitchell and Moore were seeking a theory which would explain all the characteristics of the whole cycle (not just upturns or turning points as in early theories of the business cycle which derived from crises theories), and both relied on masses of statistical data in their search. Yet here the similarities between Mitchell and Moore end. Mitchell's cautious use of statistical data and empirical approach to analysing the business cycle are as clearly distinguishable from Moore's careless but imaginative econometric work as Juglar's work was from Jevons'.

2.2.2 *The problems in Mitchell's cycle analysis*

In 1921, the National Bureau of Economic Research, under Mitchell's direction, took up his statistical business cycle programme. Among many results of this programme was a second book by Mitchell *Business Cycles: The Problem and its Setting*, in 1927, which is best described as

[5] Mitchell was not trained as a statistician and apparently thought of himself as a theorist who used statistical evidence and later, only by necessity, statistical tools. Seckler (1975) discusses Mitchell's attitude towards evidence and Veblen's influence in this respect.

Mitchell's search for the definition and the correct analytical approach to the business cycle: a survey of the problems in the field rather than a new empirical study solving these problems. He again discussed the role of theory, of institutional descriptions, of statistics and of commercial history (records of business annals) in the analysis of business cycles. As in his 1913 book, all these elements seemed to have equal status in his investigation.

Throughout the book runs an attempt to define the concept of the business cycle. The more Mitchell tried to define the phenomenon, the more it seemed to slip away from his grasp:

> The more intensively we work, the more we realize that this term [business cycles] is a synthetic product of the imagination – a product whose history is characteristic of our ways of learning. Overtaken by a series of strange experiences our predecessors leaped to a broad conception, gave it a name, and began to invent explanations. (Mitchell (1927), p. 2)

He sought help with the conceptual problem in an analysis of business cycle theories, but once again he found that there were too many theories and all were plausible:

> the theories figure less as rival explanations of a single phenomenon than as complementary explanations of closely related phenomena. The processes with which they severally deal are all characteristic features of the whole. These processes not only run side by side, but also influence and (except for the weather) are influenced by each other ... Complexity is no proof of multiplicity of causes. Perhaps some single factor is responsible for all the phenomena. An acceptable explanation of this simple type would constitute the ideal theory of business cycles from the practical, as well as from the scientific, viewpoint. But if there be one cause of business cycles, we cannot make sure of its adequacy as an explanation without knowing what are the phenomena to be explained, and how the single cause produces its complex effects, direct and indirect.
> (Mitchell (1927), p. 180)

Mitchell now had a clearer notion of the roles of data and theory. Data were useless without theory:

> the figures are of little use except as they are illuminated by theory,
> (Mitchell (1927), p. 54)

and, he argued, data in the form of economic statistics and commercial records could be used to

> suggest hypotheses, and to test our conclusions. (Mitchell (1927), p. 57)

But the issue of how to 'test the conclusions' remained problematic in Mitchell's eyes. On the one hand, he rejected the idea of testing theories individually against the evidence as being repetitious and indeed restrictive if the theories were complementary. On the other hand, he believed there was a danger of hopeless confusion if an empirical classification, such as the phases of the cycle (i.e. depression, revival and prosperity), were to provide the framework for investigation.

Of particular interest in this book is the long section in which Mitchell discussed the merits and disadvantages of the statistical techniques which were then in popular use on business cycle data. He thought that the two main statistical techniques, data decomposition (discussed later in this chapter) and periodogram analysis (discussed in Chapter 1) were both inadequate because neither measured the business cycle directly. The data decomposition technique involved, in Mitchell's view, the dubious separation of an economic time series into cyclical, trend, seasonal and accidental elements. (The cycle element was what was left when the others had been removed.) Of the alternative periodogram analysis, Mitchell noted that this had technical difficulties and, like most economists, he clearly doubted the periodicity of business cycles. Indices of the cycle (made up of several variables) were treated more leniently by Mitchell because he believed they measured the business cycle directly, although he felt that none of the indices available at the time was entirely satisfactory. Correlation of one time series with another (with or without lagging) was another popular tool of data analysis in the 1920s. Correlation techniques were not rejected outright, but Mitchell distrusted the technique because he believed correlations could easily be manipulated. In addition, contemporaries tended to take high correlations as evidence of causal connections, an inference Mitchell believed to be invalid.

This book, with its exhaustive discussion of current statistical methods, fully established Mitchell's reputation as the preeminent figure in statistical business cycle research of the interwar period. The book was an impressive survey of the field of business cycle research in the 1920s and was justly praised by his contemporaries, both for his insights into the phenomenon and for the rich discussion of business cycle research methods, particularly the statistical approaches.[6] Yet Mitchell's book contained almost nothing of direct importance to the development of econometric work on the business cycle in the 1920s and 1930s, and was, if anything, negative towards further statistical research on cycles.

The circumspection evident in the book might easily strike the modern reader as Mitchell's inability to make up his mind. He had rejected the idea of using any one theory (but conceded the necessity of theories in general); had rejected currently available methods of

[6] See, for example, H. Working (1928).

analysing statistical data (but depended on statistical data as evidence); and he also expressed doubts about the role of business records (though once again he wanted to use their evidence). One reason for this equivocation was that the difficulties of any particular approach or method loomed large in Mitchell's mind.[7] Another reason was that Mitchell could not, or would not, deal in abstract notions. Much of the statistical analysis by this time, assumed that there was some 'normal' cycle: a hypothetical cycle which would exist if there were no short-term disturbances or long-term influences. Mitchell, on the other hand, was interested in the 'typical' cycle: the cycle which was characteristic of business cycles in the real world. Mitchell's priorities in business cycle research were description and measurement of real cycles.

2.2.3 *Mitchell and Burns' cycle measurements*

The third of the major volumes in Mitchell's research programme on cycles was *Measuring Business Cycles*, written jointly with A. F. Burns and published by the NBER in 1946.[8] Their aims were to study the fluctuations in the individual sectors of the economy and to use the individual data series available to provide good measurements of the cycle. This was the programme which had been laid down at the end of Mitchell's 1927 book, but the problem of measuring business cycles turned out to be far more difficult than Mitchell had envisaged and took a long time to solve.

Measurement was seen by Mitchell and Burns as a prior requirement for testing the theories of other economists, of which they wrote:

> Their work is often highly suggestive; yet it rests so much upon simplifying assumptions and is so imperfectly tested for conformity to experience that, for our purposes, the conclusions must serve mainly as hypotheses. Nor can we readily use the existing measures of statisticians to determine which among existing hypotheses accounts best for what happens. Satisfactory tests cannot be made unless hypotheses have been framed with an eye to testing, and unless observations upon many economic activities have been made in a uniform manner. (Burns and Mitchell (1946), p. 4)

[7] Dorfman's judgement on the influence of Veblen on his pupils seems particularly apt in describing Mitchell's approach to business cycles: 'Several of Veblen's students chose a concrete field of economics for detailed inquiry and found themselves moving gradually into ever-widening realms. Veblen had had the effect of stripping them so thoroughly, though unconsciously, of their complete confidence in the old way, that they tended to base their inquiries closely upon the existing facts' (Dorfman (1949), p. 450).

[8] The initial collaborator at the NBER was Simon Kuznets; Arthur Burns took over as researcher when Kuznets left the project to work on national income accounts.

In order to test theories, the statistical data had to be reformulated into measurements of the business cycle itself. Measurement in turn required a prior definition of the phenomenon to be measured. Mitchell's definition of business cycles had by this stage become a description of observable characteristics, and since 'equilibrium', 'normal' and 'natural' were theoretical notions, not observable characteristics, they did not form part of this definition.

In his 1927 book, Mitchell had criticised statistical workers for their failure to provide direct measurements of the cycle. In this new book, Mitchell and Burns evolved a new set of statistical measures of the business cycle called specific cycles and reference cycles. Specific cycles were cycles specific to an individual variable, dated on the turning points of that variable and divided into the nine phases of that variable's cycle. As a benchmark for comparison of the different variables and to trace their interrelationships and timing, each variable also had its own reference cycle based on the timing of the business cycle as a whole (measured on the basis of business annals and the conjunction of important variables of the cycle). They wrote of the reference cycle idea:

> This step is the crux of the investigation; it involves passing from the specific cycles of individual time series, which readers not embarrassed by experience are likely to think of as objective 'facts', to business cycles, which can be seen through a cloud of witnesses only by the eye of the mind.
> (Burns and Mitchell (1946), p. 12)

Both reference cycles and specific cycles were averages of the individual cycle data in which seasonal variations had been excluded and the data were deviations from the average for each cycle (to avoid interference from trends). In practice, these cycle measurements were arrived at by a mixture of mechanically applied rules and ad hoc judgements.

The specific and reference cycles were presented in numerous tables and in graphs. An example of these graphs (the one used by Burns and Mitchell as an illustration, with their explanations) is reproduced in Figure 4. The Burns–Mitchell measurements of specific and reference cycles were quite difficult to understand. Although Burns and Mitchell gave hints on how to read them, the very newness of the graphs made it difficult to gain a clear picture of each variable's characteristics. (For the importance of graphs in general in statistical economics see Addendum to this chapter.) The position was made more difficult for the reader because the graphs were not only used to present measurements of each element in the business cycle but the relationships between the variables had to be understood largely from comparison of the graphs.

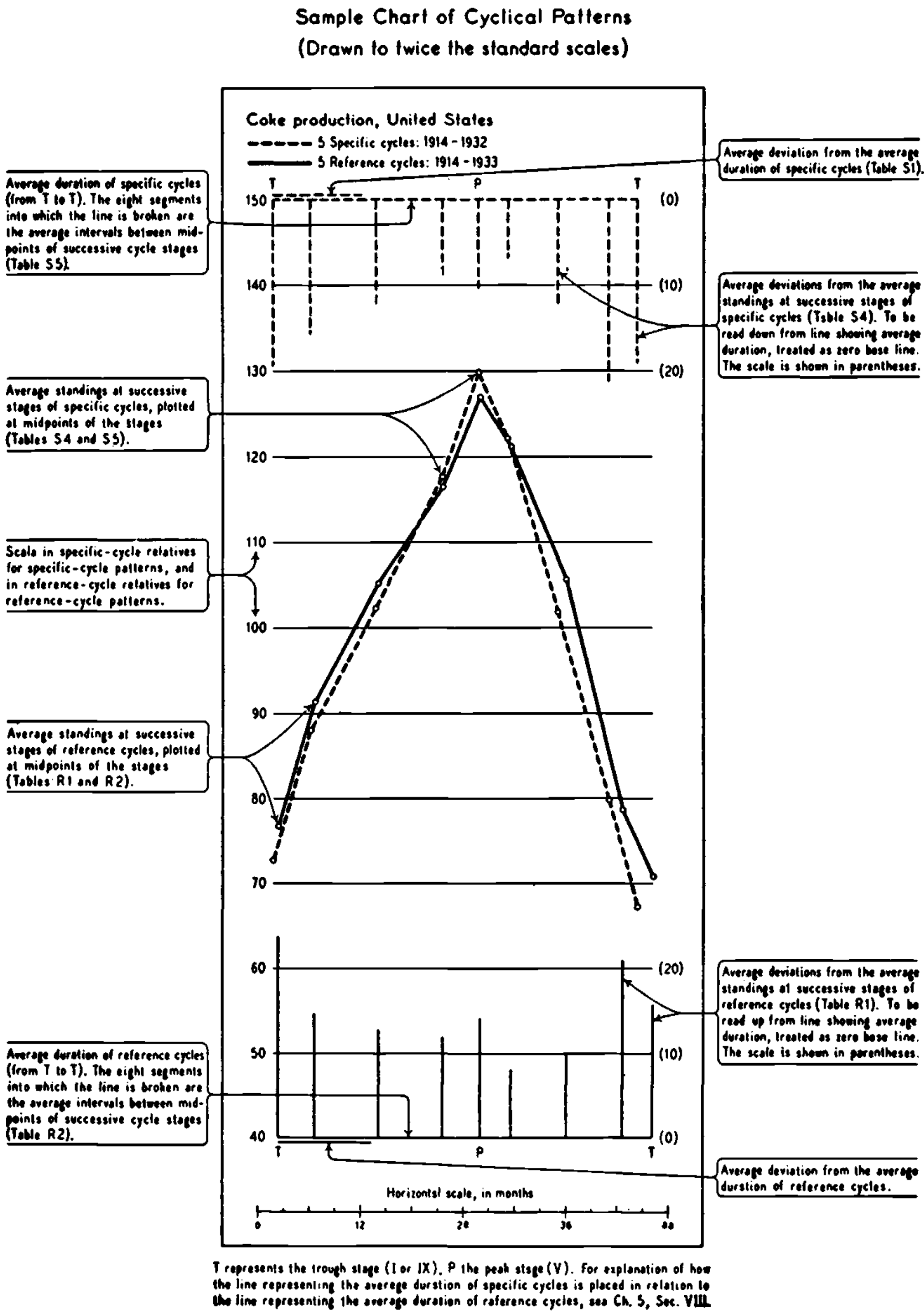


Figure 4 Mitchell's specific and reference cycle chart
Source: Burns and Mitchell (1946), p. 35, Chart 2

Mitchell and Burns did not test any theories that purported to explain cycles, but did test a number of hypotheses about the long-term behaviour of economic cycles using their measurements. For example, they tested Schumpeter's hypothesis about the relationship between different cycles, which suggested that there were three 40-month cycles (Kitchins) to each cycle of 9–10 years (Juglars) and six of these to each long wave (or Kondratieff cycle). They found it very difficult to fit their own specific cycles into the pattern proposed by Schumpeter for there was no consistent agreement of timing and dates and they therefore refused to accept Schumpeter's hypothesis. They also investigated one of Mitchell's own hypotheses, that there was a long-term secular change in cyclical behaviour. This test was more interesting for they examined whether their specific cycle measurements exhibited stability, first by fitting trend lines to their cycle durations and amplitudes, and secondly by dividing their cycles into three groups and testing for differences in the variance in amplitudes and durations. They used correlation tests and F-tests of significance, which failed to provide support for their hypothesis. They concluded that such changes as were observed were irregular or random, rather than consistent and due to secular change. Similar tests on their reference cycle measurements also failed to support the hypothesis, but these results did give them further confidence in their measurements.

Their own explanations of what happens during business cycles and how the various parts of the economy fit together were to be held over until the next volume:

> Later monographs will demonstrate in detail that the processes involved in business cycles behave far less regularly than theorists have been prone to assume; but they will demonstrate also that business-cycle phenomena are far more regular than many historically-minded students believe...
>
> Our theoretical work leans heavily on cyclical averages, such as are described in this book, and we hope that others will find them helpful. Instead of setting out from the dreamland of equilibrium, or from a few simple assumptions that common sense suggests ... our 'assumptions' are derived from concrete, systematic observations of economic life.
>
> (Burns and Mitchell (1946), p. 491)

Mitchell and Burns had codified their 'concrete observations' into representations of the cycle. This necessarily imposed some uniformity or regularity on the cycles. By thus retreating from Mitchell's earlier (1913) position that there was no uniformity in the patterns or elements in the business cycle, they left the way open for theoretical explanations of the cycle. Unfortunately, Mitchell died before the important final volume discussing their theoretical conclusions was complete. Burns carried on the work at the NBER, but the impetus of the programme was lost.

Mitchell's business cycle programme had begun in 1913 when he had prophesied that

> it is probable that the economists of each generation will see reason to recast the theory of business cycles which they learned in their youth.
> (Mitchell (1913), p. 583).

He had himself pointed out the change from crises to cycles. Yet he had been unable to foresee the future. The cycle was no longer, by Mitchell's death in 1948, the defining unit of analysis. During the course of Mitchell's working life, cycle theories had been recast twice: the theory of crises had given way to the theory of business cycles and this in turn had given way to macroeconomic theory. At the same time, the fashions in methodology also changed. Mitchell was an empirical economist primarily concerned with describing, classifying and measuring the 'typical' economic cycle.[9] His use of statistical evidence had made him a pioneer in 1913, but his lack of analysis made his statistical economics look increasingly dated as the econometric programme grew more sophisticated.[10] While Mitchell was still struggling with raw business cycle data in the 1930s, Tinbergen was busy formulating macroeconometrics. By the time of the Burns and Mitchell volume, the yawning gap between statistical economics and the more technically advanced and theoretically minded econometrics programme was evident in a highly critical review of their book by Koopmans, one of the leading econometricians of the day.

Tjalling Koopmans was a member of the Cowles Commission, which had been set up in 1932 to undertake econometric research and was *the* centre of econometric research in the 1940s.[11] Koopmans' 1947 review initiated the famous 'Measurement Without Theory' debate in which

[9] His rejection of mechanical analogies in favour of evolutionary and organic ones combined with his descriptive and classificatory approach suggests that whereas many economists looked to nineteenth-century physics for their model science, a more apt comparison for Mitchell is with the descriptive and categorising activities of nineteenth-century naturalists.

[10] It is not clear exactly when these programmes became distinct to contemporaries. The econometric programme must have been fairly clear by 1929 when the Econometric Society was formed, but because of the lack of econometric work on business cycle models in the late 1920s and early 1930s, this did not necessarily help to distinguish the business cycle programmes.

[11] The Cowles Commission for Research in Economics was set up in 1932 and funded by Alfred Cowles specifically to undertake econometric research. The journal *Econometrica* was run from the Commission, and there were close links with the Econometric Society. (Details of the history of the institution and its activities are described in Christ (1952), Hildreth (1986), Epstein (1987) and in the annual reports of the Commission for 1943–8.) Koopmans' review sparked a reply by Vining and a further response from Koopmans published in 1949, see Vining and Koopmans (1949). On Koopmans, see Chapter 8 n. 6.

he accused Burns and Mitchell of trying to measure economic cycles without having any *economic* theory about how the cycle worked. Koopmans' group had advanced a Walrasian model based on the aggregation of individual economic agents as units and in which the cycle was seen as a deviation from an underlying equilibrium level. In reply to Koopmans, Vining (1949) (a visiting research associate at the NBER) charged the Cowles group of being guilty of the same sin of measurement without economic theory in their own work. Vining took the position that the theory of how agents behaved had not been sufficiently worked out, and that the Cowles model was therefore a 'pretty skinny fellow' upon which to base so much high-powered statistical estimation: in essence, measurement without economic theory. Further, Vining argued in defence that Mitchell took a holistic view of the economy in which behaviour of the whole economy was more than a simple aggregate of individuals' behaviour and in which cycles and trends could not be unravelled because both were subject to the impact of institutional and evolutionary factors.

There was a second serious charge made by the econometricians: that the NBER group were trying to undertake measurement without *statistical* theory. Following Haavelmo's work in the early 1940s, econometricians believed that economic theories should be explicitly formulated as statistical hypotheses, so that methods of inference based on probability theory could be used to measure and test the relationships. Klein was in the process of testing a cycle (or macroeconomic) model for the Cowles group, using their newly developed methods of estimation and inference. It is true that Mitchell and Burns had not tested theories which explained the cycle, because they believed such theories had not yet been properly formulated for statistical testing; but they had (as noted above) used probability inference in their tests of theories about long-term behaviour.

This debate of the late 1940s showed the clearly irreconcilable differences both in theoretical conceptions of the cycle and in methodological approaches between Mitchell's statistical programme and that of the econometricians. The 'measurement without statistical theory' aspect of the debate is particularly important, but we need to know more about the development of the econometric programme before we can deal with it in detail in Chapter 8.

2.3 Persons and business barometers

In 1919, soon after Mitchell had produced his first statistical study of business cycles, another empirical economist, Warren M. Persons published his first main results using an alternative analytical approach to cycles based exclusively on statistical evidence.[12] This second main strand of statistical business cycle research was largely descriptive in aim and was associated with the production of time-series representations of cyclical activity called business barometers. Persons is much less well known than Mitchell, but his work was more important for econometrics because he was primarily responsible for developing data preparation and data adjustment methods which quickly became and remained standard in applied econometrics.

The methods Persons used to decompose economic time series into components and expose the business cycle were not entirely new, but had been used piecemeal in economic statistics from the late nineteenth century. Commercially produced indicators of economic activity, based on statistical series and used to forecast future business conditions, were of a similar age.[13] However, Persons developed the methods to a much higher state of sophistication and was particularly responsible for initiating serious academic study of these methods. His initiative temporarily gave academic respectability to business cycle barometers. This academic respectability led in turn to the setting up of a number of university business cycle research institutes in the 1920s which constructed their own national barometers. The barometers suffered a subsequent decline from respectability, but their descendants live on in the field of commerce and of government statistical services which publish leading and lagging series of economic statistics called indicators.

Persons was already well known in the field of economic data analysis when he was asked in 1917, by the newly established Harvard Committee for Economic Research, to undertake a study of the 'existing methods of collecting and interpreting economic statistics'. Two years later, the Committee established the *Review of Economic Statistics*,[14] with the aim of providing

> a more accurate record of economic phenomena than is now available, and [it] will also supply a method of interpreting current economic statistics which will make them more useful and significant than they are today

[12] Warren M. Persons (1878–1937), statistician and economist, was educated at the University of Wisconsin. He joined the Harvard faculty in 1919 and resigned in 1928 to become a consultant economist. Persons had produced an earlier barometer, but it was a limited affair based on little data or analysis (see Persons (1916)).

[13] This was a profitable business in 1913 (according to Mitchell), but Persons' work stimulated further commercial activity and by 1925 (see I. Fisher (1925)) there were 40 successful commercial forecasting agencies in the USA.

[14] The Harvard Committee was set up in 1917 and started the *Review* in 1919. (In 1949 the journal's name was changed to the *Review of Economics and Statistics*.) These three quotations on aims and methods come from the 'Prefatory Statement' in the first issue of the *Review* by C.J. Bullock.

and the method of interpreting economic statistics was to be developed by the application of

> modern methods of statistical analysis which have hitherto been utilized more extensively in other sciences than in economics.
> (Bullock (1919), Preface)

Persons' contributions to the first year's issues of the *Review* in 1919 included two major articles, one on the method of analysis of economic time series and the second on the development of business indicators. The first of these is more interesting from the point of view of the development of econometric thought. Persons' stated aim in these papers was to find a method to determine the significance of statistical data in indicating current and future business conditions. Persons believed that the first priority (addressed in his first paper (1919)) was to obtain long data series of reliable (accurate) and homogeneous data in order to make comparisons. His approach to the problem of which methods to use in the investigation of data was experimental in the sense that every procedure that he suggested was tried out on a number of data series. He strived for methods which would be objective and leave no room for subjective judgement for, as he said,

> No method is thought worth while testing that cannot be made entirely objective, that cannot be reproduced exactly by an individual working according to the directions laid down. The primary requirement in the argument is that each step be tested; the primary requirement in the method evolved is that the operations should be fully explained and reproducible. (Persons (1919), p. 7)

The basic conceptual framework, from which Persons proceeded, was that fluctuations in economic series occur as a result of various forces: secular (or long-term) forces, seasonal forces, cyclical (or wavelike) forces and accidental (or irregular) forces. These all occur simultaneously:

> Those fluctuations are thus a confused conglomerate growing out of numerous causes which overlie and obscure one another.
>
> In view of this complexity the simple comparison of items within time series, in what may be termed their crude state, is of but little significance.
> (Persons (1919), p. 8)

The aim of the method was to eliminate the unwanted fluctuations and reveal the business cycle:

> the problem is to devise a method of unravelling, in so far as may be, the tangle of elements which constitute the fluctuations in fundamental series; that is, of distinguishing and measuring types of changes; of eliminating

certain types; and thus of revealing in the corrected data the quantitative effect of those forces which underlie the business and industrial situation.
(Persons (1919), p. 8)

Persons attacked the problem of measuring the secular trend first, since he needed a benchmark from which to measure the cyclical changes. Ideally, he thought, the secular trend should be determined by the data, rather than *a priori*, because he had no foreknowledge of which periods were homogeneous. He tried two empirical means of determining the trend, one by the method of moving averages and one by curve fitting. For the former, rather than take a centred moving average, each observation was taken relative to the preceding average period. He tried this with different numbers of terms in the moving average period (e.g. 5 years, 10 years, etc.) to provide different trend-adjusted series. But this proved unsatisfactory, because when he fitted a straight line to each of the new (detrended) data series he found that these lines had different slopes including, in some cases, slopes of opposite signs, whereas he felt that they should have produced the horizontal lines of detrended data.

After further experimentation Persons decided that the direct trend-fitting method, in which simple curves or straight lines were fitted to the raw data, would be less cumbersome and more reliable than the moving average method. However, this direct method was far from mechanical. It required a number of judgements about the period to be used, with little guidance from Persons. Was it homogeneous? Did it cover a full cycle or was it a peak to trough measure? Was the function to be fitted linear? Persons advocated that there was no mathematical substitute for the 'direct analysis of homogeneity'. By this, he meant fitting trends to a number of different periods and looking for radical changes in the slope of the fitted line, which he took to indicate non-homogeneous periods. For example, he found evidence of a change in the trend component in various business cycle data between the two periods 1879–96 and 1896–1913 and therefore fitted two different trends to the subperiods.

It was inevitable that Persons should fail to find adequate mechanical rules for this problem of trend fitting. The lack of natural base periods (the open-endedness of the time unit) meant that once the time period had been chosen, the technical criteria of success in trend fitting (that is, the appearance of horizontal trend lines in trend-adjusted data) could always be fulfilled. But the question of how to choose a homogeneous time period remained to a certain extent arbitrary and depended on individual judgement in each case.

These problems were not evident in the second component of time series, seasonal variations, which did have a natural time unit. Persons' work in this area was correspondingly more successful. His procedure was: first to establish that such fluctuations existed, then to measure them and finally to eliminate them from the data series. Again after a number of experiments on different data series he settled on link relatives, month to month changes, to give the basis for measurement of the seasonal variations. He looked at the frequency distribution of these percentage monthly changes for each month of the year and chose the median value as the best measure of the seasonal pattern. Following an adjustment for trend effects over the year, these medians were then formed into an index which began at 100 and returned to the same value for each January. This index could then be used to correct the original data series.

Lastly, Persons went on to study the other two elements in the time series, the irregular and the cyclical fluctuations. He did not regard cyclical fluctuations as being exactly periodic or regular. Irregular fluctuations might be large or small and isolated or occurring in succession like the cyclical fluctuations. The cyclical and the accidental components might therefore have similar data characteristics. Persons took this to imply that the forces behind the two elements were interconnected and that it would be impossible to distinguish the two elements:

> No distinction, therefore, between 'cyclical' and 'irregular' fluctuations may be obtained simply by study of the statistical data.
>
> (Persons (1919), p. 33)

Later he used a short period moving average to smooth away the irregular fluctuations. By a process of subtracting the seasonal and trend elements, he was then left with the remaining cyclical fluctuations.[15] He tried out these measurement methods on 15 economic series of monthly data covering 15 years. Persons concluded that his study of economic time-series data revealed considerable support for the 'realistic character of the concepts' of the four components.

In his second major article for the *Review* (1919a) Persons' aim was to construct an index of general business conditions with particular emphasis on the cyclical aspects:

> It is the cyclical fluctuations which have received especial attention. Further, in dealing with the cyclical fluctuations the aim has been to measure them, rather than to utilize the results in constructing a theory of business cycles. (Persons (1919a), p. 115)

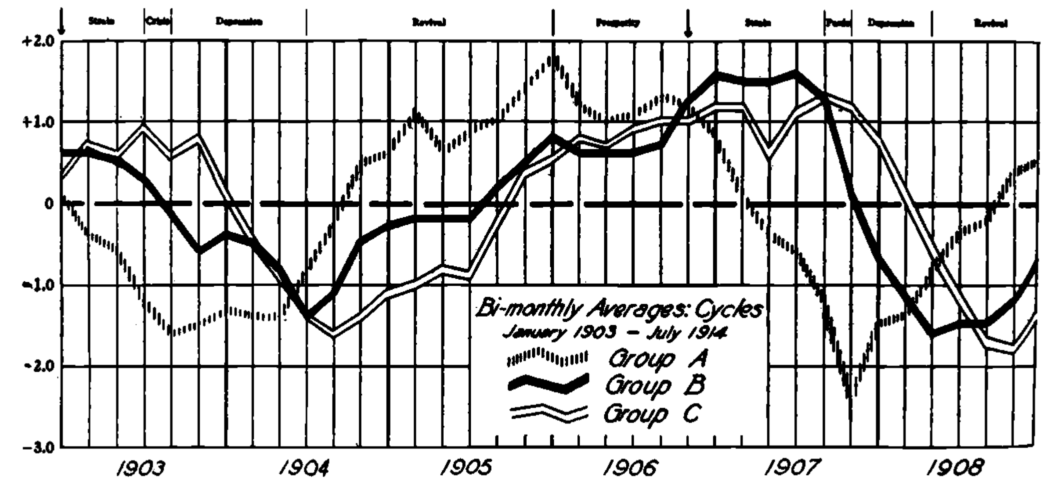
[15] In order to make the cyclical fluctuations comparable within and between data series, he calculated 'normal' (or standardised) fluctuations by dividing the percentage deviations from the secular trends by their standard deviations.

Constructing an index of the cycle raised the important question of how variables were related to each other. For Persons this was a question of fact not theory, to be answered by empirical research. He began by studying 50 data series and chose 20 of these which had similar periods but differently timed turning points. Persons relied on the methods he had already established in his previous paper to decompose the data and derive the individual cyclical series, but the problems of determining the sequence of the variables and combining them to make indices remained open.

His first step was to compare the variables and classify them according to the timing of their fluctuations to find the 'typical or average sequence' of the variables in the business cycle. With 20 data series, he needed to make 190 comparisons to cover all possible cases. These 190 comparisons were each carried out by three 'observers' with the use of an illuminated screen or light box on which were laid graphs of the two data series being compared. One graph was then moved back and forth on top of the other to try and match the movements in the two series. (Each series was expressed in terms of its standard deviations to make such comparison possible.) Each 'observer' was supposed to record the observed level of correlation (no arithemetical calculations: high, medium or low correlations were judged by eye), the sign of correlation (positive or negative) and whether the relationship was lagged forwards or backwards and by how many months. There was considerable disagreement, as might be expected, between the 'observers' as to whether a lag was positive or negative, and a correlation high or low.

Persons worked out the correlation coefficients for the lag periods suggested by his 'observers' and fixed the lag length according to the highest value of the correlation coefficient. This gave him the time sequence of the variables, and the timing of changes in the cycle. The variables were then grouped together to reflect this sequence and synthesised into three indices which would 'epitomize the business situation' and act 'as a guide to the relations that exist among economic phenomena' (Persons (1919a), p. 117).[16] Persons' indices of the business cycle were called the Harvard A-B-C curves, representing

[16] On this final aspect, of combining those series which the correlations showed to have similar timing and turning points into indices, there was no discussion and the data series in each group appear to have been averaged with no weighting involved. This omission is surprising in view of the known and much discussed statistical problems of weighting components in, for example, price indices. There was no discussion either, in this article, of the problem of forecasting from these indices, also strange given the purposes of barometers to forecast the turning points of the cycle. Both these points were discussed briefly in Person's earlier paper (1916).



BIMONTHLY AVERAGES OF CYCLES OF GROUPS A, B, AND C

Figure 5 Persons' business barometer
Source: Persons (1919), p. 112

respectively: speculation; physical productivity combined with commodity prices; and the financial markets. (Index B seemed to be rather a strange collection, made up of pig-iron production, bank clearings outside New York, indices of commodity and retail prices and the reserves of New York banks.) The A-B-C curves laid out Persons' view of how groups of variables were empirically related in cycles. An example of the A-B-C graph is given in Figure 5.

Persons had set out in his first article in 1919 to measure various components of the data so that the interpretation of economic time-series data would be more enlightening. Though, like Mitchell, he wanted to avoid the use of *a priori* theory in making his measurements, preconceived ideas were bound to enter any measurement process. First, as Schumpeter pointed out, the concept of data series being made up of four components associated with particular groups of causes or influences: cyclical, seasonal, long-term and accidental, implied underlying ideas about economic behaviour which constituted:

> a theory that was all the more dangerous because it was subconscious: they [the Harvard approach] used what may be termed the Marshallian theory of evolution ... they assumed that the structure of the economy evolves in a steady or smooth fashion that may be represented (except for occasional changes in gradient, 'breaks') by the linear trends and that

> cycles are upward or downward deviations from such trends and constitute a separate and separable phenomenon . . . this view constitutes a theory, or the backbone of one. (Schumpeter (1954), p. 1165)

Secondly, once involved in the measurement process, Persons could not avoid making some theory-based decisions, for example about what constituted a homogeneous period in economic terms and where the breaks between these periods occurred.

In his second paper of 1919, Persons had set out to show the typical empirical relationships of the cycle in the form of indices of business activity. Business cycle theory was, as already noted, fashionably new at this time, but concern about the economic theory relationships between variables was largely absent from Persons' discussion. He was concerned only with methods of representing the cycles in business activity. His process of deriving the indices which represented the cycle was based on highly subjective visual judgements about how series were related in time and which series should be grouped together. The actual measurements of the relationships were only made at the last point in his procedure when these decisions had been more or less taken by the three 'observers'. This was particularly unsatisfactory in those cases where his 'observers' disagreed as to which series were connected with others and (more problematically) what the lag length should be.

Persons was left with a number of questions as a result of his empirical approach. If one series preceded another did that mean that it caused the other? If so, Persons wondered, how similar must the cycles be? Or are both to be considered the joint effects of joint causes? These questions arose, and were unanswerable, because Persons had neither a theoretical economic answer as to how variables were related in a complex world, nor did he use any consistent statistical measurement or inference procedures which might have helped him. He used only two-variable correlation techniques which could not reveal, for example, joint connection with a third variable. This dependence on simple, rather than multiple, correlation techniques was exceedingly limiting. But despite these problems Persons' methods for the removal of trends, seasonal and irregular components quickly became commonplace in statistical business cycle research and pervaded econometrics generally. His methods (or closely related ones) for detrending data, for removing seasonal variations and for smoothing out erratic fluctuations are now such a standard part of data preparation for econometric work that they are barely given a thought.

2.4 The business cycle research institutes

The decade of the 1920s witnessed an explosion in statistical business cycle research. Institutes entirely devoted to statistical analysis and research on the business cycle were established in Germany, France, the Netherlands, Italy, Sweden, Britain, indeed throughout Europe generally, and even in post-revolutionary Russia. Though the detail of the methods adopted and the overall concept used differed from institute to institute and country to country, all relied on an empirical methodology using statistical data in which the work of both Mitchell and Persons were highly influential. Evidence suggests that Mitchell's name was invoked as the foremost exponent of statistical research of the period, although his methods were not always followed. The examples presented in the *Review of Economics and Statistics* by Persons were less acknowledged, but were at least as influential judging from the publications of the new institutes.[17] An analysis of the work of the Berlin Institute (as represented by the main work of its director, Ernst Wagemann) illustrates the way this joint influence worked. An examination of the work of the Konjuncture Institute of Moscow shows that the influence was two-way.

Wagemann's 1928 book of business cycles was translated into English as *Economic Rhythm* (1930) at the instigation of business cycle scholars in the USA and prefaced by Mitchell. Wagemann in turn much admired Mitchell's work on business cycles which seemed to him to be the ideal synthesis of the theoretical, statistical and historical schools of economics. Unfortunately, the nineteenth-century *methodenstreit* between the theoretical deductive and the empirical inductive approaches was still sufficiently alive in Germany to stifle synthetic work (particularly quantitative and econometric developments) in the 1920s.[18] These factors made Wagemann self-conscious about his own attempts to combine the theoretical and empirical methods and whilst

[17] Mitchell (1927) himself credited Persons with setting the trend for statistical analysis, both in the US and in Europe. As an example of their influence, Snyder (1927) dedicated his book jointly to Mitchell and Persons with the descriptive tag: 'pioneers in the quantitative study of business cycles'. See Schumpeter (1954) for his contemporary observations on these influences.

[18] Economists descended from the historical school believed that there were no constant laws in economics and, like Mitchell, that business cycles were individual historical occurrences, therefore it was not possible to analyse the business cycle as a uniform event. German theoretical economists scorned any empirical work and maintained a detached indifference to any actual economic event. This was true, as Wagemann (1930) pointed out, even to the extent of ignoring the infamous German inflation of 1923 and the 'miracle' of the subsequent currency stabilisation.

he acknowledged the influence of American thought, he tried to strike out afresh. For example, he claimed not to follow the Harvard (Persons') decomposition method, in the first place because he saw their classifications as statistical rather than economic. Instead, he classified movements as changes in structure (isolated changes which were continuous or discontinuous) and periodic fluctuations (changes with either a fixed or free rhythm). But despite his claims, the description of these reveal just how close his categorisation was to Persons'; continuous changes were more or less equivalent to secular trends, discontinuous changes were large irregular changes (closer in conception to changes in structure than to small random events), fixed rhythm changes were, for example, seasonal fluctuations and free rhythm changes were business cycles.

The second problem of the Harvard method, according to Wagemann, was that having eliminated the unwanted series components, this provided only a negative definition of the business cycle, that is, as the residual element in the data. In addition, Wagemann judged both the single and multiple cycle indices, such as the Harvard A-B-C curves, inadequate. The Berlin Institute developed a more sophisticated system of indices (as far as economic content was concerned) consisting of eight sectoral barometers covering production, employment, storage, foreign trade, transactions, credit, comparative prices (in security, money and commodity markets) and commodity prices. Wagemann's aim was to forecast three months ahead with these barometers once they had established a model of how they fitted together. Successful forecasting depended, however, on correct diagnosis of the events which in turn depended on there being some typical movements (or 'symptoms') in the data.

Wagemann's book is full of interesting speculations. For example, he thought that forecasting might be improved by examining the plans of individual firms (by questioning the firms) and by using statistics of orders or of raw material inputs. On the one hand, he argued, firms' plans meet obstacles, which would reduce the accuracy of forecasts based on barometric diagnosis. But on the other hand, the reactions to obstacles may be reflected in the barometers themselves, which would help forecasting. In any case, he thought, completely successful forecasts would never be possible because movements are never identical (variables are subject to structural changes), and because a static system of classification underlies the whole.

While his applied work and its definitions reflected the influence of Persons' work, Wagemann's methodological statements and conceptual definitions were extraordinarily reminiscent of Mitchell's empirical (but non-operational) ideas. Two examples make this point neatly. One is the positive definition of business cycles as

> the total complex of economic reaction phenomena
> (Wagemann (1930), p. 68)

that Wagemann adopted in preference to the negative, but practical, definition of the cycle as the residual series. The second is his call for a study of the facts before fitting business cycles into a deductive theory; to theorise without detailed study would be, he said,

> as though a doctor were to try to find a pathogenic agent before he had formed a proper notion of the disease itself by a study of the symptoms. To look for the causes of a phenomenon before acquiring a fairly adequate knowledge of its external manifestations is an absurdity; how can one search for causes before one has a grasp of the subject itself?
> (Wagemann (1930), p. 217)

Wagemann made the interesting suggestion that approaches to statistical business cycle study could be characterised nationally: German economists viewed the economy from the point of view of a doctor dealing with a human patient while the Americans saw the economy as 'a powerful piece of machinery' (Wagemann (1930), p. 9).[19] Wagemann claimed that the Russians viewed the economic system rather like the planetary system and adopted the statistical approach used in astronomy. As a result, some of their statistical work had a different flavour, and involved stochastic theory to help analyse the movements of economic variables.

The Konjuncture Institute of Moscow was founded in 1920 under the direction of Kondratieff. The institute fed information directly into the state planning organisation to help produce the 'Gosplan' (the national plan) and an important part of their analysis was concerned with more general developments of the economy. The Moscow Institute produced a number of statistical publications which contained statistical data and analysis of the sort standard from other cycle institutes. From 1925, it also published a theoretical journal to which workers at the Institute, such as Oparin, Ignatieff, Slutsky, Konüs and Kondratieff, contributed.[20] Judging by the English summaries of the

[19] There may be some truth in this latter observation in relation to economic theory but the three American economists discussed here, Persons and Mitchell, and more interestingly Moore, showed marked preferences for sea metaphors such as 'ebb and flow', 'drifting' and 'buoyant', rather than mechanical ones, in describing the cycle.

[20] The journal, called ***Problems of Economic Conditions*** (in its English translation), was published from 1925 to 1928 in Russian with English-language summaries.

articles and the works referenced by the Russian authors, they were keenly aware of, and much of their work was closely related to, the work done elsewhere. For example, the journal published some of Mitchell's work on the use of business records in cycle research.

Workers at the Moscow Institute used their theoretical journal to publish experiments on cycle data which sometimes involved novel hypotheses and unusual techniques. These works became available to the other business cycle institutes and some proved very influential. One important work, which quickly became widely known, was by Slutsky, in which he examined the ways in which stochastic processes could form cyclical patterns in the data. (This article is sufficiently important to be discussed in its own right in Chapter 3.) Another important article contained Kondratieff's first applied results on his long wave thesis. (This 1925 paper was not his original paper on the subject, but it was the first to be translated and published in the West.) Kondratieff's long wave theory was highly controversial, even within the business cycle Institute, and he was removed from his position as director. In 1928 the Moscow Institute was closed down and Kondratieff was sent to Siberia, his long wave theory dismissed as 'wrong and reactionary'.[21]

The Harvard business cycle work suffered the far less dramatic fate which comes to those in the West whose forecasts have failed, namely loss of credibility. The Great Depression was commonly believed to have seen the decline from academic respectability (and perhaps a temporary wane from commercial use) of business barometers.[22] There were other reasons why these empirical approaches, so strong in the 1920s, lost their dominant role in academia in the 1930s. One was the work by Yule and others critical of the methods of statistical analysis used in cycle research. The second reason was the development of the alternative quantitative programme in econometrics. After a temporary lull following Moore's work, the econometric approach to business cycle analysis had moved on in the early 1930s to the development of small macrodynamic models of the business cycle and these by the late

[21] Röpke (1936) claimed that the Institute was closed because it was 'reactionary' and 'the staff was sent to Siberia or shot'. This report seems to have been overstated since Slutsky continued to hold an official university mathematics post until his death in 1948, while another member of the group, Konüs, was recently made a Fellow of the Econometric Society. The story is told in Garvy (1943).

[22] Schumpeter claimed that in fact the Harvard barometer successfully predicted both the 1929 crisis and subsequent fall into the Great Depression, but 'the trouble was that the interpreters of the curves either would not believe their own methods or else would not take what they believed to be a serious responsibility in predicting depression' (Schumpeter (1954), p. 1165). Schumpeter indeed argued for a kinder reassessment of the whole barometer episode than that which had been offered by other contemporaries.

1930s grew into full-scale macroeconometric models of the business cycle.

Despite the loss of academic credibility, statistical business cycle study continued to be an important government and professional activity in the 1930s. The Dutch Statistical Office, for example, began their own statistical business cycle journal in 1929, under Tinbergen's editorship. Further, cycle indices (as leading and lagging indicators) have continued their useful role in commercial and public sector economic analysis and forecasting services. The statistical study of business cycles has continued because, as Mitchell, and more particularly, Persons and the work of the Institutes have shown, a great deal of understanding about the economy can be gained from the rich statistical descriptions of business cycles which they presented.

2.5 Statistical economics and econometrics

The parallel approaches of statistical economics and econometrics shared two traits: both recognised economic cycles in the statistical data and rejected the notion of crises as independent events caused by local circumstances. But economists involved in the two programmes differed in their aims, their beliefs about the cycle and their use of statistical methods. The statistical research programme of Mitchell and Persons can be summarised as follows. They both started from the position of wanting to describe and measure business cycles as a way of finding out all about them. Their desire to find the facts required that they first defined what the business cycle was, so that they knew what data to look at and how to set about isolating or measuring the cycles. Having no theory of the cycle, and thus no theoretical definition of what it was, both were inclined to interpret the phenomenon rather broadly. Persons was untroubled by difficulties of definition for he avoided them by turning them into questions of statistical method: the business cycle was defined by the methods he used to isolate and represent its fluctuations in barometer form. Defining and describing the phenomenon and producing measurements of it proved more difficult for Mitchell since theoretical ideas were a part of his process. As an aid in definition and a way of providing measurements, Mitchell and Persons developed different visual representations of the cycles: Mitchell's (and Burns') reference cycle charts and Persons' A-B-C indices. For Mitchell and Persons, then, statistical work was concerned with constructing representations of the business cycle itself, although for Mitchell this was but a prelude to later understanding. A comparison may make this point clearer. Jevons and Moore (and as far as aims go, we should include Juglar here) did not worry about defining the cycle as a phenomenon and describing it because their theories told them what facts or data to study and their aim was to explain what caused the cycle.

Beliefs about the nature of economic cycles determined the extent to which each economist used statistical analysis. For Jevons and Moore (the econometricians), the cyclical regularity was hidden in the statistical features of the data set, whereas for Juglar and Mitchell (the statistical economists) the regularities were visible in the pattern of events of the cycle, not in statistical features of the cycle itself such as average cycle lengths or periodicities. Thus Mitchell and Juglar treated the cycles as individual, irregular events and sought typicality of behaviour within the cycles. Both were precluded from using more sophisticated statistical methods by their belief in non-uniformity in the statistical data. Persons did not believe in regular cycles either, but he did believe that certain other regularities or constants of behaviour would be reflected in statistical data (for example, seasonal variations and correlations between variables), so he made use of analytical techniques on these aspects. By contrast to the work of these statistical economists, Jevons, and more especially Moore, used statistical analysis not only to determine the nature of the underlying cycle, but to try and establish and verify the causal relationships which made up their theories as well. That is why I have labelled their work 'econometrics'.

The question of what 'theory testing' meant is more complex and can not be answered simply by knowing whether an economist was working in the econometrics or in the statistical economics tradition. Persons was at one extreme: a pure empiricist within the statistical programme, interested only in presenting facts and not at all in theory testing. At the other extreme in the econometrics programme, Moore and Jevons held strongly to their own pet theory of the cause and course of the cycle. These two believed that theory verification, not theory testing, was required; but this meant different things to each. For Jevons, evidence was not particularly important; verification was achieved primarily through an induction resting on probability inference. For Moore, statistical evidence which verified the theory was important, but he only sought confirming evidence in favour of the relationships of his theory not negative evidence which might refute it.

There was considerable muddy ground between these two positions, suggesting a complex partition of beliefs on theory testing. Juglar has been defined in this chapter as Mitchell's forerunner in the statistical tradition of analysis because of his use of statistical evidence, yet he fits most neatly with Moore in his attitudes to theory validation. Mitchell was also in the middle, an empiricist but not anti-theory. He had thought hard about the difficulties of finding out which were useful theories and had concluded that of the large number of possible cycle theories, some were complements and some were rivals. His attitude towards testing theories was rather natural in the circumstances: if many theories were possible, they could not all be verified, some would necessarily have to be rejected. (Mitchell, remember, did examine – and reject – some theories concerned with the appearance of cycles (the changing shape or time period of cycles) over the long term, but did not test theories about the cause and process of the economic cycle itself.) Further, as Chapter 4 will show, Tinbergen, though clearly in the econometrics tradition, was in this respect like Mitchell, for he also saw that theory testing in the face of a multiplicity of theories involved both refutation and verification. Before we move on to Tinbergen's macro-econometrics, Chapter 3 explores how the idea of random shocks came to infiltrate formal business cycle analysis. But first, a brief aside on graphic methods.

Addendum: graphs and graphic methods

Graphs have played an important part in both statistical economics and econometrics. In the first place, they formed one of the main ways of presenting evidence. The aim and effect of graphing time-series data was to reduce a mass of figures to a single picture showing characteristics such as cycles more clearly than in tables. (Juglar's table-graphs (see Figure 3) were a halfway house, which give vivid demonstration to this point.) Discussions of the best ways to achieve these aims can be found in the economic and statistical journals of the late nineteenth century (documented by Funkhouser (1938) and Spengler (1961)).[23] Early statistics textbooks used by economists typically included a chapter on methods of graphing data (for example, Mills (1924)) and the interwar period saw a rash of books entirely concerned with advising economists on how to present data in graph form. Secondly, graphs often formed an important element in statistical analysis in the early twentieth-century work (for example, in Bowley (1901)). Graphs were used to help explain statistical methods, and graphic methods were even used as alternatives to aglebraic methods of manipulating variables and arithmetic methods of calculation. For example, graphs were used to explain and to calculate correlation and regression coefficients (see Ezekiel (1930) and Davis and Nelson (1935)). Thirdly,

[23] See also J. L. Klein (1987a) for an interesting analysis of the early use of graphs in economics.

graphs were used to report the results of statistical work, and various forms of graph were devised especially for this purpose.

In the work of business cycles reported in the last two chapters, graphs were used extensively. In the Jevons/Moore tradition they were used primarily to show regularity in the data on individual variables (as in Figure 1), to provide evidence of relationships between variables and to report the results of analysis (see Figure 2). The Mitchell/Persons tradition relied on graphs and graphic methods to a greater degree. Persons used every opportunity to use graphs to illustrate his methods of data decomposition and the experiments he carried out in arriving at those methods. Graphs were more fundamentally important to his work on indices; the relationships between the individual variables, and thus the content of the indices, were determined from studying graphs of the data series. Graphs were used in a standard way to represent data in Mitchell's 1913 book, but played a more central role in his 1946 volume with Burns, where their new measurements of cycles were dependent on pictorial representations of the data and presented in graph form. Though these Burns–Mitchell graphs (see Figure 4) did reduce the mass of figures to one diagram, the novelty of the measurements and their presentation made them difficult to understand. In comparison with the standard representations of the cycle, of the sort given by Persons (see Figure 5) which showed the cycles as a time series, the meaning of Mitchell and Burns' standardised reference and specific cycles was opaque. Comparison of these graphs underlines the point made earlier, that Mitchell and Burns gave representations of the typical cycle, while Persons gave representation to the cycles over time.

In the next two chapters there are further examples of the use of graphs. In Chapter 3, there are examples of how graphs were used to express the results of data experiments (Figures 6 and 7) and of methods of data analysis (Figures 8 and 9). In Chapter 4, there is an example of Tinbergen's use of graphs (Figure 10), used both to analyse the relationships (and thereby reduce the amount of arithmetic) and to report the results of his analysis. These sorts of graph were distinctive to Tinbergen's work, and were copied by relatively few other economists of the period,[24] even though they contained more information than the conventional algebraic reports.

Graphs were a particularly important element in the analysis of time-series data and business cycles; but it was commonplace throughout the econometric work of the period to give the data used

[24] Two notable exceptions who did use Tinbergen-type graphs were Stone (1954) in his work on consumers' expenditure and the US macroeconometric model by L. R. Klein and Goldberger (1955).

either in graph or table form (or both) and to use graphs to help explain methods and to report results in conjunction with algebraic methods and representations. Although graphs have continued as an important element in the *statistical* programme, algebraic representation of both methods and results had come to dominate the *econometrics* programme by the 1950s.

CHAPTER 3

Random shocks enter the business cycle scene

The introduction to this book suggested that econometrics developed as a substitute for the experimental method in economics, and that the problems which arose were connected with the fact that most economic data had been generated in circumstances which were neither controlled nor repeatable. By using statistical methods, econometricians also obtained access to an experimental tradition in statistics. This tradition consisted of generating data artificially, under theoretically known and controlled conditions, to provide a standard for comparison with empirical data or to investigate the behaviour of data processes under certain conditions. Such experiments now form a significant part of the work in econometric theory (under the title Monte Carlo experiments) and are sometimes used in applied work (model simulations), but their use dates from the early years of econometrics. Experiments played a particularly important role in the work of the 1920s which is discussed here.

This chapter deals with technical issues of data analysis and associated explanations of the generation of economic cycle data as investigated by Yule, Slutsky and Frisch. The account begins with the work of Eugen Slutsky and George Udny Yule in the 1920s both of whom made considerable use of statistical experiments.[1] Yule criticised the methods of analysing time-series data described in the previous two chapters. Slutsky explored the role of random shocks in generating cyclical patterns of data. At the same time (the late 1920s) Ragnar Frisch was experimenting with his own method for analysing economic time-series data. Although his method turned out to be a dead end as an analytical tool, Frisch used his understanding of economic time-series data, together with the suggestions of Yule and Slutsky, in his 1933 design for macrodynamic models. So it appears that the Yule and Slutsky

[1] Their work was also important in the development of theoretical work on time series. Davis (1941) has traced these developments through to Wold's (1938) analysis of stationary time series.

critique, which at first had seemed entirely negative towards the analysis of economic data, was transformed into an important element of econometric work on the business cycle.

At the same time as the statistical genesis of business cycle data was being investigated in the late 1920s and early 1930s, there were parallel developments in the mathematical formulations of dynamic theories of the business cycle. Frisch's 1933 model is the most important of these mathematical models from the point of view of the history of econometrics because he was the only economist to combine the insights of the statistical analysis with a mathematically formulated dynamics in his cycle model.

3.1 The experiments of Yule and Slutsky

3.1.1 Yule, measurement error and the pendulum and peas model

Since Hooker's paper of 1901 on the relationship between the marriage rate and trade, it had been recognised by contemporary researchers that correlations of time-series variables had to be carried out with some care. Hooker found different correlations between the oscillations of the variables and between their trends: over the short term, the marriage rate was positively correlated with the trade cycle, but over the long term the trend relationship was negative. Various methods were suggested to get at the 'true' correlation, including the variate difference method: taking differences between successive observations. George Udny Yule's 1921 survey of the problem of 'time-correlations' found that the field was full of confusions; time-series correlations were not well understood and there was disagreement about exactly what problem the variate difference method was supposed to solve.[2]

In his 1926 paper, 'Why Do we Sometimes Get Nonsense Correlations between Time-Series?', Yule set about his own analysis of the problem and discovered that correlations between two sets of data in which the observations were time related were likely to be very misleading, in fact, they would generally be biased upwards. The high correlations often found in time-series work were, according to Yule, not 'spurious' but 'nonsense' correlations. His critique naturally threw suspicion on the findings of business cycle analysts whose observations were related through time and who relied on the evidence of high

[2] George Udny Yule (1871–1951) trained as an engineer and then learnt statistics from Karl Pearson. Yule's contributions to the field of mathemataical statistics were many and varied and his statistics textbooks (1911) (later editions by G. U. Yule and M. G. Kendall, then by M. G. Kendall and A. Stuart) has been used by many econometricians. (See S. M. Stigler (1986) for a recent account of his place in the history of statistics.)

correlation coefficients to establish relationships between variables. For contemporaries, Yule's article was yet another nail in the coffin of the business barometers and time-series methods discussed in Chapter 2.

Yule defined the problem as follows:

> It is fairly familiar knowledge that we sometimes obtain between quantities varying with the time (time-variables) quite high correlations to which we cannot attach any physical significance whatever, although under the ordinary test correlation would be held to be certainly 'significant'. As the occurrence of such 'nonsense correlations' makes one mistrust the serious arguments that are sometimes put forward on the basis of correlations between times series . . . it is important to clear up the problem how they arise and in what special cases.
>
> (Yule (1926), reprinted in (1971), p. 326)

Yule rejected the usual 'spurious correlation' argument: that an unexpectedly high correlation found between two variables was due to the influence of the 'time factor', taken as a proxy for some other variable or variables, which indirectly caused the two variables to move together. He had been the originator of this 'spurious correlation' notion in 1895, but here he favoured a more technical explanation of the problem:

> But what one feels about such a correlation is, not that it must be interpreted in terms of some very indirect catena of causation, but that it has no meaning at all; that in non-technical terms it is simply a fluke, and if we had or could have experience of the two variables over a very much longer period of time we would not find any appreciable correlation between them. But to argue like this is, in technical terms, to imply that the observed correlation is only a fluctuation of sampling, whatever the ordinary formula for the standard error may seem to imply: we are arguing that the result given by the ordinary formula is not merely wrong, but very badly wrong. (Yule (1926), reprinted in (1971), p. 328)

On considering the assumptions underlying the calculation of the standard error formula for the correlation coefficient, Yule decided that there were two of these which did not hold good in economic and social time-series data. The two conditions breached were, first, that each observation of the sample was equally likely to be drawn from any part of the aggregate population (when in fact successive observations were drawn from successive parts of the aggregate), and, second, that each observation in the sample was independent of the observation drawn either before or after it.[3] Since social and economic time-series data did

[3] Persons had also by this stage realised that the standard error formula did not apply, but he did not go on, as Yule did, to analyse how far wrong the normal correlation coefficients would be. Instead, as we shall see in Chapter 8, he believed that since the data did not behave according to the assumptions of probability laws, then probability methods could not be applied to business cycle data.

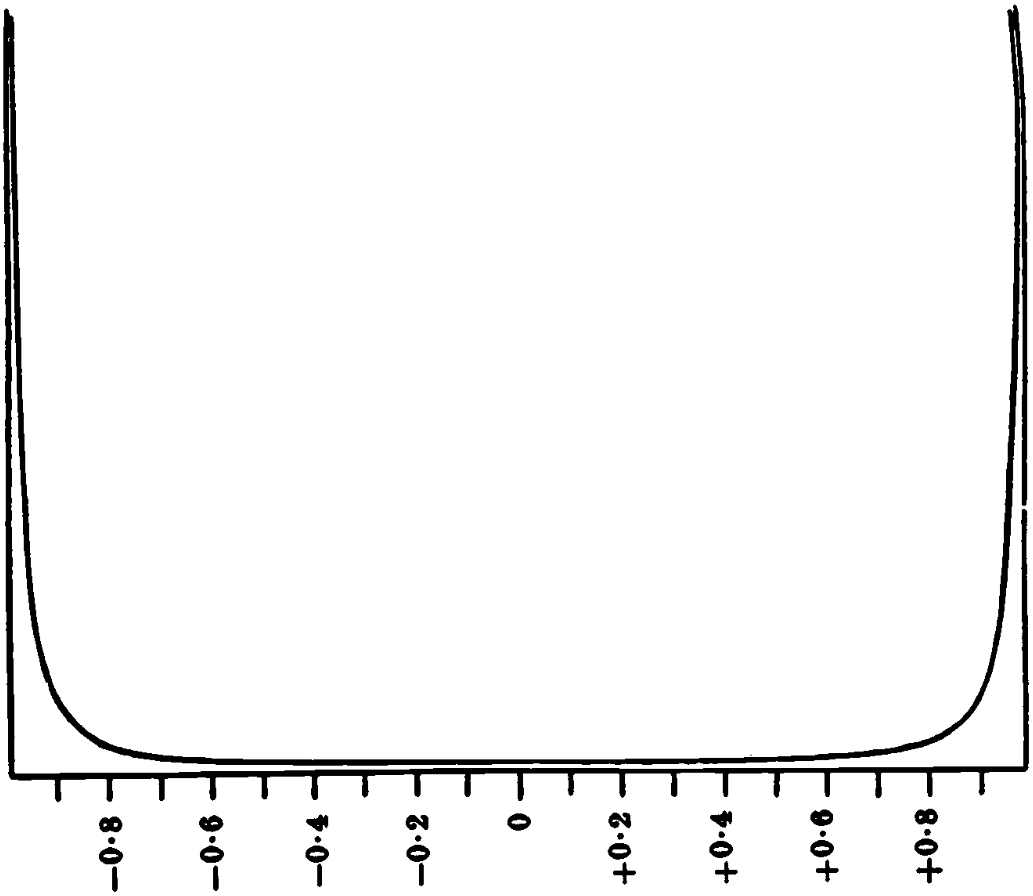
not constitute independent series of observations but series in which successive terms were related, Yule argued that

> the usual conceptions to which we are accustomed fail totally and entirely to apply. (Yule (1926), reprinted in (1971), p. 330)

Yule investigated how such time-related series of observations gave spuriously high correlations by considering two sine waves, differing by a quarter period. These two series have a correlation of zero over their whole length; however, over any short period of time, the correlation between the two series will be either +1 or −1. In fact as the centre of a short observation period moves along the curves, the correlation coefficient between the two series switches between +1 and −1. This gives a U-shaped frequency distribution of the correlation coefficients obtained from simultaneous observations on the two curves, always giving values furthest away from the true value (zero) as the most frequent. One of Yule's frequency distributions is reproduced here (Figure 6).

Yule analysed what sorts of series would produce the nonsense correlations found in his experiments on harmonic curves. His results were partly based on experimental work on artificial series (a random number series, and series derived from that), and partly on existing data series. He found that he could predict which series would give nonsense correlations on the basis of their serial correlation characteristics. Samples of observations from two positively serially correlated series (a 'conjunct' series), which had random differences, would produce a higher standard error for the correlation coefficient than that obtained for a random series. The distribution of correlation coefficients, though not definitely misleading, was far from bell-shaped and might possibly be bi-modal. On the other hand, two positively serially correlated series with positively serially correlated differences (a conjunct series with conjunct differences) would produce a U-shaped distribution or correlation coefficients like those of the harmonic curves and would therefore lead to nonsense results.

Yule found it more difficult to reach any conclusions about the case of oscillatory series: series where the serial correlation changes sign often but which are not well-behaved sine waves. (He used Beveridge's series of wheat prices as an example of this common type of series.) Given these difficulties, Yule carefully cautioned against the comparison of artificial series (possessing known or engineered characteristics) with real series of unknown characteristics:



Frequency-distribution of correlations between simultaneous elements of the harmonic curves.

Figure 6 Yule's graph of the distribution of correlation coefficients for time series
Source: Yule (1926), reprinted in (1971), p. 334

it is quite possible that what looked a good match to the eye would not seem at all a good match when subjected to strict analysis.
(Yule (1926), reprinted in (1971), p. 350)

In 1927, the year following his analysis of the problem of nonsense correlations, Yule published an investigation of the frequency analysis of sunspot data, 'On a Method of Investigating Periodicities in Disturbed Series'. By the obvious association of sunspots with business cycles (following Jevons' work), this paper also dealt indirectly with the use of frequency analysis to decompose economic time series. The focus of the paper was on the role played by errors when combined in different ways with an harmonic process. Yule believed that frequency analysis of cyclical data usually began with the hypothesis that the periodic function was masked solely by added-on errors. He showed that if small measurement errors were added to an harmonic process then the resulting data series would look less regular, but the wave pattern would still be clear and frequency analysis of the series would always recover the underlying process. If such errors were large the graph would appear very irregular and jagged and the harmonic process might not be clearly visible to the eye but, Yule said, it could always be recovered.

Yule argued that there might also be other sorts of errors:

> If we observe at short intervals of time the departures of a simple harmonic pendulum from its position of rest, errors of observation will cause superposed fluctuations ... But by improvement of apparatus and automatic methods of recording, let us say, errors of observation are practically eliminated. The recording apparatus is left to itself, and unfortunately boys get into the room and start pelting the pendulum with peas, sometimes from one side and sometimes from the other. The motion is now affected, not by *superposed fluctuations* but by true *disturbances*, and the effect on the graph will be of an entirely different kind. The graph will remain surprisingly smooth, but amplitude and phase will vary continually. (Yule (1927), reprinted in (1971), p. 390)

Yule found by experiment that true disturbances which feed into the process itself, and whose effects are carried through to successive time periods, result in a data picture with smooth cycles which looked very similar to that produced by sunspots. These results may seem to be counterintuitive since superimposed (measurement) errors gave Yule a jagged data picture which looked less like an harmonic function than the smooth picture produced by disturbances.[4]

The obvious question was whether the underlying harmonic function could be revealed in the case of true disturbances? Using ordinary harmonic analysis led to results which were liable to error and to be misleading. Yule found that the best way to replicate such data, that is the best model to use to describe the data, was the regression of a linear difference equation:

$$X_t = B_1 X_{t-1} - B_2 X_{t-2}$$

He fitted this regression equation to his experimentally obtained disturbed harmonic process, and the sunspot data, and found that both difference equations had as their solution a heavily damped harmonic

[4] A physical experiment with a galvonometer was carried out in the 1930s by Davis at the Cowles Commission to replicate Yule's pendulum being hit by peas (see Davis (1941)). The oscillations were set to match the periodicities observed in the Dow Jones Industrial Average index and then subjected to shocks. This produced very irregular data, closer in appearance to economic cycle data than the smooth picture of Yule's demonstration.

movement. Yule claimed that the role of the disturbances 'deduced' from the fitted sunspot regression equation (the residuals) was to maintain the amplitude of the harmonic period in the observed data:

> The distribution of the disturbances seems to me to have some bearing on the question whether we may perhaps, tentatively regard the damped harmonic formula at which we have empirically arrived as being something more than merely empirical, and representing some physical reality. As it seems to me, the disturbances do occur just in the kind of way that would be necessary to maintain a damped vibration, and this suggests that broadly the conception fits the facts.
>
> (Yule (1927), reprinted in (1971), p. 408)

Yule's conclusions suggested that any variable subject to external circumstances would be affected by disturbances:

> many series which have been or might be subjected to periodogram analysis may be subject to 'disturbance' in the sense in which the term is here used, and that this may possibly be the source of some rather odd results which have been reached. Disturbance will always arise if the value of the variable is affected by external circumstance and the oscillatory variation with time is wholly or partly self-determined, owing to the value of the variable at any one time being a function of the immediately preceding values. (Yule (1927), reprinted in (1971), p. 417)

Since economic variables were undoubtedly subject to external circumstances and were probably self-determined in Yule's sense, economic time-series data probably contained disturbances of the sort Yule had in mind. We have already noted Yule's paper in connection with the decline in periodic business cycle analysis which followed Moore's work: if disturbances were present, the use of harmonic analysis would be an inappropriate way of finding the underlying economic cycles. The implication was that harmonic analysis, like correlation analysis, seemed a dangerous tool for econometricians to use. But, although Yule had alerted economists to another danger, he had also proposed a solution to the problem in the sense that he had found a model (the linear difference regression model) to describe the vagaries exhibited by such disturbed data.

3.1.2 Slutsky's random waves

In the same year (1927), the Russian economist Eugen Slutsky suggested a more deeply worrying possibility than the presence of measurement errors or disturbances in economic data.[5] His idea was

[5] E. E. Slutsky (1880–1948), a Russian economist of considerable reputation in the field of economic theory, but whose publication record was more heavily weighted towards

that cycles could be caused entirely by the cumulation of random events. The article, 'The Summation of Random Causes as a Source of Cyclic Processes', was published in the theoretical journal of the Moscow Institute for Business Cycle Research in 1927 (with a summary in English). Though it appeared in an unusual source, Slutsky's paper was immediately widely reported in the West via academic books and journals.[6]

Slutsky was responsible in this paper for one of those crucial experiments much beloved by historians and philosophers of science. Crucial experiments are usually taken to mean decisive experiments but I prefer to follow Hacking's (1983) suggestion and use this term in the sense of a signposting experiment which points out the ways. Slutsky did not indulge in repetition of his experiment. There was no need, for he had made his point and he preferred to use his resources to carry out several different experiments. As already noted, these sorts of experiments in statistical theory rely on artificially generated data; the outcome of such an experiment is an insight into what might be the process generating some real data series. I will return to this point following a discussion of Slutsky's work and his results.

Slutsky wanted to know whether the combination of random causes would be sufficient to generate regular cycles. He posed the question as follows:

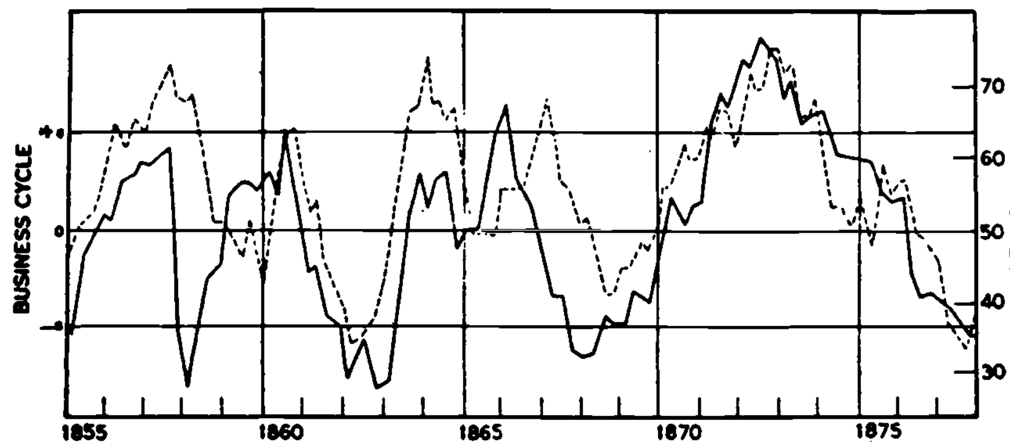
> is it possible that a definite structure of a connection between random fluctuations could form them into a system of more or less regular waves? Many laws of physics and biology are based on chance, among them such laws as the second law of thermodynamics and Mendel's laws. But heretofore we have known how regularities could be derived from a chaos of disconnected elements because of the very disconnectedness. In our case we wish to consider the rise of regularity from series of chaotically-random elements because of certain connections imposed upon them.
>
> (Slutsky (1937), p. 106 (quoting from the 1937 English version))

His method was to be experimental:

> Generally speaking the theory of chance waves is almost entirely a matter of the future. For the sake of this future theory one cannot be too lavish with experiments: it is experiment that shows us totally unexpected facts,

statistical work of an experimental and theoretical kind on stochastic processes and correlation. He worked on time-series problems at the Moscow business cycle institute in the 1920s and later used sunspot data as the raw material for some of his work (see the entry in the *International Encyclopedia of the Social Sciences* and Allen (1950)).

6 Slutsky's 1927 article was fully reprinted in English in 1937 at the instigation of Schultz and Frisch. The first five sections of the 1937 article were similar (but with some revisions) to the original 1927 article; the latter part of the 1937 paper draws on results he published in



———An index of English business cycles from 1855 to 1877; scale on the left side. ------Terms 20 to 145 of Model I; scale on the right side.

Figure 7 Slutsky's random series juxtaposed with business cycles
Source: Slutsky (1927), p. 38, Chart 3; reprinted in (1937), as Figure 3, p. 110

> thus pointing out problems which otherwise would hardly fall within the field of the investigator. (Slutsky (1937), p. 107)

and his raw material consisted of a data series of numbers drawn in the People's Commissariat of Finance lottery, which he took to be a random series (that is, a series with no serial correlation).

Slutsky then generated different models of empirical data processes using the basic data series, and defining a model in the following words:

> Any concrete instance of an experimentally obtained chance series we shall regard as a model of empirical processes which are structurally similar to it. (Slutsky (1937), p. 108)

The first model was based on a cumulative process – each new observation (X_t') was based on a simple 10–item moving summation of the basic random number series (X_t):

$$X_t' = \sum_{i=1}^{10} X_{t-i}$$

He plotted a section of the data of this first model next to a chosen section of an index of business cycles in England, reproduced in Figure 7. A close similarity between the two graphs was readily apparent. Further experiments, using models with different weighting patterns in the summation, reinforced Slutsky's first results and he claimed

Russian, French and Italian journals in the intervening period. This activity suggests that he avoided the miserable fate of some of his colleagues at the Moscow Institute (see Chapter 2.4).

> an inductive proof of our first thesis, namely that the summation of random causes may be the source of cyclic, or undulatory processes.
> (Slutsky (1937), p. 114)

The rest of his article was concerned with the regularity of the cycles involved and with a frequency analysis of the experimental series. This led him to make the following generalisation:

> The summation of random causes generates a cyclical series which tends to imitate for a number of cycles a harmonic series of a relatively small number of sine curves. After a more or less considerable number of periods every regime becomes disarranged, the transition to another regime occurring sometimes rather gradually, sometimes more or less abruptly, around certain critical points. (Slutsky (1937), p. 123)

That is, random terms can generate data series which appear to consist of a number of sine waves and are thus amenable to frequency analysis; whereas in fact the periodic components are not well behaved but irregular.

Slutsky's juxtaposition of the random cumulated series with the business cycle data was not supposed to be a proof that random events had caused the business cycle index shown. Nor was any such claim made by Slutsky. He neither established that business cycle data were generated by random events, nor that they were not generated in some alternative way. The experiment was an artificial 'as if' experiment, rather than real experiment. The inference was therefore more limited: the graph merely suggested that such a data generation process could give very similar data to that produced by economic activity.

Slutsky was not responsible for the idea that random events have a role in business cycles, for it was a long-held view that accidental or outside disturbances formed the immediate impetus behind economic crises or turning points. No statistical model of how this occurred had been advanced prior to the work of Yule and Slutsky. Yule's model (the pendulum being pelted with peas) suggested a cyclical mechanism, whose oscillations were maintained by random shocks. Slutsky's hypothesis went further by giving random causes sole responsibility for the complete cyclical movements in economic activity, rather than for just the turning points or the maintenance of oscillations.

It seems that Slutsky's insight into how random events might cause business cycles captured the imagination more than the partial role for disturbances suggested by Yule. Slutsky's idea was rapidly reported and discussed in business cycle circles. It was discussed briefly by Mitchell in his influential 1927 book. Then Holbrook Working (1928), in reviewing Mitchell's book, seized on this brief report of Slutsky's work and suggested that, although he was not prepared to accept it as a complete explanation, there might be considerable elements of truth in the idea. In 1929 Kuznets discussed Slutsky's work at length and reproduced the crucial graph (Figure 7) in the context of carrying out some of his own experiments on time-series data. He wrote of Slutsky's theory:

> If cycles arise from random events, assuming the summation of the latter, then we obviously do not need the hypothesis of an independent regularly recurring cause which is deemed necessary by some theorists of business cycles. Indeed, if one can explain how in certain processes of economic life, the response to stimuli is cumulative, then the whole discussion of the cause of business cycles becomes supererogation. (Kuznets (1929), p. 274)

As Kuznets pointed out, Slutsky's work not only removed the necessity for periodic cause of economic cycles (an idea which had been persistently followed by Jevons and Moore, but which had made most economists feel distinctly uneasy), but might also, if the cumulative mechanism were understood, make any further discussion of the cause of business cycles superfluous.

Later in 1939, with the benefit of some hindsight, Schumpeter gave a different interpretation of the signposts which pointed from Slutsky's experiment:

> that proof did two things for us: first, it removed the argument that, since our series display obvious regularities, therefore their behaviour cannot result from the impact of random causes; second, it opened an avenue to an important part of the economic mechanism, which has since been explored by R. Frisch in a powerful piece of work.
> (Schumpeter (1939), p. 181)

This interpretation recognised both the possibility of random causes (by rejecting the notion of their impossibility) and the possibility of a new sort of business cycle mechanism. Schumpeter referred here to Frisch's 1933 business cycle model (discussed later in this chapter), but the avenue of thought which Frisch was actually exploring in the late 1920s was the composition of economic time-series data.

3.2 Frisch's time-series analysis

Ragnar Frisch was one of the leading practitioners of econometrics in the 1920s and 1930s.[7] His leadership of the econometric movement was exercised through his personal style whilst Editor of *Econometrica* (from

[7] Ragnar Frisch (1895–1973), one of the founders of the Econometric Society, was a very influential figure in the development of econometrics (see, for example, Tinbergen's (1974) sketch of his role). His first degree was in economics and his doctorate was in mathematical

1933 to 1954), at meetings of the Econometric Society and through his writing and teaching. He was a prolific writer: a number of his articles quickly became classics; but some of his econometric papers, particularly those on econometric methods and ideas, are as hard to comprehend today as they were for his contemporaries. Despite his dominant role, Frisch's work on time-series analysis has been largely forgotten. It is important to understand something of this work because his analysis helped him to understand economic time-series data and thus to produce an innovative model of the business cycle.

In the middle 1920s Frisch, dissatisfied with the available methods, decided to invent his own method of time-series analysis.[8] He wanted to develop a totally objective method of analysing time-series data which was more flexible yet more rigorous than the methods in current usage. Despite a lengthy and massive research effort, his results were reported in just three papers in 1927 (unpublished), 1928 and 1931.[9] This description of his method is based on all three.

According to Frisch, the problem of existing time-series decomposition methods (of the sort used by Persons, for example) was that different methods were used to isolate each component, and there was

> no logical relation between the various methods ... no general principle from which these various methods may be derived.[10]
>
> (Frisch (1927), p. 4)

Frisch thought that if all the usual components (trends, cycles, seasonal and erratic variations) were regarded as cycles, then trends were merely part of a long cycle and erratic elements were short cycles which

statistics. He spent his career at the University of Oslo (though he travelled widely) and his interests ranged over the whole field of economics. J. C. Andvig (1985) gives a comprehensive account of Frisch's work in building macroeconomic models which parallels his work described here in econometrics. (See also Andvig (1978) and (1981).)

8 It is not clear exactly when Frisch started to work on time-series problems for he gave no hint of the impetus behind it. But this was not unusual with Frisch; he often did not deign to fit his own work into the literature or to say specifically who he was arguing against. Davis, who worked with Frisch on these problems in the 1930s at the Cowles Commission, stated that it was Yule's (1927) ideas on disturbed harmonic processes and his analogy of the pendulum and peas which first stimulated Frisch into working on the decomposition problem. Andvig maintains from his study of Frisch's papers that the work started earlier in 1925.

9 Frisch produced a mimeographed paper (circulated privately with the help of Mitchell) on his time-series methods in 1927. The two published articles are (1928) and (1931). Andvig (1985) also reports a mimeographed paper in Norwegian.

10 Wald (1936) agreed with this criticism in an interesting critique of the decomposition method, undertaken at the request of Morgenstern, head of the Vienna business cycle research institute (see Chapter 8 n. 5). Wald suggested that since there were no generally consistent and complete ways of defining the components of economic data series according to the outside groups of causal forces, a more fruitful approach was to do a complete frequency analysis of the components of a series and then identify those internal components found with groups of external causes. Wald's work on the subject was and is

appear jagged because the data were not available in short enough time units. Time series could then be regarded as a suite of cycles in which only the middle waves can be traced. Frisch called each component a 'trend' denoting a changing, probably cyclical, form. (This is a good example of Frisch's confusing habit of changing the meaning of existing terminology.) A low order 'trend' (such as seasonal variation) involved a short period length and a high order 'trend' was one of long length (for example, a Kondratieff long wave). This, he claimed, enabled a unified approach to all the components of a time series, instead of the mixed approach adopted by Persons.

Frisch also proposed to abandon the standard periodogram analysis which involved the assumption of constant and fixed periodicities because he believed economic cycles were variable (both in length and shape) rather than constant phenomena. He proposed instead a 'moving method' which would show how component cycles ('trends') evolved and trace these changing cycles in the historical data series. His analysis was concerned with local properties rather than the total properties of the whole series. That is, Frisch wanted each component of the time series at any given point of time to be determined from the data in the vicinity of that point rather than by the course of the time series in all years. He described his method as a 'principle of moving fit' or 'curvefitting without parameters' (Frisch (1928), p. 230).

In his first paper on the subject in 1927, Frisch began by assuming that a time series W is a complicated function made up of many different 'trends', Y_i, each of which are assumed to be changing sine waves (i.e., waves of changing period length):

$$W = Y_0 + Y_1 + Y_2 + \ldots + Y_n$$

He claimed that these component Ys could be unravelled relatively easily if the order of each 'trend' were very different from the others. That is, provided the period of one 'trend' component was, say, 7 to 10 times greater than that of the next 'trend', both components could be found. Frisch's method of unravelling the different 'trends' he called the 'method of normal points'. It was based on the observation that if the 'trends' were of different orders, the curvature of Y_0 would be very much greater than that of Y_1, which would be that much greater again than the curve of Y_2 and so on.[11] Figure 8 illustrates Frisch's idea (but it

little known (an extract is printed in the forthcoming volume of classic readings by Hendry and Morgan) but is cited in connection with the 1960s revival of frequency analysis in econometrics (see Morgenstern (1961)).

11 Frisch also suggested a second method, called the 'method of moving differences', to be applied when there were several component 'trends' of the same order of magnitude in each series. The general approach was the same but it involved the higher derivatives of W,

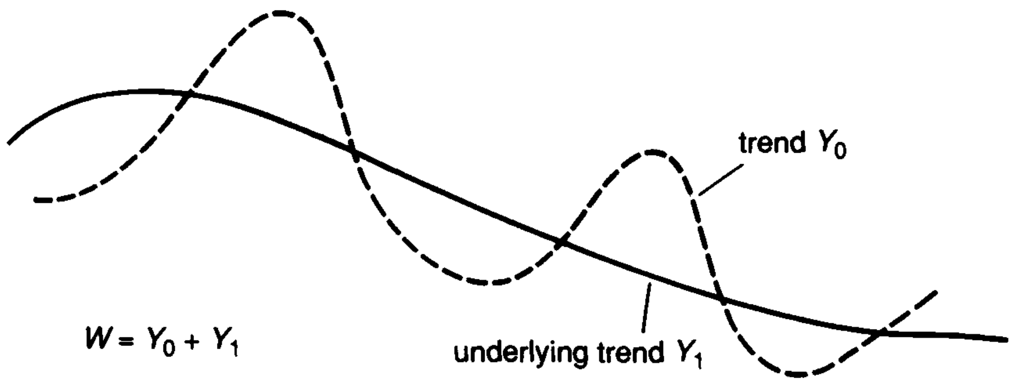


Figure 8 Frisch's time-series method – graph 1

is not an illustration from Frisch, for he used none): a time series W, consisting of two components, in which the slope of the shorter period component, Y_0, is steeper in general than that of the longer period component, Y_1.

Frisch argued that if you took the second derivatives of the data series that is, W'', the dominant element in it would be Y_0'' except at the 'normal point' defined as the point where Y_0'' vanishes. At a slight displacement from its normal point, however, Y_0'' would continue to dominate, since at the normal point, the first derivative Y_0' is at its maximum (or minimum). In practice, if the period of Y_1 is several times greater than that of Y_0, then the dominance of Y_0 is such that the normal points of the lowest order (smallest period) 'trend' Y_0 would be found where W'' is zero. These normal points could then be used as observations in a new time series W_1 where the 'trend 'of lowest order, Y_0, had been eliminated. This is shown in Figure 9 (which is adapted from an illustration of Frisch's method given in Schumpeter (1939), p. 469), where the original series W_0 consists of three 'trends', Y_0, Y_1 and Y_2 and the new series W_1 consists of only the two longer period 'trends' Y_1 and Y_2.

Following the elimination of Y_0 from W_0, the 'trend' Y_1 could be eliminated from the new composite series W_1 to give the remaining $W_2 = Y_2$ series. The deviations between the old series W_0 and the first new series W_1 would give the 'trend' Y_0, and successive Y_1 components ('trends' of higher order) could be found in the same way.

In practice, the second derivatives could not be found because the data were discrete, so Frisch used the second differences of the data series, denoted V, as an estimate of the second derivatives W''. That is, he calculated the individual data points from:

required more reliable and regular material than the first method and was considerably more complicated.

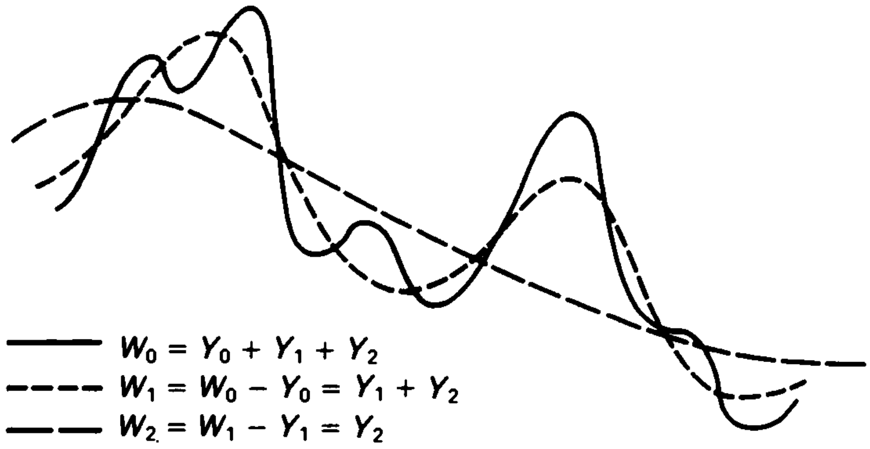


Figure 9 Frisch's time-series method – graph 2

$$\hat{W}_t'' = V_t = W_{t+1} - 2W_t + W_{t-1}$$

Plotting this second difference series revealed the estimated 'normal points' (where V_t equalled zero) providing, Frisch argued, that two conditions hold. First, the distance between the three data points used in calculating the second difference of W must be small in relation to the distance between the normal points of the lowest 'trend'; secondly, accidental errors must not dominate in the data and lead to false zeros. So, if the graph of V_t changed signs frequently within short intervals, Frisch suggested that the 'trend' was of too low an order (relative to the data frequency) to be investigated. Such a 'trend' should be treated instead as accidental components, eliminated by taking a moving average of the original series W. These problems might arise, for example, in the attempt to isolate a seasonal component using monthly data.

In his 1928 article, Frisch presented a generalised and condensed version of his method which he described as the use of linear operators to measure changing harmonics in time series. These linear operators could be derivatives, but were more likely to be difference operators or weighted moving averages of various kinds. In a further short paper in 1931, Frisch turned his attention to the Slutsky effect and experimented with different linear operators acting on random terms. He regarded Slutsky's cycles as spurious and was interested in finding a way of telling whether cycles in data were spurious (caused by the Slutsky effect: the cumulation of random terms) or not. He found, as Slutsky had done, that certain linear operators acting on random shocks would produce changing harmonic waves.

Frisch's time-series research programme continued, but despite his promises and the considerable research input into his method by research assistants at the University of Oslo, by students at Yale and at the Cowles Commission during the 1930s, none of this further work appeared. During these years, Frisch was research consultant to the Commission, advising on the general direction of research and more particularly on a project on the time-series analysis of stock market prices which involved the participation of H. T. Davis.[12] As a result of his econometric work with the Cowles Commission, Davis wrote a massive survey of time-series methods published in 1941, which contained almost the only serious contemporary discussion of Frisch's work. In his 600-odd pages, Davis devoted only one and a half pages to Frisch's method of changing harmonic analysis. This is not because Davis disagreed with Frisch's view of the nature of economic time series. On the contrary, he rejected periodogram analysis on exactly the same grounds that Frisch had done. Davis himself worked out a more straightforward way of treating moving components involving a small modification of the standard periodogram method. He carried out empirical work to demonstrate that his own method obtained results similar to those obtained by Frisch, and that both were more successful in reducing the residual variance of a series than standard harmonic methods using a small number of harmonic terms. After all Frisch's hard work on the topic, it is ironic that Davis arrived at his quicker method because of a hint from Frisch.[13]

Given his prestige, it is perhaps surprising that Frisch's theoretical work on time series has received so little attention (Davis apart) from those assessing his work.[14] Yet, as a practical method of time-series analysis, Frisch's work was a failure. Moreover, this was evident at the time, for in marked contrast to other innovations that Frisch proposed, contemporaries did not adopt his time-series methods. With hindsight, there are a number of convincing reasons why this was so. One reason was the communication difficulties generally evident in Frisch's writing

[12] Harold T. Davis (1892–1974), a mathematician, became associated with econometrics through his connection with Alfred Cowles. Davis worked with the Cowles Commission research staff particularly on mathematical problems and on the analysis of time-series data. He wrote one of the first econometrics textbooks (1941a), mainly on mathematical economics. He also published widely on pure and applied mathematical problems.

[13] Davis' hint from Frisch was acknowledged. Cargill's (1974) survey paper claims that the contribution of Davis' book on time series was the treatment of changing cyclical components, but he ignores Frisch's work.

[14] Schumpeter (1939) is the other contemporary exception to this rule. Recent assessments by, for example, Arrow (1960) and Edvardsen (1970) also ignore this work. Andvig (1985) is, of course, comprehensive in this respect.

on statistical methods. Frisch usually presented his work in this field as if it were a brand new approach to a problem and invented his own new terminology. In this case for example, he confused the reader by using the term 'trend' for all the components of the data. He was particularly interested in the problems of computation and sometimes invented new algorithms for calculating terms, but, as here, he failed to give proper examples illustrating his new methods. He rarely rewrote his results in order to make them comprehensible, preferring instead to move on to a new problem. He often failed to relate his analytical techniques to economic problems so their relevance was not always obvious. Because of these characteristics, much of Frisch's written work in this area is therefore very difficult to understand and his analysis of economic time series illustrates these communication problems vividly.[15]

A more potent reason for the failure of Frisch's time-series method to make much impact on econometrics was that his methods were not particularly original, although Frisch's written style obscured this fact. Frisch's work appeared novel because his analysis was based on a continually changing series of components. But the techniques he used, linear operators of various sorts, were also used in different guises by Persons and in the variate difference approach and, most important, they suffered from the same problems as other business cycle methods of the 1920s. Specifically, despite all his mathematics, Frisch had failed in his desire to invent a totally objective method; in practice, some judgement was needed. Frisch's method therefore possessed no real advantages over Persons' decomposition approach.

Further, as Davis pointed out, any method of treating changing harmonics, either his own moving periodogram method or Frisch's method of linear operators, raised questions about degrees of freedom and significance of the results. The ideal representation of data should account for as much variation in the data with as few parameters or functions as possible. Any method which fitted each segment of data (or each observation) individually, as in Frisch's method or in fitting a polynomial of order nearly equal to the number of data points, is not a very efficient representation of the data. Frisch's method accounted for more variation than the standard method, but just how many degrees of

[15] They are also evident in another research project of Frisch of the late 1920s and early 1930s. This was on confluent relationships: several relationships holding between a set of economic variables at the same time. He viewed both as problems of unravelling components from the data; in one case these were cycles and in the other relationships. His method of attacking these problems and some of the techniques he used were similar. (See Chapter 7 Letter 11 for a discussion of confluence analysis.)

freedom were left, and thus how significant were his results, would be difficult to determine.

A further comparison with another contemporary approach is pertinent here. In Frisch's method, each cycle of each component 'trend' was treated as an individual event. His method therefore suffered from the same difficulties as beset Mitchell's analysis, namely the inability to support general explanations of economic data or explanations based on constant relationships. These problems were the inevitable result of theory-free data analysis applied at the level of the cycle. It is the cycle unit which is the real problem here, not the theory-free analysis. Remember, Yule had analysed the problem of changing components in terms of a stochastic process which generated the data: this enabled theory-free analysis, but allowed for a generalised model of the process. But, unlike Yule's analysis or Slutsky's work, Frisch's unravelling of time-series data produced no general model of how time-series data were generated.

In his 1931 paper, Frisch had defined the general problems in the analysis of economic time-series data. The first was the question of the composition of economic time series. Frisch's programme to solve this problem and

> find out on more or less empirical grounds what is actually present in the series at hand, that is to say, what sort of cømponents the series contains (Frisch (1931), p. 74)

has been described in this section. He believed he had succeeded in this task and although he failed in his attempts to revolutionise the field of economic time-series analysis, he had learnt a considerable amount about the behaviour of economic time-series data from his investigations. The crucial issue which remained unexplained was how the time-series components, which Frisch had spent so much energy trying to unravel, came to be combined together in the first place. His solution to this puzzle is discussed next.

3.3 Frisch's rocking horse model of the business cycle

In marked contrast to his work on time series, Frisch's 1933 paper on business cycle models presented few communication problems. It was an ambitious paper but was highly successful and quickly became a classic in its field, for it both told economists how to do macrodynamics and showed what an econometric model of the complete cycle should look like. As we shall see later, Frisch's small theoretical model provided the design for Tinbergen's full-scale applied macroeconometrics in the later 1930s.

The advent of mathematically formulated, dynamic models of the business cycle was, as I said earlier, a parallel story to the development of statistical work on business cycle data. The focus of endeavour was to produce an economic theory to explain business cycles and, particularly by the early 1930s, to explain the Great Depression. This required a dynamic model of the whole economy, a task in which econometricians (as mathematical economists) played a very important role. During the 1920s econometricians had explored the cycles that occurred in prices and outputs of agricultural goods and modelled these as systems of lagged relationships. Frisch developed these ideas further in the business cycle field to form a model of the cycle in its entirety. But he was not the only economist to do so in the late 1920s and early 1930s. As Tinbergen's 1935 survey of the field showed, apart from the cycle models of other econometricians such as Vinci, Roos and Tinbergen himself, there was also the work of Kalecki.

The development of small mathematical dynamic macromodels was very important for the history of econometrics. But the unique importance of Frisch's paper lies in another aspect of his model, namely the integration of random shocks into the cycle model as an essential part of that model. In this chaper, I shall concentrate on those aspects of Frisch's model design which are of particular interest from the econometric point of view. An analysis of the macroeconomics of the model are contained in the definitive study of Frisch's work during the interwar period by Andvig (1985) (see also Andvig (1978)).

Frisch's paper, entitled 'Propagation Problems and Impulse Problems in Dynamic Economics', was concerned with the following problem: what should a model look like which accounts not only for cycles in economic variables but also does so in a way that can be reconciled with observed economic data? As we have seen from the reactions to Moore's cycle work, business cycle theories which specified regular forced oscillations (exogenous theories such as the sunspot theory) were thought by reviewers to be unsatisfactory since observed cycles were clearly not of exactly regular length and appearance. At the same time, although economists liked theories which suggested that a cycle was generated from inside the economic system (endogenous theories such as changes in credit availablity) and which involved a tendency to return to equilibrium, such theories failed to explain why it was that in observed data, the cycles in economic activity were maintained.

By the time Frisch wrote his paper in 1933, there had been two statistical demonstrations of how outside disturbing forces could generate the type of time-series data seen in economic variables. Yule had shown, in his 1927 investigation of sunspots, how the presence of shocks hitting an harmonic process would give rise to an irregular harmonic series. Slutsky's experiment of 1927 had shown how in the extreme case, the summation of a random series on its own would result in data series of the business cycle type. Frisch took note of these examples in his own attempts to postulate a mechanism that would generate the sort of economic time-series data which were observed.[16] Since Frisch himself believed that observed business cycle data consisted of a suite of changing cyclical components, in effect he sought an explanation of how these components were combined to form the economic time series.

Frisch suggested that the original idea of his model was due not to Slutsky or Yule but to Wicksell:

> Knut Wicksell seems to be the first who has been definitely aware of the two types of problems in economic cycle analysis - the propagation problem and impulse problem - and also the first who has formulated explicitly the theory that the source of energy which maintains the economic cycles are erratic shocks. He conceived more or less definitely of the economic system as being pushed along irregularly, jerkingly . . . these irregular jerks may cause more or less regular cyclical movements. He illustrates it by one of those perfectly simple and yet profound illustrations: 'If you hit a wooden rocking-horse with a club, the movement of the horse will be very different to that of the club.'
>
> (Frisch (1933), p. 198)

This idea, termed here the 'rocking horse theory' of the business cycle (but usually known more prosaically as the Cassel paper, after the volume in which it appeared), provided a general explanation of the generation of economic time-series data. It had appeared in other guises in cycle analysis, for example, in Jevons' description of the economy as a vibrating ship being hit by waves and in Yule's analogy of boys pelting a pendulum with peas, but the rocking horse analogy, through Frisch's work, proved the most influential.

The particular attraction of Frisch's rocking horse model was that it allowed for the free and damped oscillations desired by economic theoreticians (the normal movement of the rocking horse which would gradually return the horse to a position of rest if left to itself) and yet was compatible with observed business cycle data which were irregular and undamped (the rocking horse movement disturbed by the random blows). As Frisch observed at the start of his paper:

[16] The influence of Yule and Slutsky on Frisch is clear in this case for he took unusual care to reference their work in his (1933) paper and later (1939) stated explicitly that one of the aims of his (1933) paper was to explain how Slutsky's 'shocks' came to be summed by the economic system.

> The most important feature of the free oscillations is that the length of the cycles and the tendency towards dampening are determined by the intrinsic structure of the swinging system, while the intensity (the amplitude) of the fluctuations is determined primarily by the exterior impulse. An important consequence of this is that a more or less regular fluctuation may be produced by a cause which operates irregularly. There need not be any synchronism between the initiating force or forces and the movement of the swinging system. This fact has frequently been overlooked in economic cycle analysis.
>
> If a cyclical variation is analysed from the point of view of a free oscillation, we have to distinguish between two fundamental problems: first, the *propagation* problem; second, the *impulse* problem.
>
> (Frisch (1933), p. 171)

Frisch first of all turned his attention to the propagation problem and the question of dynamic economic models of the economic system:

> The propagation problem is the problem of explaining by the structural properties of the swinging system, what the character of the swings would be in case the system was started in some initial situation. This must be done by an essentially dynamic theory, that is to say, by a theory that explains how one situation grows out of the foregoing. In this type of analysis we consider not only a set of magnitudes in a given point of time and study the interrelations between them, but we consider the magnitudes of certain variables in different points of time, and we introduce certain equations which embrace at the same time several of these magnitudes belonging to different instants. This is the essential characteristic of a dynamic theory. (Frisch (1933), pp. 171–2)

Frisch distinguished between the microdynamic models which had been developed of particular markets and his own macrodynamic analysis. Yet it is clear that his macromodelling was based on contemporary micro-models such as the cobweb model: a two-equation demand and lagged supply model first used by Moore in 1925 (and discussed in Part II). Although his model was to be of the whole economy, Frisch stressed that it had to be a simplified model otherwise it would not be possible to study

> the *exact time shape* of the solutions, the question of whether one group of phenomena is lagging behind or leading before another group, the question of whether one part of the system will oscillate with higher amplitudes than another part and so on. But these latter problems are just the essential problems in business cycle analysis. (Frisch (1933), p. 173)

So he set up a small macroeconomic model, formulated as a determinate system of mixed differential and difference equations, and showed how it could give rise to oscillations.

Because of his earlier research on the nature of economic time series Frisch naturally wanted to investigate

> whether the system is satisfied if each of the variables is assumed to be made up of a number of *components*, each component being either an exponential or a damped oscillation. (Frisch (1933), p. 183)

To do this, he first assumed that each of his three main economic variables ('consumption', 'capital starting' and 'carry-on-activity') were composed of a number of component cycles or waves. The period and form of these component cycles were dependent in turn on the structural parameters of the system that he had set up. In order to study the nature of the time-series solutions to these equations (between the structural parameters and the harmonic terms) he inserted his own guessed values for the structural parameters. These guesses he described as:

> numerical values that may in a rough way express the magnitudes which we would expect to find in actual economic life. At present I am only guessing very roughly at these parameters, but I believe that it will be possible by appropriate statistical methods to obtain more exact information about them. I think, indeed, that the statistical determination of such structural parameters will be one of the main objectives of the economic cycle analysis of the future. If we ask for a real *explanation* of the movements, this type of work seems to be the indispensable complement needed in order to co-ordinate and give a significant interpretation to the huge mass of empirical descriptive facts that have been accumulated in cycle analysis for the past ten or twenty years. (Frisch (1933), p. 185)

The mathematical solution of the equation system with the inserted guesses was not easy.[17] Frisch used a system of 'numerical and graphical approximation' which resulted in a solution for each variable, consisting of a trend plus three component cycles. These cycles were a primary cycle of 8.57 years (the business cycle), a secondary cycle of 3.5 years (the short 42–month business cycle) and a tertiary cycle of 2.2 years. All of these were heavily damped. The first two cycle lengths were close to those found in actual business cycle data for the late nineteenth and early twentieth centuries. Further, these cycle lengths were fairly insensitive to changes in all of the chosen structural parameter values, except that representing the length of time required for the production of capital goods (the technical coefficient of 'carry-on-activity'). In order to show more clearly how the structural parameters determined the time shape of the solutions, Frisch also provided a step-by-step computation of the variables, assuming that certain

[17] The solution method was ad hoc but the problem prompted further research and a paper by Frisch and Holme (1935) gave a formal method of solving mixed differential-difference equations.

initial values of the system were given.

Frisch was clearly very pleased with his results and their closeness to observed average cycle lengths:

> Of course, the results here obtained with regard to the length of the periods and the intensity of the damping must not be interpreted as giving a final explanation of business cycles; in particular it must be investigated if the same types of cycles can be explained also by other sets of assumptions, for instance, by assumptions about the saving–investment discrepancy, or by the indebtedness effect etc. Anyhow, I believe that the results here obtained in particular those regarding the length of the primary cycle of 8½ years and secondary cycle of 3½ years, are not entirely due to coincidence but have a real significance. (Frisch (1933), p. 190)[18]

Remember that Frisch had simulated his economic system based on guessed values for the structural parameters rather than using real economic data in his work. He must have been very confident in his model, and in his experiment, to predict that a new cycle would be discovered in empirical data:

> I want to go one step further: I want to formulate the hypothesis that if the various statistical production or monetary series that are now usually studied in connection with business cycles are scrutinized more thoroughly, using a more powerful technique of time series analysis, *then we shall probably discover evidence also of the tertiary cycle, i.e. a cycle of a little more than two years.*
> (Frisch (1933), p. 190)

Having discussed the internal economic cycle mechanism (the propagation problem), Frisch turned his attention to the impulse problem and to the question of consistency between his theory and economic time-series data. How could the damped component cycles he found for his determinate dynamic system be reconciled with the absence of damping and smoothness in the data?

> There are several alternative ways in which one may approach the impulse problem and try to reconcile the results of the determinate dynamic analysis with the facts. One way which I believe is particularly fruitful and promising is to study what would become of the solution of a determinate dynamic system if it were exposed to a stream of erratic shocks that constantly upsets the continuous evolution, and by so doing introduces into the system the energy necessary to maintain the swings. If

[18] The dangers of making such inferences were discussed by Haavelmo (1940). Like Yule, Haavelmo discussed both the role of superimposed errors and disturbances in estimated economic relationships and pointed out the dangers of letting verification rest on periodicities equal or similar to those of observed cycles, for when shocks are added to the model, these periodicities may change or may not even exist. This point is discussed further in Chapter 4.4.

> fully worked out, I believe that this idea will give an interesting synthesis between the stochastical point of view and the point of view of rigidly determined dynamical laws. (Frisch (1933), pp. 197–8)

Frisch pointed out that both Yule and Slutsky had suggested how economic time series might be generated by irregular events. Yet Frisch felt that the central problem still remained. How are the time shapes of observed economic variables actually determined, that is, how do Slutsky's random terms come to be summed in economic activity or how are Yule's shocks absorbed into the system; and what economic interpretation can be given to the processes?

Frisch's explanation linked his two fields of enquiry: the time shape of the components and their generation. His earlier analysis of economic time-series data had made him characterise such data as a combination of *changing* component cycles and he had used linear operators to unravel the components in each variable. Later he had turned this method round to experiment with linear operators acting on shocks as Slutsky had done. Now he used this idea again to suggest that the observed time series of an economic variable was the result of

> applying a linear operator to the shocks, and *the system of weights in the operator will simply be given by the shape of the time curve that would have been the solution of the determinate dynamic system in case the movement had been allowed to go on undisturbed.* (Frisch (1933), p. 201)

That is, the economic system has its own time path (consisting in his model of the combination of three damped, but *fixed*, component cycles) which provides a set of weights. This acts as a linear operator on the random shocks in a cumulative fashion to produce a combination of *changing* component cycles like those he had observed in real economic variables. Frisch gave a 'proof' of this suggestion with another simulation which showed that a linear operator acting on erratic shocks produced a curve with a changing harmonic, a not quite regular cycle where

> the length of the period and also the amplitude being to some extent variable, these variations taking place, however, within such limits that it is reasonable to speak of an *average* period and an *average* amplitude. (Frisch (1933), p. 202)

These are precisely the sort of component cycles Frisch had analysed in his time-series work and which he believed were typical of economic time-series data. He felt that this provided further empirical support that his model design was on the right lines.

Thus the system and the shocks are both required to maintain economic oscillations and to produce data consisting of changing component cycles:

> thus by connecting the two ideas: (1) the continuous solution of a determinate dynamic system and (2) the discontinuous shocks intervening and supplying the energy that may maintain the swings - we get a theoretical set-up which seems to furnish a rational interpretation of those movements which we have been accustomed to see in our statistical time data. The solution of the determinate dynamic system only furnishes a part of the explanation: it determines the *weight system* to be used in the cumulation of the erratic shocks. The other and equally important part of the explanation lies in the elucidation of the general laws governing the effect produced by linear operations performed on erratic shocks.
>
> (Frisch (1933), pp. 202–3)

Frisch's finding was similar to Yule's result that the best way to model the process of a disturbed harmonic function was by a simple difference equation (a linear operator) with an added disturbance. It is interesting that, although Yule postulated the process for such a time series, estimated the process and then solved for the cycles, Frisch himself continued to shy away from estimating the process that generated the cycles. Thus, despite recommending that structural parameters of the equations in the system should be estimated, Frisch had preferred instead to guess these parameters. This might seem a puzzling decision, but there were several reasons for it. In the first place, despite Frisch's modern terminology of 'macrodynamics', the cycle still formed the central concept and unit of analysis in his work. This is reflected both in his practice of solving the system into the time-series domain to find the cycle lengths and in the novelty of his appeal to the resulting cycles as confirming evidence of the model. It was the shape and periodicity of the time-series solutions to the macroeconomic system, and not the structural parameters themselves, which interested Frisch.[19] His views in this respect were conventional since the intellectual context of economic thought in the early 1930s was still that of business cycles not macroeconomics.

But Frisch had other good reasons for avoiding the structural parameters. He was at this time involved in two separate investigations into econometric methods. One was his time-series work discussed earlier and the second was a study of parameter estimation methods in

[19] I am indebted to John Aldrich for this suggestion which is consistent with Frisch's apparent change of heart towards estimation of such cycle models by the time of the European Meeting of the Econometric Society in 1936. By that time he advocated estimating the final form equations (similar to Yule's difference equations) not the structural system of equations because of his worries about collinearity and this led naturally on to his 1938 paper on autonomy (see Chapter 4.4 and Aldrich (1989)).

the presence of confluent relationships: several relationships holding between the variables at the same time. He thought there would be a particular danger of confluent relationships in business cycle research because of the parallel movements in the data series, and that the usual least squares estimates of structural parameters would therefore be untrustworthy. He was heavily critical of the standard regression methods, but he had not yet (by 1933) worked out his own bunch map techniques to deal with this problem. On the other hand, his time-series research programme was well advanced and, in these circumstances, he perhaps felt more at home in seeking the solution to his system, in terms of component cycles than in working with the structural parameters.

Good intellectual reasons apart, Frisch, unlike Tinbergen, preferred to work on problems of the methods and methodology of econometrics rather than on applied econometrics using real data. Although he had not 'applied' his rocking horse model to real data, and had therefore made no direct recourse to evidence, Frisch had made two indirect appeals to evidence to convince himself that his design was on the right lines. The first was that his simulated system had certain numerical results on cycle lengths in common with those from standard business cycle analysis. This was highly satisfactory, but was insufficient for Frisch. It was the further fact, that by putting together his economic model and the shocks he could reproduce the features he believed characterised economic time-series data, which finally convinced him. Frisch's model design did provide an answer to his question of how the changing cycle components came together to form economic data series.

Frisch's 1933 paper was highly successful; it was referenced by all the main writers on business cycles working in econometrics and mathematical economics. As a dynamic macromodel, the rocking horse model of the business cycle was successful because it had all the 'correct' features. The models of Jevons and Moore had not been accepted by other economists, not primarily because the chain of causation was so tenuous, but because economists were unwilling to believe in an economy which was totally reliant on regular exogenous forces for its motion. Frisch's rocking horse theory rested on an essentially mechanical system, a system, with a natural tendency towards an equilibrium or position of rest; the motion of the underlying model was endogenous and had damped oscillations. Frisch's paper has been credited by Samuelson with causing a revolution in economics equivalent in effect to the revolution from classical to quantum mechanics in physics (Samuelson (1947), p. 284). The revolution in thought that Samuelson referred to, however, was not in econometrics but in economic theory, and the transition in question was from the method of comparative statics to what might be called comparative dynamics.[20] Frisch had shown economists how to manipulate dynamic macromodels.

Yet there was more involved than suggested by Samuelson's judgement. Non-mathematical models of the trade cycle could not usually be translated into macromodels, let alone into potential econometric models, because such theories were often indeterminate or incomplete and the time relations were rarely specified. Frisch's model was dynamic, determinate and complete, a model which theorists could explore and manipulate for insights into how the economy might work, but one which was also amenable to econometric analysis. Econometricians such as Frisch and Tinbergen played a crucial part in developing mathematical macroeconomic models that could be made operational.

The rocking horse model was also a landmark in econometrics. Statistical work on the business cycle, such as that by Persons and Mitchell, had mostly been concerned with attempts to measure or isolate the cycle rather than to provide explanatory models of the cycle or test the many theories of the cycle. Moore had worked out a viable explanation of the business cycle and estimated it by a process of defining successive relationships, but it was limited in its design to fit the dynamic pattern of a particular time and place. Frisch's model was not built to fit any particular data set; it is not even clear that it was intended as a serious economic theory of the cycle, but rather as an illustrative exercise in method in which Frisch showed the sort of model or explanation econometricians should be using.

The importance of Frisch's paper for econometrics lies therefore in his model design. The economic dynamics are one important element of this. His model could generate economic cycles through the interactions of the equations in its system. But of course, whether the model did generate cycles depended on the parameters of the model and it was possible that with other parameters, the model might not necessarily produce cycles, the solutions might be undamped or even explosive. The second important econometric design feature was the role of random shocks in conjuction with the deterministic system. Although business cycle theory was non-stochastic, it was generally accepted in an informal way that observed cycles were influenced by non-regular outside events. Frisch recognised that these shocks were real disturbances (in Yule's sense), rather than either a measurement error (as in Yule's superposed errors) or an unexplained residual left after the

[20] See, for example, Merlin's (1950) analysis of equilibrium notions and the econometric contribution to cycle theory.

elimination of trend, seasonal and cyclical features from the data (as in Persons' work). The economic system provided the summation mechanism for these random errors as suggested in Slutsky's hypothesis about cycle data. The role of random shocks was crucial, it transformed the model from a theoretical model which could produce the underlying cyclical components to one which could lay claims to the data by producing the rather jagged appearance of economic data and by maintaining the oscillations of the cycle. The shocks provided the final important element which changed the dynamic economic model into an econometric model, a formal stochastic model of how real economic data might be produced.

Frisch's econometric model was successful and influential and yet the first attempts, by Tinbergen, to build a full-scale model of the economy and follow Frisch's suggestion of estimating the structural parameters using real data, rather than by guesswork, proved to be far more controversial.

CHAPTER 4

Tinbergen and macrodynamic models

Jan Tinbergen built and estimated the first macrodynamic model of the business cycle in 1936.[1] Amongst all the econometricians of his day, Tinbergen was ideally suited for such a task. He had been experimenting with small-scale models of the trade cycle since the late 1920s and was well versed in the ways of dynamic models. He also had a wide knowledge of quantitative business cycle research from his experience as Editor of *De Nederlandsche Conjunctuur* (the Dutch statistical business cycle journal). Yet, even for one so well qualified, it was a formidable undertaking, for there was a considerable jump from putting together a small cycle model to constructing an econometric model of the business cycle covering the whole economy, and Tinbergen was well aware that the difficulties were not merely due to the difference in scale.

Tinbergen had already given considerable thought to the problems involved in a 1935 survey of econometric research on business cycles commissioned for *Econometrica*. He had taken as his starting point Frisch's (1933) idea that a business cycle model should consist of two elements, an economic mechanism (the macrosystem):

> This system of relations defines the structure of the economic community to be considered in our theory (Tinbergen (1935), p. 242)

and the outside influences or shocks. But, as Tinbergen had pointed out, this was only a basic design for an econometric model; the scope of Frisch's new term 'macrodynamics' was unclear. Just what variables and relations should a complete model of the business cycle include? How should these relations be put together to form an adequate

[1] The Dutch economist Jan Tinbergen was born in 1903. He graduated from the University of Leiden in 1926 and received a doctorate for his thesis on physics and economics in 1929. From 1929 to 1945 he worked on business cycle research at the Central Bureau for Statistics (apart from 1936–8 spent working for the League of Nations). He was appointed the first director of the Netherlands Central Planning Bureau in 1945 where he remained until 1955. Besides being one of the great pioneers in econometrics, he has made valuable contributions to several other fields of economics.

econometric model of the business cycle? What properties should this system of relations have? None of these questions should be considered trivial or obvious, for in the 1930s even the most sophisticated theories of the business cycle tended to be incomplete and the essential characteristics of an econometric model remained undefined.[2]

A second set of problems concerned how to choose the relations to make up the system when many theories and dynamic characterisations of those theories were available. The multiplicity of theories necessitated some form of test or verification in order, as Tinbergen had stated:

> To find out whether these schemes can explain real business cycles and which of them most resembles reality. (Tinbergen (1935), p. 281)

Last, but certainly not least as far as Tinbergen's interests were concerned, was the question of how to use the model to investigate policy problems.

These were the issues and problems in Tinbergen's mind when he started work on his large model of the Dutch economy. They seemed to pull two ways. He saw that a successful applied model needed to replicate reality as closely as possible, but that the model would only be amenable to policy analysis if it were relatively simple. He gained some comfort from recognising that this tension between realism and simplicity was a standard problem of applied scientific research.

4.1 The Dutch model

Tinbergen's first macrodynamic model was built in response to a request from the Dutch Economic Association to present a paper in October 1936 on policies to relieve the depression. He was pleased to respond to this opportunity for he had switched out of his initial field, physics, into economics because he believed it was a more socially useful science. His model is a remarkable piece of work, involving not only building and estimating a model of the whole economy but also using the model to simulate the likely impact of various policies. His memorandum on the model was written in Dutch for a non-quantitative audience, and consequently avoided discussion of technical and methodological problems. But by the following February, Tinbergen had produced a full treatment of the econometric aspects of his work in an English-language version which is discussed here.[3]

The experience of building and estimating the Dutch model forced

[2] Tinbergen has recently discussed this problem in an interview (see Magnus and Morgan (1987)).

[3] The original Dutch model was published in Dutch in 1936, and is available in English in Tinbergen (1959). The revised English version was published in a French series edited by Gibrat in 1937.

Tinbergen to find practical solutions for some of the problems he had foreseen. First of all he described how he proposed to set up the system of causal relationships to form the model.

> We may start from the proposition that every change in economic life has a number of proximate causes. These proximate causes themselves have their own proximate causes which in turn are indirect 'deeper' causes with respect to the first mentioned change, and so on. Thus a network of causal relationships can be laid out connecting up all the successive changes occurring in an economic community. Apart from causal relationships there will also exist relationships of definition ... And, finally, there will be technical or institutional connections. All these relationships together form a system of equations governing the movements of the various elements in the community. Each of these equations can be looked upon as a determining equation for one of the elements, explaining what factors influence that element and how large is the effect of a given change in each factor. (Tinbergen (1937), p. 8)

He worried less about realism, aware that a model is but a stylised version of the economic system:

> I must stress the necessity for simplification. Mathematical treatment is a powerful tool; it is, however, only applicable if the number of elements in the system is not too large ... the whole community has to be schematised to a 'model' before anything fruitful can be done. This process of schematisation is, of course, more or less arbitrary. It could, of course, be done in a way other than has here been attempted. In a sense this is the 'art' of economic research. (Tinbergen (1937), p. 8)

Tinbergen explained his model building as an iterative process involving both hypotheses and statistical estimation:

> The description of the simplified model of Dutch business life used for the consideration of business cycle causes and business cycle policy commences with an enumeration of the variables introduced. The equations assumed to exist between these variables will be considered in the second place. This order does not correspond exactly to the procedure followed in the construction of the model. One cannot know *a priori* what variables are necessary and what can be neglected in the explanation of the central phenomena that are under consideration. It is only during the actual work, and especially after the statistical verification of the hypotheses, that this can be discovered. As a matter of fact, the two stages mentioned are really taken up at the same time; it is for the sake of clearness that the exposition is given in two stages. A glance at the 'kitchen' will nevertheless be occasionally made in order to avoid an impression of magic.
> (Tinbergen (1937), p. 9)

These introductory remarks portray Tinbergen's ideas and general approach to the task ahead.

The Dutch model was very much a simple model as far as Tinbergen was concerned but was huge and complex by the standards of the time. It contained 22 relationships and 31 variables (divided into prices, physical quantities and money values). The relationships were divided into technical equations, definitional equations (such as those defining value) and direct causal relationships which provided explanations of price movements, sales, competition and the formation and disposal of incomes. He estimated the 16 non-definitional relationships covering three sectors (domestic production, income and consumption, and international trade) for the period 1923 to 1935. Most of these equations involved only one or two systematic or explanatory variables. In addition, Tinbergen included time trends in the equations rather than working with variables in the form of deviations from trend values (considered to represent the long-run or equilibrium values).[4] Each of the equations was estimated separately and, for the most part, independently of the others; though in some cases, information from already estimated equations was incorporated in a new equation. Accidental influences were assumed to be small and random, leading to residuals in the regression, but these random error terms were not written into the relationships.

In contrast to the current publication conventions of econometrics, Tinbergen was not afraid to let his reader see the 'craft' element in his work. A summary of his discussion of the investment equation will give, to use Tinbergen's own words, 'a glance at the "kitchen" '. The principal factor determining investment was thought to be profit expectations, but, Tinbergen argued, these are probably based on previous profits, which can therefore safely be taken as the main explanatory variable. The influence of interest rates on investment was investigated statistically and found unimportant (as indicated by the regression coefficient). Finally he argued:

> economists would perhaps prefer a more complicated function ... Their argument would probably be that the volume of investment should be large enough to make marginal profits equal to zero. I think this argument only applies to long-run tendencies but not to the rather short-run tendencies represented by our equations. (Tinbergen (1937), pp. 25–6)

This left a simple equation in which investment depended only on profits lagged by one year and a trend term. Tinbergen reported the equation's results in graph form, showing the observed and fitted

[4] In this he was following Frisch and Waugh's (1933) result which had shown the equivalence of the methods (discussed in Chapter 5). Tinbergen also calculated unusual regression coefficients: he divided each standard least squares regression coefficient by the correlation coefficient.

values of the dependent variable and the explanatory variables on one chart. These charts (an example of which is reproduced as Figure 10) were distinctive to Tinbergen's work and their role and importance will become evident during the course of the chapter.

As is clear from this example, the formation of each individual equation and the particular choice of variables were found by iterating between theoretical ideas and empirical investigations. This iterative approach was extended to the 'verification' of hypotheses. This consisted of judging whether the estimated relationships were reasonable, both in terms of their economic sense (the first criterion) and in terms of closeness of fit, judging by eye the distance between the observed and fitted values of the dependent variable on the graph. It should be clear that Tinbergen was not claiming 'statistical testing' of his model here, but 'statistical verification'. His purpose was to show that the model was compatible with the statistical observations and provided a sensible explanation of the observed movements in the economy over the previous years.

Business cycle historians, according to Tinbergen, could gain many insights into past economic experience by comparing the variations in the explanatory variables with those of the dependent variable. These comparisons could easily be made using Tinbergen's graphs and could reveal a specific cause of a crisis or a revival. For example, he found that a particularly large residual in one equation, suggestive of some outside cause, could be interpreted as the result of the introduction of an import quota system. But Tinbergen provided a sting in the tail by reminding his readers that such an interpretation of large residuals was only possible

> in so far as we believe in the theory behind our hypotheses.
> (Tinbergen (1937), p. 46)

The next stage of the investigation was to see if the model had a cyclical pattern as its solution. Why and how this was done requires further explanation for it was difficult for some of Tinbergen's contemporaries to understand and may still be opaque to the modern reader. The question was, in the absence of shocks and disturbances, would the economic system have a cyclical path? If it did, then Tinbergen argued that the model provided a theory of the business cycle. This test of the model was in line with Frisch's 1933 paper which required that the economic system part of a macrodynamic model show a cyclical tendency. Frisch had studied the time path of each of the three variables in his model. Tinbergen's system of equations was more complicated and none of the elementary equations could represent the system of the whole economy.

In order to discover what the natural tendencies of Tinbergen's model were, his large system of 22 equations first had to be reduced to one equation in one variable. By taking arbitrary values for the international variables (such as import prices) and by a process of substitution and elimination of the domestic variables in the model, one 'final equation' was obtained. This final equation was a simple difference equation in the variable Z (which represented non-labour income):

$$Z_t = .15Z_{t-1} + .26Z_{t-2} - 34.7$$

In a later discussion of this part of his method Tinbergen admitted the difficulty of following the economic meaning of the model when the original economic relationships were converted into one final equation.[5] This had led the analysis to be called 'night train analysis', because economists had to move from the elementary equations or relationships, with which they were familiar, to 'deduced economic laws' for which it was not easy to give an account. In this case, the final equation represented the structure of the Dutch economy because the coefficients of the difference terms depended on the estimated coefficients of the system of 22 elementary equations which formed the model. So, the equation was not itself a regression equation, but it depended on all those that were.

Of course, the final equation was not derived for its own sake, but as a preliminary step in finding the time path of the system. Tinbergen proposed two methods for this. There was a mathematically simple, but tedious, method, and a mathematically advanced alternative. The simple way involved taking two initial values of the variable Z in the final equation (assuming no disturbances) and simply extrapolating further values of Z from the equation. From this, other variables' paths could also be calculated because of the interdependencies in the system of equations. Of course, with different starting values, and different values for the international variables, other time paths would be found for the system. The more advanced method of finding the time path of the system, which avoided the problems of choosing starting values, was to solve the final linear difference equation in terms of its harmonic terms. This is the route both Yule (1927) and Frisch (1933) had followed in order to find out what the system's path looked like. The final equation of the Dutch model was solved by Tinbergen to show that the economy had a nicely damped cyclical path which would tend to an equilibrium position provided there were no disturbances. This

[5] See Tinbergen (1940) and the background to this paper in n. 19 below.

solution was taken to show that the model did provide an adequate theory of the Dutch business cycle.

In practice, the process of determining the dynamic character of the model was complicated by the presence of disturbance terms. These might be represented in the initial conditions of the system or they might occur as additive terms coming into the elementary equations at specific points due to, for example, changes in government policy. As an example, Tinbergen examined how an imported international cycle could disturb the Dutch economy. This exercise confirmed his belief that for a small country like the Netherlands, the shocks from the internal economy (through new inventions or good harvests, for example) would not be very large; they would be dominated by disturbances imported from the movements of larger economies.

Extrapolation of the model to show its time path was a test of whether the model provided a theory of the business cycle, but it was also a prelude to an investigation of the optimum policy. In Tinbergen's analysis (based on analogies from physics on the movements of a pendulum) policy changes either affected the relations superficially through the additive disturbance terms or more deeply by changing the coefficients and causing a change in structure. Tinbergen investigated six policy options: public works schemes, a protectionist policy, a rationalisation policy, a lowering of monopoly prices, wage reductions and devaluation. He modelled these simply as terms added to the estimated elementary equations of the system; and for each policy option the system was again solved to find the new final form equation. Tinbergen then compared the movements in two variables, non-labour income (Z) and employment, which would result if he extrapolated the system for seven periods ahead for each of the six different policies. (He also, by putting $Z_t = Z_{t-1} = Z_{t-2}$, compared the long-run equilibrium values for all the main variables under the different policies.) Of course which policy was considered optimum depended on the criteria chosen. Tinbergen's criterion in 1936 was to obtain the lowest unemployment, and, for this purpose, devaluation proved to be the most promising policy.

Tinbergen also tried to work out the best possible policy for stabilising an imported business cycle, either through making compensatory movements in the exchange rate or in public investment. Here some of his examples involved changing the coefficients on the equations, a more complicated problem than dealing with policies as added disturbances.

To conclude his study of the Dutch economy, Tinbergen offered a résumé of the advantages of the econometric approach to business cycle problems:

> The establishment of a system of equations compels us to state clear-cut hypotheses about every sphere of economic life and, in addition, to test them statistically. Once stated, the system enables us to distinguish sharply between all kinds of different variation problems. And it yields clear-cut conclusions. Differences of opinion can, in principle, be localised, i.e. the elementary equation in which the difference occurs can be found. Deviations between theory and reality can be measured and a number of their consequences estimated. Finally, the results of our calculations show, apart from many well-known facts, that, as regards the types of movement that are conceivable, there exist a number of neglected problems and of unexpected possibilities. (Tinbergen (1937), p. 73)

Given the multitude of business cycle theories then available and the serious policy problems posed by the Great Depression, the benefits outlined by Tinbergen were potentially valuable. His comments perhaps appear overoptimistic now, but comparisons of his work on the Dutch model with Frisch's 1933 paper or with Mitchell's 1927 book on business cycles suggest that Tinbergen was justified in boasting the benefits of econometrics. He was soon given an opportunity for a deeper exploration of the subject's potential.

4.2 The first League of Nations' report

In 1936, Tinbergen was commissioned by the League of Nations to undertake statistical tests of the business cycle theories examined for the League by Haberler in *Prosperity and Depression* (1937).[6] Tinbergen worked at this task for two years and reported his results in two volumes, *Statistical Testing of Business-Cycle Theories* published in 1939. The first contained an explanation of the method of econometric testing and a demonstration of what could be achieved in three case studies. The second volume contained an ambitious macroeconometric model of the USA.

In order to air the problems involved in testing business cycle theories, Tinbergen opened his first report with some general comments on the methodology of econometric research. It is again worth quoting these more or less in full, so that we can understand Tinbergen's own ideas on testing. He believed that the empirical study of business cycle data, involving correlation and decomposition analysis

[6] I am grateful to Earlene Craver for pointing out the explanation for the apparent confusion in these dates: Haberler's book was widely circulated in draft by the League prior to publication.

of economic time series (of the sort discussed in Chapter 2), was of limited use when it came to testing theories.

> Certainly all this work had its value, especially for the *negative* evidence it afforded on the validity of certain theories. For the purpose of applying more searching tests, however, it is necessary to dig deeper. An apparently simple relation, such as that between prices and production, is often not a direct causal relation at all, but a more or less complicated chain of many such relations. It is the object of analysis to identify and test these causal relations...
>
> The part which the statistician can play in this process of analysis must not be misunderstood. The theories which he submits to examination are handed over to him by the economist, and with the economist the responsibility for them must remain; for no statistical test can prove a theory to be correct. It can, indeed, prove that theory to be incorrect, or at least incomplete, by showing that it does not cover a particular set of facts: but, even if one theory appears to be in accordance with the facts, it is still possible that there is another theory, also in accordance with the facts, which is the 'true' one, as may be shown by new facts or further theoretical investigations. Thus the sense in which the statistician can provide 'verification' of a theory is a limited one.
>
> On the other hand, the role of the statistician is not confined to 'verification' ... the direct causal relations of which we are in search are generally relations not between two series only – one cause and one effect – but between one dependent series and several causes. And what we want to discover is, not merely what causes are operative, but also *with what strength each of them operates*: otherwise it is impossible to find out the nature of the combined effect of causes working in opposite directions.
>
> (Tinbergen (1939), I, p. 12)

Tinbergen's view, then, incidentally a long-held opinion, was that statistical testing could lead to either disproof or to limited verification.[7]

Tinbergen's separation of the two roles, the statistician and economist, is disconcerting in this context. But he went on to explain how econometrics united these approaches. Tinbergen argued that economic theory must be expressed in mathematical form but that quantitative economics had chiefly been concerned with the long-run equilibrium conditions, to the neglect of short-run dynamic problems of the cycle:

> To be useful, therefore, for business cycle research, economic theory needs to be made 'dynamic'. A 'dynamic' theory, in the sense which is here attached to that ambiguous word, is one which deals with the short-term reactions of one variate upon others, but without neglecting the lapse of

[7] Tinbergen had suggested that statistics could not prove theories but could be used to help disprove theories as early as his first article in economics in 1927, see Magnus and Morgan (1987).

> time between cause and effect. The equations in which it is expressed thus relate to non-simultaneous events, and take a form which Swedish economists have described as 'sequence analysis'
>
> (Tinbergen (1939), I, p. 13)

At the same time, the conversion of theory into relationships 'capable of statistical test' required: that the economic relations be given in terms of cause and effect; that the time lag between these be specified; and that all the major causes of variation be specified rather than leaving them 'concealed in a *ceteris paribus* clause' (Tinbergen (1939), I, p. 13). The development of dynamic theory had sometimes, according to Tinbergen's view, resulted directly from statistical research rather than from purely theoretical research. Thus,

> we find that the correlation analysis suggested by statistical technique and the sequence analysis dictated by 'dynamicised' economic theory converge and are synthesised in the method employed in this study.
>
> (Tinbergen (1939), I, p. 14)

This synthesis of mathematically expressed dynamic theory and statistical method formed the ideal of the econometric method and so, in a practical way, the activities and aims of the statistician and the economist were united, as indeed they were in Tinbergen's work.

The main and novel feature of this first League of Nations' report, subtitled: *A Method and its Application to Investment Activity*, was its marked emphasis on testing, using a very wide range of procedures involving both economic and statistical criteria; the substantive material was provided by three case studies on general investment, investment in housebuilding and in railway rolling stock. Tinbergen felt constrained from the start by the amount of calculation needed to cover all the cases he wanted to investigate. But he made a virtue of his necessity and used his limited resources to create another opportunity to test the model as follows. In his first case study on general investment, he began by working with a large model which incorporated all the variables suggested by theorists as influencing investment. He applied this full model to only a limited number of countries and time periods. After studying the results from this exercise, he reduced the size of the model to include only the variables which appeared to be most important. He then applied the reduced model to a number of different countries and time periods to see whether it worked equally well, and at this second stage was able to apply a full range of statistical testing procedures.

Once again, theoretical discussions were interwoven with the applied work in a recursive treatment. Tinbergen happily showed the 'cook at work' creating his investment model from a mixture of verbally expressed theories, correlations between various variables and empirical results. Prior to estimation, Tinbergen discussed the following sorts of questions. What variables should be included? What was the exact form of a variable to be used, for example, should it be profit margins, total profits, profit rates or expected profits? What were the expected signs of the variables? Was there a danger of multicollinearity?[8] He also thought about, and tried to sort out from the correlations, the problem of two-way causality between profits and investments and dealt with the parallel problem of determination of both demand and supply functions (discussed further in Part II).

Tinbergen used his distinctive 'stacking' technique, which involved graphing the causal variables one on top of the other like plates in an attempt to 'explain' the variation in the dependent variable.[9] These graphs both helped him to choose the contents of the individual relations, and delayed the calculation of the regression equation until he was reasonably sure of its contents (thus reducing the burden of computation on his assistants to a minimum). He had used this graphing technique earlier for the Dutch model, he now produced more complex charts containing more information. One of them, showing the regression equation for the production of pig-iron, is reproduced here as Figure 10. From my earlier comments and this illustration, it should be obvious that a large amount of information can be gleaned from Tinbergen's charts. As well as helping him to choose the model, they were a useful way of reporting the results. They showed the final calculated relationship and revealed the patterns of both the individual explanatory variables and the residuals.

Tinbergen felt it was important to calculate both regression and correlation coefficients for he had suggested earlier in his study that regression coefficients measured the strength of the effect of the explanatory variables in a relationship whereas the correlation coefficient gave evidence for the verification of a relationship, that is, a high correlation coefficient verified a theory. So, in calculating the regression relations, Tinbergen followed a sort of simple to general modelling procedure looking at the sign, stability and size of the regression coefficients and total correlation coefficient as each new variable was added in. He used the evidence from his calculations as further help in deciding the correct model (and in some cases the regression

[8] Fear of multicollinearity had been engendered by Frisch's confluence analysis (1934). Tinbergen's rule of thumb was that if the correlation coefficient between any two explanatory variables was particularly high ($|r| > .80$) then one of the variables was omitted (see also n. 11).

[9] The term 'plate stacking' was used by Tinbergen himself to describe his practice in an interview with him on 30 April 1980. Despite their usefulness, few copied his graphic methods, though see Chapter 2 n. 24.

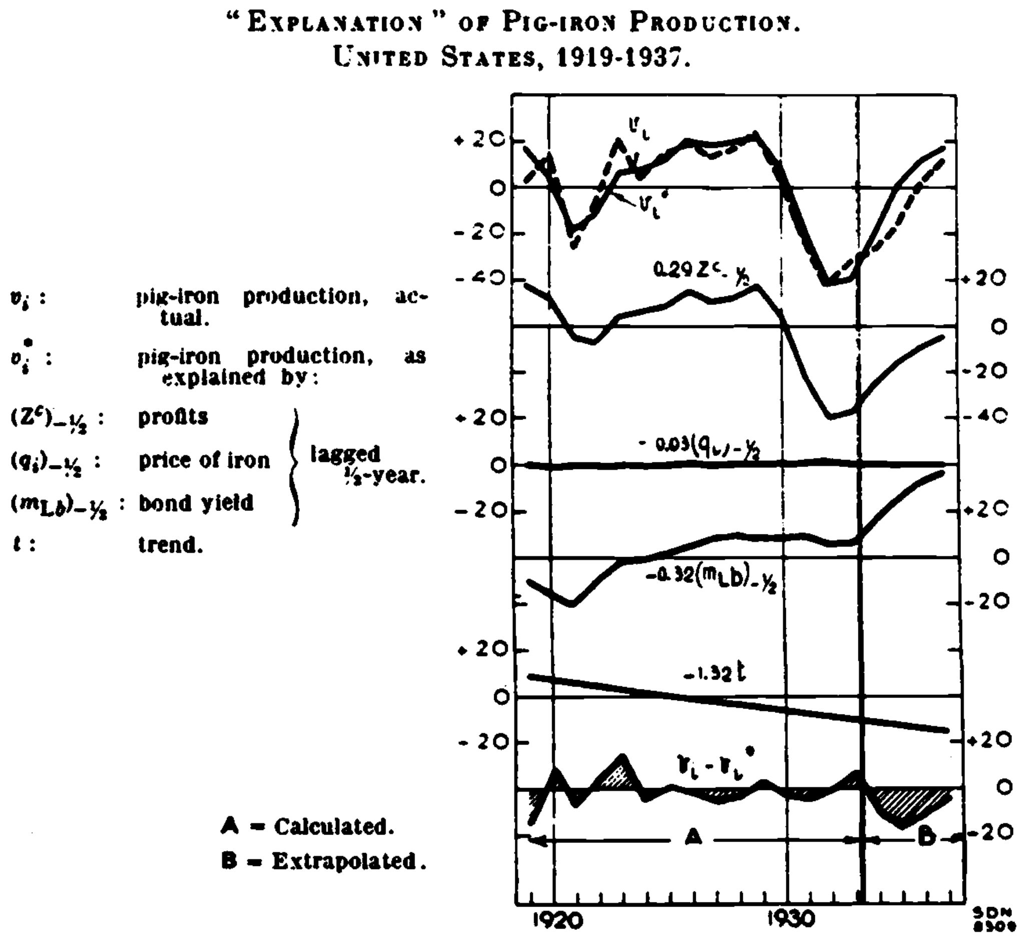


Figure 10 Tinbergen's plate-stacking graph
Source: Tinbergen (1939), I, p. 73, Graph III.8

calculations were carried out ahead of the graphing study). He did not always go along with its results; for example, he would sometimes include variables which seemed ineffective in the relation but which theorists argued should be present. In addition, Tinbergen tried a number of different lags lengths to obtain the best time relation.[10]

Once estimated, the equations were subjected to several different sorts of testing procedures. First, there was the already mentioned test of the models on new countries and time periods. Second, Tinbergen tried the models out on different subperiods to test for the possibility of structural changes or non-constant coefficients and for different coefficients on the up-phases of the cycle compared to the down-phases.

[10] Tinbergen's initial assumption was that the explanatory variables entered the equations with a half-year lag. He also experimented with no lag and a one-year lag. In addition, he plotted out the frequency distribution of the coefficients on the lagged profit term in the investment equation and found the average lag to be eight months.

Third, he carried out prediction tests by extrapolating the fitted equations for the next two years of the period (an example of this practice is evident on the graph reproduced above: 'B' is the forecast period) and he also checked the assumption of linear trends by looking at the partial scatter diagrams.

Last, but not least, Tinbergen followed a complete programme of statistical significance calculations. First, this involved testing the residuals. He had characterised the omitted and non-economic influences in the relationship as being many, small and of accidental character – traditionally the characterisation associated with the 'law of errors' or normal curve. Testing for first order serial correlation of the residuals, he wrote,

> serves to test the hypothesis ... that the residuals are to be considered as sample drawings from a 'normally distributed universe'. At the same time, it gives information as to whether the regression chosen satisfies the scheme of the shock theory. (Tinbergen (1939), I, p. 80)

In this he equated the shock scheme of Frisch's 1933 model with the assumption of a normal distribution of the residuals. The shock model was therefore in Tinbergen's treatment the same as the classical regression model. Tinbergen used the classical tests on ordinary least squares and Frisch's confluence analysis as complementary checks on the relations he had found. So his second statistical test was to look at the standard errors of the coefficients to check their significance, and thirdly, he looked at the bunch maps to check for collinearity and to get some idea of the error due to weighting. He even worked out limits for the errors of weighting in line with Koopmans' improvements to Frisch's confluence analysis scheme.[11]

Tinbergen's first example of macrodynamic modelling had been concerned with explaining the past behaviour of the Dutch economy and with judging the effects of various policies rather than with testing. In this first report for the League of Nations he concentrated wholeheartedly on the problem of testing at the level of the individual equations. He carried out an enormous range of different tests (16 in all) to demonstrate the extent to which a theory could be confirmed or refuted by using the econometric approach. Although it is probable that none of these tests was an innovation, the sheer range of tests that Tinbergen presented was entirely without precedent in econometrics in the 1930s. Other econometricians were beginning to get interested in testing models but few used many tests or understood what the

[11] Frisch's confluence analysis (1934) and Koopmans' 1937 critique and application of some of R. A. Fisher's ideas to Frisch's work are presented in Chapter 7 Letters 11 and 15.

problems were. Tinbergen clearly led econometrics in this field, but he still had to tackle the problem of the statistical testing of business cycle theories.

4.3 The second League of Nations' report

Tinbergen's commission, remember, was to test the theories of the business cycle which had been surveyed by Haberler. This did not mean that Tinbergen was faced with two or three well-prepared competing macroeconometric models. On the contrary, he was faced with a large number of verbally expressed economic theories in a form not immediately appropriate for statistical measurement or evaluation. In his second report for the League of Nations, Tinbergen therefore developed a three-stage procedure for evaluating theories of the business cycle. The first stage was to find out whether a verbal model could be translated into an econometric model. For stages two and three, Tinbergen returned to the approach adopted in his earlier Dutch model of 'statistically verifying' the relationships of the model first, and then deriving and testing whether the final equation had a cyclical solution. To provide material for his evaluation process he built the first large-scale macroeconometric model of the USA; and the report was consequently subtitled: *Business Cycles in the United States of America, 1919–1932*.

Tinbergen's first test of a verbal theory was whether it could be translated into a mathematically expressed econometric model. It was only while working on the US model that he began to understand and clarify what this meant in the business cycle context.[12] Tinbergen required that the theory should be able to form a model which was complete (as many relationships as variables to be explained) and determinate (the causal factors in each relation to be fully specified) and dynamic (the time lags fully specified). All three properties were needed to take sensible measurements of the model, but the first two properties were also required for the system of equations to be solved for the final equation while the shape of the time path of the system also depended on the dynamic aspects of the relationships. The qualities needed to make an economic model into an adequate econometric model were not widely appreciated at the time.

According to Tinbergen's view, most of the verbal theories of the business cycle failed at this important first hurdle because they were

[12] Tinbergen found this idea of a model very important and now sees it as one of the main contributions that econometrics has made to economic theory (see Magnus and Morgan (1987)).

incomplete or were indeterminate, while the dynamic aspects of a theory would almost certainly be couched in vague terms, if at all. It was certainly the case that most of the theories of the business cycle in Haberler's survey were likely to be incomplete since they dealt with only one or two relations of the economic system. Some of these theories could be considered complementary since they dealt with different parts of the economic system while others were directly competing theories. Like Mitchell, Tinbergen rejected the notion of testing each business cycle theory one by one as being too repetitious. The most efficient way to test these theories seemed to Tinbergen to combine the theories into one system of relations to form a model. Indeed, since most existing theories failed the first test, it was obviously *necessary* that some theories be combined in order to make a complete model. The theories, and the equations which represented them, became inter-related by their place in the whole model. Inevitably this process also involved making some choices between competing theories.

In the second stage of evaluating business cycle theories, the individual estimated equations were to be tested for their correspondence with economic theory relations. But, because Tinbergen needed to combine theories to form the model, it was not really possible to test the individual theories separately. Instead, Tinbergen specified the individual equations in his model of the USA in accordance with one or more of the economic theories concerned with those variables. He then estimated each equation and examined these statistical relationships between the economic variables in the light of the various theories. He used his normal iterative approach to estimation and model choice so that in the process of building the model he was also 'testing' the theories, combining some and rejecting others not supported by his applied work.

The completed model of business cycles in the USA incorporated 71 variables, 48 equations and covered the period 1919 to 1932. It was a considerable advance on the Dutch model not only in size but also in economic interest. He exhibited considerable skill as an applied econometrician in juggling so many different partial verbal theories, different variables and other pieces of information. For example, he paid particular attention to the peculiar difficulties of the period, notably the Great Crash of 1929, with crafty modelling of cash hoarding and the stock market boom.

Tinbergen relied on two different types of assessment at this second stage test: one used economic criteria and the other relied on historical explanation. The use of economic criteria involved examining the sizes (both absolute and relative) and signs of coefficients in the measured relations compared to those expected from the theories. A good example of this was on an equation involving the marginal propensity to consume. Here he rejected cases where the propensity was greater than unity and where it was lower for low income earners than for those with high income. In this same section he also rejected negative coefficients where positive were expected and occasionally retained coefficients for theoretical reasons when they proved statistically insignificant. Although standard errors of the coefficients and multiple correlation coefficients were reported, Tinbergen carried out no independent tests of the model on other data, nor did he subject his equations to the battery of tests used in the first report for the League.

The other criterion used was to see whether the individual equations offered reasonable and adequate explanations of the historical record for the US for 1919 to 1932. This entailed using the stacking graphs to trace which were the important variables causing variation (turning points, rises or falls) in the dependent variables. Tinbergen proffered some explanations about the specific turning points on the basis of this examination of the graphs, but of course these results could not be generalised to other countries or time periods but were specific to time and place.

In the third stage of evaluating the model, the interrelated equations of the model were then combined to make the final equation representing the system. This derived final form equation was examined to see whether the model as a whole would generate a cyclical pattern. Tinbergen was now very much more articulate about why he was doing this. His rationale was as follows: the model consists of a network of causal connections or equations between a number of economic variables. Although these variables fluctuate over time, none represents the business cycle itself. It is the causal connections, he argued, which form the mechanism of the business cycle and must be able to explain and represent the cycles. He demonstrated what he meant with a small three variable, three equation, dynamic model and showed, with worked examples, that different coefficient values would lead to different patterns in the single final form equation for the system. Some were cyclical and some not.[13] Tinbergen reasoned from this example, that

> In fact, it seems difficult to prove by pure reasoning alone – i.e., without knowing anything about the numerical values of the coefficients – whether or not any given theory explains or does not explain cyclic movements.
> (Tinbergen (1939), II, p. 18)

[13] In his later discussion (1940) of the mathematical structure of the models, Tinbergen emphasised the importance of knowing not only the constant coefficients of the model but

This in turn explained why the second stage of testing was so very important: you needed to have some confidence that the coefficients were reasonably correct before you could tell whether the causal connections would result in the cyclical pattern required of a business cycle theory. But, Tinbergen also warned later that different estimated equations which seemed to fit the data equally well under the second testing stage could lead to different period and damping ratios in this third stage. In this case, he said, the choice between the models would have to depend on other considerations.

In practical terms, the simplicity of a model was regarded as being an advantage at this stage. If the model was too complicated, the necessary elimination process to get to one final equation could not be carried through. On the other hand, the model had to be sufficiently complex to reflect the most important relationships in the economy in a form which provided a workable system of equations. As Tinbergen later described it:

> The performance of the elimination process exhibits very clearly one fundamental difficulty in business cycle theory. In order to be realistic it has to assume a great number of elementary equations and variables; in order to be workable it should assume a small number of them. It is the task of business cycle theory to pass between this Scilla and Charybdis. If possible at all the solution must be found in such simplifications of the detailed picture as do not invalidate its essential features
>
> (Tinbergen (1940), p. 78)

These were the practical implications of the trade-off between simplicity versus realism. In addition, since there were always a number of different ways of deriving the final form equation, it was important to choose the correct place to start so that the elimination process could take place smoothly.

Faced with the large US model, Tinbergen was forced to indulge in a certain amount of approximation of the individual equations in order to be able to carry through the elimination process. This process resulted in a group of several equations which he called the 'strategic relationships' forming the

> kernel of relations which can more easily be treated. It is, of course, not by chance that we are left with these equations and these variables. The logical structure of our system of equations, which after all is nothing but a reflection of the structure of the business-cycle mechanism, is such that they play the central role. (Tinbergen (1939), II, p. 133)

also the lag structure in order to determine whether a model would generate a cyclical pattern from its final form equation.

Having derived the final form equation (a difference equation in one of these strategic variables), he then went on to examine the characteristics of that equation and the solution path of the single strategic variable under different regimes (such as the presence or absence of cash hoarding).

It is worth examining in more detail Tinbergen's explanation of one of these cases: the final equation in the case of absence of a stock exchange boom or cash hoarding (Tinbergen (1939), II, Equation 6.31, p. 137):

$$Z_t^c = \sum_{i=1}^{4} e_i Z_{t-i}^c + (AU + HO + F + R)_t$$

where Z_t^c was the net income (profits) of corporations. The other four variables represented external (exogenous) factors in the system; each had its own explanatory equation, except for R, which was a conglomerate of little disturbances and regarded as random. AU represented influences from changes in the gold stock and central bank policy. HO represented developments in the housing market which showed almost autonomous cycles. F stood for external (climatic) influences on crops.

The first four coefficients (e_i) of this derived final equation depended on all the regression coefficients of the individual elementary equations of the model (because the final equation had been obtained by substitution and elimination from the individual equations). The coefficients

> describe in an abbreviated form the structure of the economic mechanism with regard to business cycles; they will be different in other countries, or under another regime, where the economic structure of society is different.
> (Tinbergen (1939), II, p. 137)

The second consequence of the elimination process was that, since each elementary equation contained a group of unsystematic influences (treated together as one random variable), the final equation contained a great number of these random terms originating in the individual equations. These, of course, were the 'shocks' of Frisch's model. In this case, the four external forces were more or less independent of the general position of the business cycle and therefore had the same effect as shocks on the path of Z_t^c.

Tinbergen illustrated how Z_t^c depended on earlier values of the exogenous and random forces using another of his distinctive visual aids,[14] reproduced in Figure 11 (where R here represents the

[14] Tinbergen's arrow scheme had made its first appearance in his small book on statistics in 1936. It was not absolutely new to econometrics, for it had appeared in Sewall Wright's

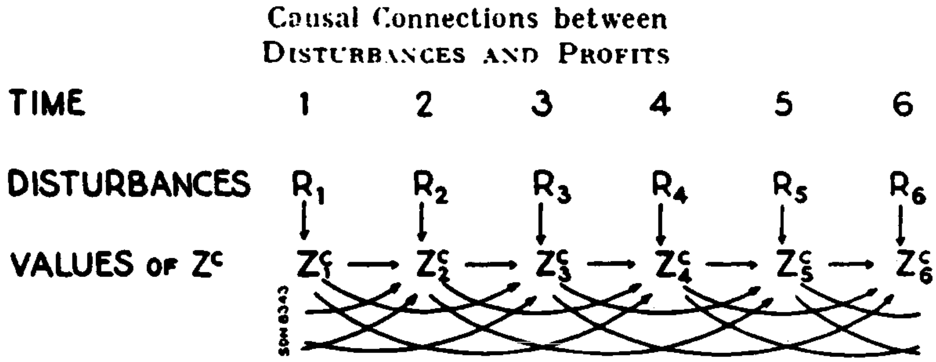


Figure 11 Tinbergen's arrow chart showing the role of disturbances
Source: Tinbergen (1939), II, p. 138, Graph 6.31

sum of the four external variables and the shocks) with the commentary:

> The arrows indicate causal connections. Each value of Z^c depends immediately on certain disturbances, but it depends also on the earlier ones through its connection with the Z^c-values for one, two, three and four years back. (Tinbergen (1939), II, p. 138)

The diagram showed how the internal and external forces of the business cycle mechanism came to be woven together and how the shocks (already cumulated into the final equation) are carried along affecting future values of the systematic variable. It was the disturbances from the external variables and the cumulated shocks (and perhaps, conceded Tinbergen, non-linearities) which

> make it possible that one cycle is completely different from another, and that it is yet, in both, one mechanism that links the variables together. (Tinbergen (1939), II, p. 162)

Tinbergen's careful discrimination between the role of internal and external forces meant that he could categorise policies into those which attempted to change structure (through changing the relationships or their coefficients), or those which tried to change the average level of variables (for example, minimum wage legislation) and those which affected shocks. Policy could be addressed to stabilising one elementary equation or to act on the strategic equations at the kernel.

Tinbergen's critical appraisal of the theories of business cycles surveyed by Haberler was a little disappointing following the richness

discussion of corn and hog cycles in agricultural econometrics in 1925. However, Wright's work was not well known, and Tinbergen has only recently (1979) recognised the similarities.

of his model building. But there were several difficulties in the way of making any very definite inferences. First, there was the problem of assessing economic theories which mostly failed to pass Tinbergen's first test: that of completeness and determinateness. The lack of explicitly specified dynamic relationships in the theories formed a second serious drawback, since these, together with the structure and coefficients of the model, were paramount in determining whether the model passed the final test of providing a cyclical pattern. Thirdly, the appraisal was considerably limited by the shortness of the estimation period of only 13 years.

Tinbergen's general conclusion was that a depression would result from prior 'disproportionalities' (Haberler's term) in the economic system, provided there was unchanged economic structure and no exogenous shocks. Policy changes or shocks might both, according to Tinbergen, intervene to prevent the continued rise of a boom or the continued fall of a depression. This conclusion at least refuted the claims of the periodic models of Moore and Jevons and at most suggested that government policy might have successfully intervened to halt the Great Depression. Tinbergen also used the statistical results that he had found in working with his Dutch, his US and his UK models (see below) to carry out a more direct appraisal of Haberler's theories; this appeared later in 1942.[15] But by this time, war had vanquished the economic depression and business cycle theories were fast going out of fashion.

One further work completed Tinbergen's large-scale macroeconometric modelling programme. This was his model and report on business cycles in the UK covering 1870–1914. The work had been carried out in 1939 and 1940, but the volume was not published until 1951. It followed a similar format to his work for the League of Nations, but by the time the report came out, Tinbergen had new competitors using more sophisticated statistical techniques provided primarily by the work at the Cowles Commission in the 1940s. The econometricians at the Cowles Commission were great admirers of Tinbergen's econometric work and at one stage they thought of applying their methods to Tinbergen's own US model. Instead, they hired L. R. Klein to formulate a new model and this step began the post-war generation of macroeconometric models. The title of Klein's book, *Economic Fluctuations in the United States 1921–1941* (1950) signifies the continuity of thought from Tinbergen's earlier econometric work on business cycles.

[15] But this exercise proved no more definitive because Haberler's theories were grouped into categories which Tinbergen (like Mitchell) viewed as providing complementary rather than rival explanations of the cycle.

4.4 The critical reaction to Tinbergen's work

In contrast to his work on the Dutch model which had made little impact on his fellow economists, Tinbergen's first report for the League of Nations proved highly controversial. It was circulated in 1938, prior to publication, and provoked an interesting and long-lasting discussion on the role of econometrics in theory testing. The second report containing the US model was received more calmly. Because of the differences in content of the two reports, it is important when considering the arguments which follow, to bear in mind which volume is being discussed.

It was J. M. Keynes' (1939) famous critique of Tinbergen's first League of Nations study, 'Professor Tinbergen's Method', which sparked off the debate about the role of econometrics.[16] It was unfortunate that, while Keynes attacked the subject with his ususal rhetorical flourish, he had clearly not read the volume with any great care. Some of his criticisms also revealed his ignorance about both the dynamic economic models of the business cycle developed in the previous decade and the technical aspects of econometrics. For example, Keynes supposed that business cycle theory was still at the stage of Jevons' sunspot theory and he failed to see how cyclical patterns could occur except through periodic causes, and certainly not through a system of linear relations.[17] In another example, Keynes supposed wrongly that trends were measured by joining up the first and last observations of the series, whereas Tinbergen had used a moving average trend or a linear trend term in the multiple correlation. (It is perhaps just a little surprising that Keynes should have been so unaware of the econometric literature in question since he had been on the editorial board of *Econometrica* from its first issue and had been a member of the Council of the Econometric Society since 1935.[18]) To those economists who failed to read Tinbergen's report for themselves and who remained ignorant of the developments of econometrics since the mid-1920s, Keynes' criticisms of Tinbergen's first volume must have been devastating. This was a pity since Tinbergen's econometric work demonstrated how much more advanced the subject had become.

[16] The Keynes–Tinbergen debate remains a popular topic (see, for example, Stone (1978), Hendry (1980), and Pesaran and Smith (1985)). The account here deals not only with the immediate criticism and reply but the contemporary debate which followed.

[17] It is clear also from Keynes' correspondence with Harrod about Tinbergen's work (Keynes (1973), pp. 284–305) that Keynes did not recognise the dynamic mathematical models or Frisch's shock model which Tinbergen used.

[18] Of course, editorial board members are not necessarily aware of everything which appears in their journal. Keynes remained on the Council until his death, and was the Econometric Society's President in 1944–5. It is worth pointing out that many economists had joined the Society at its formation who were not sympathetic to the statistical side of the econometric approach; Robbins was another case in point.

In fact, most of Keynes' specific points of criticism proved invalid since, in his model building and applied work, Tinbergen had dealt with the particular problems raised and carried out the various tests or procedures which Keynes criticised him for omitting.

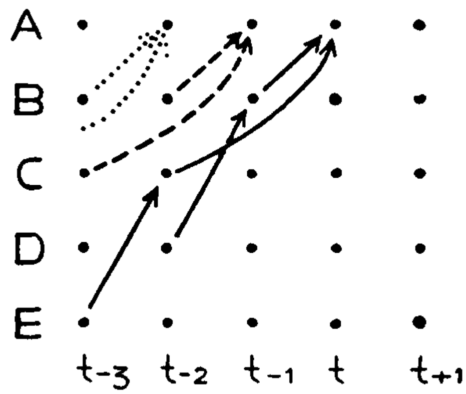
Keynes also made a number of weightier criticisms of the methods and methodology of econometrics. These points were not new but were already the subject of concern to thoughtful econometricians like Tinbergen. For example, Keynes complimented Tinbergen on his discussion of the method of multiple regression, but he felt strongly that the prior 'logical problem' had not been worked out; that is, Tinbergen had not explained

> fully and carefully the conditions which the economic material must satisfy if the application of the method to it is to be fruitful.
>
> (Keynes (1939), p. 559)

These conditions, according to Keynes, were that econometric techniques could only be applied where there is a correct and complete list of causes, where all the causal factors are measurable, where these factors are independent, where the relationship is linear and where problems such as time lags and trend factors are adequately dealt with. Keynes claimed that these conditions had not been satisfied in Tinbergen's work.

Tinbergen defended his work against the general difficulties raised by Keynes, not only in direct reply but in an additional paper in 1940.[19] Tinbergen argued that provided the list of explanatory variables held the most significant or important ones, this was sufficient for the method to provide good measurements of relationships. Further, as econometricians knew, for statistical purposes explanatory factors did not need to be statistically independent of each other but only reasonably uncorrelated. Tinbergen distinguished this statistical dependency from his idea of economic dependency which he represented by an arrow scheme, showing the immediate causes and the secondary causes (the causes of the first causes) which made up a causal chain system of relations. This arrow scheme is reproduced in Figure 12. Tinbergen argued that for estimation purposes, each elementary equation (or statistical explanation) should only represent one level of causes; the economic dependency between the different levels of causes was inherent in the relationships between the equations in the system. Tinbergen used these arrow schemes as a pedagogical tool to represent his model and to help readers understand the logical structure of his

[19] Tinbergen's direct reply (see Keynes and Tinbergen (1940)) and his introduction to (1939), II dealt with many of the issues Keynes had raised, and tried to correct some of his errors. (But Tinbergen's modesty softened the force of his defence (see Magnus and



Symbolic representation of logical structure of dynamic economics (sequence analysis).

Figure 12 Tinbergen's arrow scheme showing causal structure
Source: Tinbergen's reply to Keynes in Keynes and Tinbergen (1940), p. 144, Graph 1

models and their lagged schemes.[20] Keynes had also quibbled about where the time lags in equations come from and wanted to insist that their specification came from the theorist. But theorists gave no help on such matters, so Tinbergen was forced to choose the time lags between causes and effects from a process of estimation by trial and error.

There was a second major area of methodological criticism which followed on from Keynes' concern that the necessary conditions be fulfilled. This was the issue of inductive claims. Keynes wrongly supposed that Tinbergen was primarily interested in measuring the strength of various factors. This was because Keynes believed that the method of multiple correlation

> is one neither of discovery nor of criticism. It is a means of giving quantitative precision to what, in qualitative terms, *we know already as the result of a complete theoretical analysis* – provided always that it is a case where the other considerations ... are satisfied. (Keynes (1939), p. 560; my italics)

Keynes, like so many economists, was unwilling to concede to econometricians any role in developing theory and regarded the measurement

Morgan (1987)). In addition, following his debate with Keynes, Tinbergen was invited by the editors of the *Review of Economic Studies*, to give further explanations of his method. In the resulting article (1940), Tinbergen explained at greater length many of the points which contemporaries had found difficult to understand.

[20] Tinbergen's arrow schemes and discussion indicate he thought in terms of recursive models, but as the equation on p. 118 above shows, his systems could not always be reduced to one single difference equation, suggesting non-recursiveness of some parts of the system. To discuss the topic in these terms is to leap ahead to the later 1940s when the idea of recursiveness was proposed by Wold as an alternative to the simultaneous equations model of Haavelmo (see Chapter 7, Letters 12 and 14).

of already known theories as their only task.[21] Because Keynes 'knew' his theoretical model to be correct, logically, econometric methods could not conceivably prove such theories to be incorrect. Therefore, if a theory was not confirmed by the applied work, then the reason according to Keynes' position must be that the conditions necessary for econometric measurement, as laid out by Keynes (and referred to above), had not been met. More crudely stated, this common argument runs: if the results are not in accordance with theoretical preconceptions, then blame the method and the data but not the theory (an argument reminiscent of the proverb: a bad workman blames his tools, and in this case his raw materials as well).

Keynes had rejected the possibility that econometrics could be innovative about theory or claim to test theory, and regarded Tinbergen's work as 'a piece of historical curve-fitting and description' (Keynes (1939), p. 566). So when Keynes criticised Tinbergen for his lack of inductive claims, it was for lack of foretelling the future. But, Tinbergen had, as a matter of course, forecast two periods ahead with many of his equations as a test of the model. Keynes had also warned, quite correctly, that in order to forecast from the model, the periods must be homogeneous and recommended Tinbergen to investigate this matter by testing the model on different subperiods of the data series. Again, this is precisely what Tinbergen had done, and had found evidence of changes in structure in his comparison of the regression coefficients! Tinbergen believed that changes in structure did not necessarily invalidate the usefulness of the whole model since the changes might be localised or might be subject to estimation. Alternatively, the changes might be in the values of variables rather than in the functions themselves. In any case, he thought that the more interesting problems were the variation problems (tracing the effect of alternative policy changes) rather than the problem of short-term forecasting.

Tinbergen, in his reply to Keynes, defended the wider inductive claims that he made for his own work. For Tinbergen, econometrics was concerned both with discovery and criticism, as should be clear from the discussion of this chapter. He believed, contrary to Keynes, that if the theory was not confirmed by the results, then the inference was that the theory was wrong or insufficient. Thus, for example, statistical work might result in the introduction of new variables, or

[21] Tinbergen has a delightful example of this attitude in which Keynes 'knew' the numerical value of the parameter in his theory. I have not discussed details of the model content in this account, but Neil de Marchi points out that one of the reasons Tinbergen found Keynes' comments peculiar was that he was influenced by Keynes' macroeconomic ideas. On both points, again, see Magnus and Morgan (1987).

forms of variables, not previously considered as important by theory. Theory might also be criticised by statistical evidence, if, for example, little or no influence was found in the regression relationship for a variable considered important in the theory. It is clear that Tinbergen saw econometrics as a way of evaluating different theories; a much wider frame than that viewed by Keynes.

There was another very important criticism of Tinbergen's first report not published at the time though it was distributed in mimeographed form: this was a memorandum written by Frisch for a conference held in 1938 to consider Tinbergen's first League of Nations' volume.[22] Frisch's memorandum discussed the relationship between the economic relations of theory and those obtained by fitting curves to data. He suspected that there was a considerable gap between Tinbergen's applied results and economic theory, due to a large measure of arbitrariness in the solutions to his final equations. Frisch argued that in finding the component solutions of time-series data, the values found for amplitudes and timing of phases were only relative; they were fixed absolutely only by the initial conditions. A further element of arbitrariness was due to the model chosen: different choices of lags, for example, would give different solutions. In general, the solutions were not unique and different equations could give the same solutions. Those equations which were not 'reducible' (that is, those equations with the lowest element of arbitrariness) for the system as a whole were the only ones which could actually be discovered from the data.

Unfortunately, according to Frisch, these 'coflux' (irreducible) equations often had a low degree of 'autonomy'. Autonomy is a concept concerned with economic behaviour and theory: relations with a high degree of autonomy were defined as ones which remain unchanged when other relations alter. There is thus a sort of 'super-structure' in the economy consisting of those relations with a high degree of autonomy. These relationships are more fundamental and give deeper insights into the behaviour of the economic system than ones of low autonomy. If they can be found, they are closest to being a 'real explanation' of the economy. Frisch gave the following example to illustrate his concept:

[22] Although both volumes were officially published by the League of Nations in 1939, the first volume was circulated in advance in 1938 and evaluated at a special conference in Cambridge (England) in July of that year. Neither Keynes nor Frisch attended this meeting but Frisch produced a written critique which arrived after the event. This memorandum, Frisch (1938) (to be published for the first time in Hendry and Morgan (forthcoming)) gave a new stimulus to the work on identification by Koopmans and others at the Cowles Commission in the 1940s (see Epstein (1987)), and was crucial in the development of the concept of structural relations which is my concern here (and see Aldrich (1989)).

> The coflux relations that can be determined by observation of the actual time shapes may or may not come near to resembling an autonomous relation, that depends on the general constitution of the phenomena studied. To give two extreme examples: the demand function for a consumers commodity as depending on price and income and perhaps on some secondary variables will, if the coefficients can be determined with any degree of accuracy, come fairly near to being an autonomous relation. It will not be much changed by a change in monetary policy, in the organisation of production etc. But the time relation between the Harvard A, B, and C curves is a pure coflux relation, with only a small degree of autonomy. (Frisch (1938), p. 17)

Frisch accused Tinbergen of finding only coflux equations, which could not be used to refute theory since they did not necessarily express the autonomous structural system which was equivalent to theory.

Tinbergen, in immediate reply to the 'Autonomy' Memorandum, interpreted Frisch's criticisms as being concerned with multicollinearity. It is important to note that this term had been earlier defined by Frisch to mean the existence of linear dependence between variables (including the explanatory variables of an equation), other than the relationship of interest. Multicollinearity could also therefore in Frisch's definition denote identification problems; and indeed in the 1930s these concepts were not separate but interlinked principally because of Frisch's work on confluent relationships. For example, in the context of business cycle work, multicollinearity might mean that the structural relationships were obscured by parallel cyclical movements in all the variables. Tinbergen interpreted Frisch's view of the problem as being that

> all business cycle curves are more or less sines or waves, and that therefore the danger of multicollinearity is permanently present,
> (Tinbergen, 1938, see Frisch (1938/48), p. 20)

and argued that in this case, bearing in mind his model, parallel data series were not particularly a problem.

In a further response to Frisch (at the beginning of his second report to the League), Tinbergen pointed out that he had started out from theoretical relationships, and the individual equations he estimated reflected the direct causes (rather than the secondary causes). He believed this would give his relationships the maximum degree of autonomy, that is,

> relations which are as little as possible affected by structural changes in the departments of economic life other than the one they belong to.
> (Tinbergen (1939), II, p. 14)

He thought that his relationships would have high autonomy compared to the observed A-B-C relationships found by the Harvard method. Further, Tinbergen argued that the use of economic theory put a constraint on the choice of lags, reducing the danger of arbitrary solutions. So, although Tinbergen took note of Frisch's criticisms in his second report, he did not view them as seriously undermining his work on the US model.

Because of the controversial reception accorded to the first report, there was some anxiety at the League of Nations about publishing the second. In fact, Tinbergen's second volume received less critical attention, possibly because its publication was overtaken by the outbreak of war. The most critical of these reviews was by Freidman (1940), who also discussed the issue of testing. He complained that no statistical tests had been carried out, and that none could be, because the estimated equations were all the result of correlation hunting (using trial and error to obtain high values of the multiple correlation coefficient). Friedman went on to quote Mitchell in support of his view that the only way to test such an empirically derived model was to see how it performed on other data sets. (At this point, it must be remembered that Friedman was reviewing the second report, and that it was in his first report that Tinbergen had tested his model on other time periods and on data from other countries.) Friedman himself experimented in this direction by using Tinbergen's final equation to forecast five years ahead. He described the result as 'unimpressive' ($R^2 = .68$ between actual and forecast amounts) but the cyclical pattern was reasonably good.

Friedman's closing statement was that

> The methods used by Tinbergen do not and cannot provide an empirically tested explanation of business cycle movements. His methods are entirely appropriate, however, for deriving tentative hypotheses about the nature of cyclical behavior; and from this viewpoint students of business cycles may find Tinbergen's volume of considerable interest.
>
> (Friedman (1940), p. 660)

Tinbergen had provided a statistically verified explanation of the business cycles in the US for a particular period and had checked to see that the model would generate cycles in the absence of shocks. He had not made a test of the model against new data, so Friedman's judgement is justified. But, Friedman, unlike Keynes, was willing to see Tinbergen's work as a useful source of further theorising.

A more technical criticism was advanced by Haavelmo (1940).[23] In

[23] A further technical problem was pinpointed by Orcutt (1948) who argued that Tinbergen had a serious problem of autocorrelation in the data used for his US model. This criticism

this paper he dealt only with the final testing stage of Tinbergen's work and criticised him (and others) for neglecting the effect of the errors in determining the time path of the system. He showed that it was possible for a system of equations with a non-cyclical solution to explain observed cycle movements once the effects of the errors were taken into account. (This was a reminder of Slutsky's result that cycles can occur solely through the summation of random error terms.) It therefore followed, according to Haavelmo's work, that a cyclical solution to the economic system part of a business cycle model could not be taken as test of adequacy for business cycle models, as both Frisch and Tinbergen had proposed. This, of course, was a serious problem for Tinbergen's testing programme, but, like Keynes' criticisms, Haavelmo's observations had little effect on the practice of econometrics in this case because the business cycle programme of study was near its end, at least for the time being.

Tinbergen's work had aroused strong criticism, but he also had many staunch supporters who joined in the debate about the role of econometrics. Some, like Allen (1940), used their own reviews of Tinbergen's League of Nations reports to defend him. Marschak and Lange (1940) wrote a stronger reply than Tinbergen's for the *Economic Journal*, but it was not published at the time.[24] Particularly important is Haavelmo's (1943) response to the critique of Keynes. This paper provided a highly articulate discussion of the role of econometrics in testing theory. Haavelmo defended the econometric approach (and Tinbergen's work) but he also wanted to insinuate probability theory as the essential missing component which would make the approach fruitful. (This had not, by any means, been thought a necessity by econometricians, as we shall see later.) Haavelmo stated:

> When we speak of testing theories against actual observations we evidently think of only those theories that, perhaps through a long chain of logical operations and additional hypotheses, lead to *a priori statements about facts*. A test, then, means simply to take the data about which the a priori statement is made, and to see whether the statement is true or false.
>
> Suppose the statement turns out to be true. What can we then say about the theory itself? We can say that the facts observed do not give any reason for rejecting the theory. But we might reject the theory on the basis of other facts or on other grounds. Suppose, on the other hand, that the

marks the start of new developments, which are not treated in the present study, but see Gilbert (1988).

24 Keynes, as editor of the journal, perhaps felt enough had already been said. The paper is available in Marschak's papers, (UCLA Special Archive Collection), and is to be published in Hendry and Morgan (forthcoming).

> statement turns out to be wrong. What we can then say about the theory depends on the characteristics of the theory and the type of a priori statement it makes about the facts. (Haavelmo (1943), pp. 14–15)

For example, Haavelmo argued, no economist would work with a theory which implied that next year's national income would be exactly $X million, since this theory would always be rejected. That is,

> If the statement deduced is assumed to be one of necessity, we would reject the theory when the facts contradict the statement. But if the statement is only supposed to be true 'almost always,' the possibility of maintaining the theory would still be there ... If the statement is verified, should we accept the theory as true? Not necessarily, because the same statement might usually be deduced from many different constructions. What we can say is that an eventual rejection of the theory would require further tests against additional facts, or the testing of a different statement deduced from the same theory. Each test that is a success for a theory increases our confidence in that theory for further use.
> (Haavelmo (1943), p. 15)

Haavelmo's position had two implications. One was that, in answer to Keynes, simple models were legitimate, since if the

> simplified theory also covers the facts, the discovery is an addition to our knowledge (Haavelmo (1943), p. 15)

and might provide constraints additional to a priori knowledge. The second implication was that a stochastic formulation was necessary for workable and fruitful econometrics because otherwise inference would mean that we would either always reject a theory or never reject a theory, depending on whether it was sharply or broadly specified. Haavelmo clearly differentiated Tinbergen's pioneering macroeconometric models from previous statistical work on business cycles and approved of his work. But he criticised him for not going far enough in his use of statistical reasoning to warrant the description of 'deriving relations with "inductive claims" ' (as Keynes had described the aim) versus the alternative of 'historical curve fitting' (Haavelmo (1943), p. 17). The proper formulation of the problem required to assess these inductive claims was still lacking in econometrics. That was the role, Haavelmo argued, of probability theory. This important development is discussed in Chapter 8.

Abstracting from the immediate debate enables us to evaluate Tinbergen's achievements and put them into perspective. When the work of Tinbergen is compared to that of Mitchell, Persons, Jevons and Moore, it is clear that there had been real advances in the quantitative treatment of the business cycle. First, like Moore, Tinbergen relied on quantitative theory, but he had the advantage of Frisch's general mathematical model designed especially for business cycles. Tinbergen used this model to develop complex multi-equation econometric models to represent the cycles of the whole economic system.[25] In the process of building and using his models, Tinbergen increased econometricians' understanding of the nature and properties of econometric models and of associated concepts such as the structure of the system and causal chain models. Second, this was not a period of technical progress in statistical techniques. Correlation and regression analysis (in various guises) were still the main analytical methods; but a comparison points up the greater skill and finesse in Tinbergen's econometric work compared to that of Moore. His development of testing procedures for theory evaluation entailed a broadening in scope and clarification of purpose rather than the introduction of new techniques. Tinbergen's contributions to econometrics came through his practical work: he developed best practice for applied econometrics, and he deepened econometricians' understanding of their approach to economics.

[25] In this context, we should note that Tinbergen's models are generally regarded as the first macroeconometric models. This is undoubtedly correct, but to see them only in this light is to ignore the general structure of his models (and particularly his concern with dynamics) and the historical and theoretical business cycle context from which they sprang. Tinbergen's models owed more to the development of dynamic cycle theory, both by the econometricians themselves (and particularly Frisch's shock model) and by the Stockholm school of economists (for the development of sequence analysis), than it did to the development of macroeconomic theory. Conventional accounts of the development of macroeconomic theory and data in the 1930s often neglect the business cycle theory context from which they emerged and ignore econometrics entirely. Patinkin (1976), for example, concentrates on the advent of national income accounting (for which, see Kendrick (1970)) and Keynesian macroeconomics as the two great achievements of interwar economics. The renewed interest in business cycles may lead to some revision of this view (see, for example, Lucas (1980) whose brief historical account supports my view of the importance of Tinbergen). We should also note that it was for developing and applying macrodynamic models that Tinbergen and Frisch were awarded the first Nobel Memorial Prize in Economics in 1969.